

The Ψ - $\Delta_{I \rightarrow G}$ Framework: V -Structure Coupling and Structural Gaps under Finite Resolution

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Abstract

This paper formulates the Ψ - $\Delta_{I \rightarrow G}$ framework as a package-conditioned diagnostic of structural gaps in multi-level systems. Its guiding claim is that finite descriptions do not report objective structure from an exterior position. Reportability is produced through a recursively declared package structure: substrate packages lock operative state spaces and reachability conventions; perspective packages lock admissible readouts; a relation-substrate package locks the complete instance-pairing structure when that structure is not canonically derived; and a relation-diagnostic package locks reconstruction, discrepancy, aggregation, query, domain, and reporting resolution. The generic symbol $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ denotes the unchanged four-slot package schema at any node of this dependency structure.

No package creates the underlying organization, but each package node determines the form in which its operand structures become reportable. Changing a substrate, perspective, pairing, or relation package changes which distinctions are visible, which instances correspond, which reconstructions are admissible, which residuals are measurable, and which comparisons are licensed. Every carrier entering a diagnostic must therefore be package-locked or explicitly derived from package-locked carriers; no state, reachability, optimization, or pairing domain is treated as a bare background space.

Functionally, the framework distinguishes carrier, report, and bridge roles among the typed components produced or licensed by these declarations. This role decomposition is orthogonal to package-dependency depth: packages and declarations are typing and locking specifications rather than members of the role classification, and a single package may therefore contain clauses relevant to more than one functional role.

Within the report role, the framework separates two diagnostic quantities of different arity. The intra-level structural tension readout Ψ is unary at the perspective level: it records infimal residual tension within one package-locked perspective. The fixed-slice gap $\Delta_{I \rightarrow G}$ is an ordered relational readout: two package-locked perspectives and a declared pairing structure are its operands, while the relation package supplies the admissible class $\mathcal{U}_{I \leftarrow G}$, the I -side discrepancy D_I , the aggregation rule, and the relation threshold ε_R . The pairing structure fixes which operand instances correspond but does not select a reconstruction direction. The subscript $I \rightarrow G$ names the compression or reporting passage being audited; its sufficiency is tested by admissible reverse reconstructions $G \rightarrow I$ with discrepancy evaluated on the I side. The Ψ - Δ pairing is therefore an arity-shifted diagnostic duality within the report role, not an algebraic duality, not the full functional architecture of the framework, and not a claim that the two quantities occupy the same package depth. The familiar notation $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ is retained as the total-readout, shared-substrate, identity-pairing, minimax flattened specialization.

To describe persistence under declared evolution, refinement, or scale change, the paper introduces a minimal dynamical interface. Along a declared evolution parameter γ , $\Psi_{\text{run}}(\gamma)$ is treated as a slice-local running readout, and a typed accumulated readout $\Delta_{\text{acc}}(\gamma)$ is

updated by a declared type-preserving accumulation rule:

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(\gamma') d\gamma'.$$

When Ψ_{run} is locally integrable, the accumulated quantity is absolutely continuous and the running readout is its derivative almost everywhere along the declared axis. The report type of Δ_{acc} is inherited from the declared accumulated-report package and maintained by its transport clauses; the integral supplies a compatible increment rather than creating a fixed-slice relational Δ by itself. This is a definition-level reporting interface, not an unconditional physical law and not an identification of every local residual carrier with Δ_{acc} . Comparison with a separately completed fixed-slice endpoint relation report requires a licensed endpoint–path comparison bridge.

The main contribution is a minimal reportability framework in which substrate identity, instance pairing, and reconstruction sufficiency are separated and independently auditable. Threshold binding is local to the package that produces the reported quantity; thresholds at distinct dependency nodes are independent by default and require an explicit alignment clause only when they are jointly consumed or compared. The resulting discipline also separates what is structurally carried, what is reported, and what requires a separately licensed bridge: carrier transport does not by itself propagate a scalar report, and a theorem-derived sufficient budget is not a constitutive threshold declaration. The paper supplies formal definitions, dependency-aware package cards, admissibility and comparability gates, resolution-aware reporting semantics, explicit undefinedness diagnoses, and a fixed-target resolution-feasibility module. Detailed macro-closure mechanisms, residual-complement licensing, closure-restoring extensions, and local fiber-activity diagnostics remain companion modules rather than primitive assumptions of the present framework.

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1 Introduction: V -Structure Coupling and Cross-Level Reportability

Across logic, physics, computation, learning, and coarse-grained modeling, a recurring difficulty appears: a compressed or upper-level description is often operationally useful and is then over-read as if it were structurally sufficient. The object may be an endpoint, a macro-label, a compressed state, a quotient variable, or a protocol-visible report. Such objects can be legitimate within a declared task, but their legitimacy does not by itself imply reconstructive sufficiency, cross-level closure, or ontological identity.

The problem is not compression itself. The problem is undeclared structure around compression. A cross-level diagnostic becomes determinate only after four type-distinct questions have been answered: which substrate is being described; how each perspective reads that substrate; which instances are paired; and whether the declared G -side readout can reconstruct the paired I -side readout under an admissible reconstruction class and discrepancy rule. Reporting resolution is a further package-local clause applied to the resulting report. Without these declarations, endpoint agreement, macro-label stability, approximate equality, instance correspondence, and reconstructive adequacy can slide into one another without a licensed relation structure.

This paper formulates that dependency through the generic package schema

$$V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C}).$$

The symbol V does not denote one universal flat bundle. It denotes the same four-slot schema at recursively related nodes: substrate packages $V_{\Sigma_I}, V_{\Sigma_G}$ lock the operative carriers and reachability conventions used by the two operand perspectives; perspective packages V_I, V_G lock readouts of those carriers; a relation-substrate package V_{Σ_R} locks the complete pairing structure when it is not canonically derived; and a relation-diagnostic package V_R locks reconstruction, discrepancy, aggregation, task, domain, and relation resolution. The two operand substrate packages may coincide, as in the shared-substrate core, but they need not do so in the general relation type. Dependency depth is relative rather than absolute. A package may cite previously locked structures as operands, but it may not silently redefine them.

Three notions are kept distinct throughout. A descriptive level concerns the representation/state distinction introduced below. The package-dependency axis records which locked operands a declaration depends on and therefore its relative dependency depth. The functional-role axis classifies the typed components produced or licensed by those declarations as carriers, reports, or bridges. Packages and declarations themselves are not assigned one of these three roles: they are typing and locking specifications, and their slots may declare components serving different roles. Descriptive level, dependency depth, and functional role are therefore not used as interchangeable hierarchy labels.

Within the report role, the framework separates two diagnostic quantities. The intra-level quantity Ψ is unary: it records residual tension inside one package-locked perspective. The

fixed-slice quantity $\Delta_{I \rightarrow G}$ is relational: it records the infimal aggregated residual left when the G -side readout of paired instances is used to reconstruct the corresponding I -side readout. The pairing structure (Z_R, p_I, p_G) fixes correspondence but does not select which side must reconstruct the other. The subscript $I \rightarrow G$ records the compression or reporting passage under audit; the admissible reverse reconstruction class $\mathcal{U}_{I \leftarrow G} : \Omega_G \rightarrow \Omega_I$ and the I -side discrepancy test whether that passage is sufficient. Thus Ψ and $\Delta_{I \rightarrow G}$ form an arity-shifted diagnostic duality: one asks within a perspective, the other asks between two perspectives under a declared relation. This duality is internal to the report role and does not define the full functional architecture.

The current core retains the earlier formula $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ as a flattened specialization. In that specialization both perspective maps are total on the same package-locked reachable substrate, the relation-substrate package coincides with the shared substrate package, and $p_I = p_G = \text{id}$. Identity pairing is still a declared pairing clause; it is not the absence of pairing. More general function-type, partial, or many-to-many pairings use the same two-projection form and require a complete package-locked or canonically derived pairing structure, including the relation carrier and both projection maps.

Resolution is part of the semantics, not a numerical afterthought. A relation report $\Delta_{I \rightarrow G} \approx_{\varepsilon_R} 0$ means that the completed relation residual lies below the threshold declared by V_R . Thresholds belonging to substrate, perspective, relation-substrate, and relation-diagnostic packages are independent by default. No threshold is inherited across dependency edges unless an explicit alignment clause is declared. If the required clauses are absent, the output is not a guessed scalar but an undefined or non-licensed report.

Finally, the paper introduces a minimal dynamical interface. Along a declared evolution, refinement, hierarchy, or scale parameter γ , a slice-local running readout $\Psi_{\text{run}}(\gamma)$ may supply compatible increments to a typed accumulated readout $\Delta_{\text{acc}}(\gamma)$ under a declared transport law. The update preserves the accumulated-report type (and any pairing/correspondence clause carried by that type) rather than generating a fixed-slice relational Δ from integration alone. This interface is not asserted as a universal physical law and does not identify every local residual carrier with Δ_{acc} by default. Comparison with a separately completed fixed-slice endpoint relation report requires a licensed endpoint-path comparison bridge.

The resulting core dependency route is therefore recursive rather than flat: package-lock the substrate, package-lock the perspectives, package-lock or derive the pairing, declare the directional relation diagnostic, and only then report $\Delta_{I \rightarrow G}$ at the relation threshold. The unary Ψ report route terminates at one perspective package; the relational Δ report route requires the additional pairing and relation-diagnostic declarations.

1.1 Motivating examples: calculus, Bayesian conditioning, and RG running

The claim that structure becomes reportable only through a declared coupling specification can sound more radical than it is. The point is not that objective structure is created by a description, nor that every structural statement is merely subjective. The point is that a finite report has to declare an interface through which structure is accessed, compressed, compared, and made negligible or non-negligible. Several standard scientific languages already exhibit this dependency. The following examples are not used as proofs of the framework. They calibrate the semantic point that a reportable object is not a viewpoint-free copy of structure, but a V -conditioned form in which structure becomes reportable.

Calculus: whole structure through an infinitesimal interface. In elementary calculus, a whole object is often re-expressed through local infinitesimal reports. For a differentiable function, one may write locally

$$f(x+h) - f(x) = f'(x)h + r(h), \quad \frac{r(h)}{h} \rightarrow 0,$$

and then pass to the differential expression

$$dy = f'(x)dx.$$

The differential dx should not be read simply as a discarded small quantity. It marks the local reporting interface through which the whole curve is re-expressed as an infinitesimal linear report. Likewise, an integral re-expresses a whole quantity by a declared decomposition into local pieces and a limiting summation protocol. Thus the passage between whole object, local differential element, and reconstructed aggregate is licensed by coordinates, partition rules, limiting conventions, and negligibility clauses. Calculus therefore already displays a primitive form of V -structure coupling: the same structure is not reported in the same form when it is read as a whole curve, as a local tangent report, or as an accumulated integral. The equality notation is legitimate inside the declared calculus, but the semantic bridge is carried by the limiting interface, not by a viewpoint-free identity between all levels of description.

Bayesian conditioning: event structure under declared evidence. Bayesian conditioning gives a non-geometric example of the same pattern. The event A is not changed as a set-theoretic object when evidence B is declared, yet the report changes from $P(A)$ to

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

when $P(B) > 0$. The conditional value is not attached to A alone; it is attached to A under a declared condition, background measure, and normalization domain. In this sense, the evidence event is not a verbal annotation appended after the probability has already been determined. It is part of the reporting interface that selects the relevant slice of the probability structure. This example is useful because it prevents V from being misread as merely a scale parameter. A package can include evidence, admissible information, likelihood/readout rules, and normalization clauses. When the report is distribution-valued, Bayesian conditioning may be viewed as a conditionalized update of unresolved V -fiber candidates after a declared evidence event. This reading introduces no new primitive; it is a special probabilistic instance of the broader claim that reported values are condition-dependent structural readouts.

RG running: effective parameters under scale, scheme, and interface. Renormalization-group running gives a physical version of the same dependency. A coupling written as $g(\mu)$ is not a bare copy of objective structure. It is an effective coordinate of structure under a renormalization scale μ , regulator and scheme conventions, threshold rules, and an observable interface. Changing μ changes the reporting slice, and the parameter must flow so that physical reports remain comparable under the declared renormalization conditions:

$$\mu \frac{dg}{d\mu} = \beta(g).$$

The running of $g(\mu)$ is therefore not best understood as a constant arbitrarily changing in itself. It is the compensatory flow of a V -conditioned coordinate when the scale component of V is varied. Finite truncations, finite parameter families, or endpoint-only compression may leave distinct reportable structures on a declared scale window. In the endpoint–path application, the strict residual Γ_{path} tests whether the joint endpoint report is sufficient for a declared path-process discrimination task, whereas the geodesic excess Γ_{geo} tests whether the declared RG trajectory realizes the endpoint distance induced by the interface. These are type-distinct diagnostics; any relation between them requires an additional declared bridge [18].

Together, these examples isolate the same semantic pattern in three different registers. Calculus shows whole/local re-expression through an infinitesimal limiting interface. Bayesian

conditioning shows event/evidence re-expression through a conditional reporting interface. RG running shows physical-parameter re-expression through a scale–scheme–interface package. In all three cases, the reportable object is not obtained by simply copying structure into language. It is generated by the coupling between objective organization and declared access conditions. This is the function assigned here to V -structure coupling.

1.2 Terminology and scope: levels, perspectives, and multi-level systems

Throughout this paper, a *perspective* A means a declared representation scheme: it specifies (i) which variables or structural factors are treated as distinguishable, (ii) which background conditions are held fixed, and (iii) which aggregation/compression or identification rules are used to form the representation. A *level* is a representation stratum induced by such a declared representation together with a declared resolution regime: different levels correspond to different representation/state spaces (e.g. micro/individual descriptions versus macro/global summaries) connected by generally non-invertible maps. A system is called *multi-level* when its analysis necessarily involves at least two such levels linked by an irreversible passage (coarse-graining, completion, identification, or any other genuinely cross-level gluing).

The term *level* is reserved for this representation/state distinction. *Dependency depth* refers only to the relative position of a package declaration in the dependency graph. *Functional role* refers only to the carrier/report/bridge classification introduced in Sec. 3. These axes are distinct and are not used interchangeably.

All operational diagnostic reports and numerical comparisons in this paper are relative to a declared package dependency structure. The generic schema $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ is unchanged at every node, but the node’s operands and interface content differ by type. A substrate package locks boundary and reachability; a perspective package locks a readout; a relation-substrate package locks or derives a pairing carrier; and a relation-diagnostic package locks reconstruction, discrepancy, aggregation, task, domain, and its own reporting threshold. This package-relative stance is not stylistic: without these declarations, substrate identity, instance correspondence, and reconstruction sufficiency are not jointly well-posed.

1.3 Structural fracture law as a working meta-constraint

The structural fracture law is used in this manuscript as a motivating meta-constraint, not as a theorem and not as a premise from which the formal objects are mechanically derived. The canonical statement is recorded in Sec. 6. At the level of the present introduction, its function is narrower: it names the recurring pressure that motivates the V -structure coupling framework.

Operationally, the law points to cases in which opposing contributions, feedback, and a single cross-level continuity demand make a package-free judgment unstable. The framework does not treat that instability as a slogan. It translates it into declared objects: intra-level residual tension is read through Ψ , cross-level compression–reconstruction residuals are read through $\Delta_{I \rightarrow G}$, and missing declaration clauses are reported as **undefined** rather than replaced by implicit comparisons. Thus the law functions as orientation; the reportable content is carried by the package-locked diagnostics and the resolution-aware reporting protocol.

1.4 From a static constraint to an operational diagnostic object

The paper proceeds by separating two distinct sources of obstruction:

- **Intra-level tension:** residual conflict that persists even when one stays within a fixed level and a fixed perspective (no cross-level gluing).
- **Inter-level residual:** residual mismatch generated by passing between levels (e.g. compressing to a global description and attempting to reconstruct individual structure), even when one allows an admissible class of reconstructions.

This separation is implemented by the $\Psi\text{-}\Delta_{I\rightarrow G}$ pair. The first object is unary at the perspective level; the second is relational and requires a declared pairing structure plus a directional reconstruction interface. Their pairing is therefore diagnostic and arity-shifted, not algebraic. Every report is evaluated on package-locked carriers and at the threshold of the package node that produces that report.

1.5 Declared regime and comparability discipline (package discipline)

Every report in this framework is package-relative. The notation V is used generically for one declared package node and, where explicitly stated, as a flattened shorthand for a completed dependency record. Threshold binding is node-local. Relation formation from declared operands is distinct from comparison between already completed reports; only the latter is governed by the cross-package transport rule. The detailed semantics of resolution-aware equality, non-licensing outputs, and completed-report transport are fixed in Normative Rules 2.1–2.2.

1.6 Positioning relative to existing frameworks

The $\Psi\text{-}\Delta_{I\rightarrow G}$ object is not introduced as a competing replacement for existing domain frameworks; it is a V -conditioned *reporting discipline* that can be instantiated within them.

- **Renormalization/coarse-graining viewpoints.** The use of multi-level passages and admissible reconstructions is compatible with RG/EFT practice [7, 13, 14, 12]. Reporting endpoint quantities as such does not by itself invoke the endpoint-sufficiency criterion; the criterion applies whenever those quantities are used to stand in for, summarize, or license a conclusion about the intervening finite window. In that use, the corresponding endpoint-sufficiency claim is licensed only under a declared package fixing the task, operands, interface, discrepancy, and resolution. Cross-level identification likewise requires the relevant package clauses, with explicit undefined outputs when they are absent.
- **Information-geometric and divergence-based diagnostics.** Many diagnostics are naturally expressed using relative entropy and its induced local geometry [2, 1, 18]. The framework keeps these tools admissible as package-level choices (e.g. as a discrepancy D), while separating them from the *comparability discipline* (transport, domain gates, resolution conventions).
- **Probabilistic instantiation as a reporting interface.** Appendix B records a distribution-valued reading in which the already-declared diagnostic $\Delta_{I\rightarrow G}(\mathcal{L}; V)$ is translated into TV/IPM reporting readouts once the observation space, admissible lifting class, and discrepancy rule are locked. The instantiation introduces no new core object.
- **Transport and cross-report comparability.** When comparisons across packages are invoked, explicit transport clauses are required; when an optimal-transport style reconciliation is intended, the framework treats it as an explicit declared component rather than an implicit identification [11].

The distinguishing feature is therefore not the choice of discrepancy or the underlying domain model, but the insistence that: (i) reporting semantics is locked by a declared package, (ii) cross-package comparisons require declared transport, and (iii) under missing clauses the correct output is undefined rather than an unscoped surrogate.

1.7 Main objects, contributions, and what is proved

Fix a package-locked substrate $\mathfrak{S}_\Sigma$ and perspective package V_A for a unary report. For a relational report, fix package-locked operand substrates $\mathfrak{S}_{\Sigma_I}, \mathfrak{S}_{\Sigma_G}$, perspective packages V_I, V_G ,

a complete pairing structure $\mathfrak{P}_R$, and a relation package V_R . The two operand substrates may coincide. Two structural quantities are introduced:

- **Intra-level tension Ψ .** Within one package-locked perspective, Ψ is the infimum of a declared nonnegative conflict functional over the relevant locked reachable domain.
- **Relational reconstruction gap $\Delta_{I \rightarrow G}$.** Given paired instances, $\Delta_{I \rightarrow G}$ audits the declared passage $I \rightarrow G$ through the infimal aggregated residual obtained by reconstructing the I -side readout from the G -side readout via $\mathcal{U}_{I \leftarrow G}$ and evaluating D_I on the I side. The earlier $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ notation is the shared-substrate, identity-pairing, minimax flattened specialization.

Beyond definitions and reporting discipline, the paper establishes strictly operational non-vanishing mechanisms at the package-relative level: under metric or symmetric-pseudometric discrepancy rules, whenever a macro representation non-injectively identifies interface-distinguishable states within a single macro fiber (i.e., induces fiber non-constancy), the minimax residual $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ admits an explicit positive lower bound (Theorem 5.1 and Corollary 5.1). This supports witness-based “not resolution-negligible” reports at finite ε without invoking any ontological equality claim.

Separation of statement categories (definitions vs. proved statements vs. protocol).

To support long-term study and controlled reuse, the manuscript keeps four statement categories explicit:

- **Defined objects:** $\Psi(\mathfrak{S}_{\Sigma_A}, V_A)$, $\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R)$, their explicitly stated flattened shorthands, and the typed accumulated readout $\Delta_{\text{acc}}(\gamma)$ (Sec. 7), and the comparability/undefinedness semantics (Sec. 8).
- **Adopted semantic axioms/postulates:** resolution-aware cross-level “=” is read through the package-local $\approx_{\varepsilon_\bullet}$ predicate in reporting (Normative Rule 2.1); and genuine cross-level relations carry a non-vanishing meta-gap schema $\Delta_R^* > 0$ (Sec. 6.3).
- **Proved operational claims:** when the required package clauses are declared, certain non-negligibility reports (e.g. positive lower bounds for $\Delta_{I \rightarrow G}$ under fiber non-constancy of the coarse representation relative to the interface content) are supported by explicit witnesses (Theorem 5.1 and Corollary 5.1).
- **Protocol clauses:** package cards, admissibility/comparability gating, resolution-feasibility checks when a certified mechanism family is invoked, and explicit undefined outputs are treated as part of the reporting semantics, not as mathematical conclusions (Sec. 8).

The framework also provides: (i) a reproducibility-oriented reporting protocol (package cards, admissibility/comparability gating, and explicit undefinedness diagnoses, optionally tagged in Appendix A); (ii) a fixed-target resolution-feasibility core, formulated as typed refinements of the existing package clauses, with a full pairwise/partition module in Appendix E; (iii) a minimal dynamical interface in which accumulated residual readouts are tracked along a declared evolution coordinate γ ; and (iv) optional probabilistic, dynamical, and geometric-closure modules that remain subordinate to the same package and reporting protocol. Detailed macro-closure, residual-complement licensing, closure-restoring extensions, and local fiber-activity diagnostics are treated as companion closure modules rather than as primitive assumptions of the present framework [17, 15].

Minimal guiding example. Suppose two total perspective maps read the same package-locked reachable substrate and the relation package declares identity pairing. If the G -side map identifies two substrate states whose I -side readouts remain distinguishable, no admissible reconstruction can be uniformly exact on that fiber. The framework records the resulting relation residual as a non-negligible $\Delta_{I \rightarrow G}$ when the pairing, discrepancy, aggregation, lifting, domain, and threshold clauses are declared, or as undefined when any required clause is absent.

1.8 Roadmap

Section 2 fixes the type discipline and the resolution-aware semantics used throughout. Sections 3–5 introduce the package-relative objects, the fixed-target threshold-feasibility module, and the core diagnostics. Section 6 records the structural fracture law and its diagnostic translation. Section 7 introduces the minimal dynamical interface, makes the slice-local $\Psi_{\text{run}} \rightarrow \Delta_{\text{acc}}$ dependency explicit, and separates path accumulation from any optional endpoint–path comparison bridge. Section 8 formalizes well-posedness, licensed comparisons, package cards, resolution robustness, locked adaptation, and undefined outputs. Appendices record optional status-tag, probabilistic, dynamical, geometric-closure, and full resolution-feasibility modules. Any illustrative material retained in the appendices is optional and is not used by the core claims. None of these materials changes the core V –structure coupling semantics.

2 Conventions and Reader Guide

2.1 Symbols and type discipline (system vs. declared package)

This paper distinguishes *target-system objects* from *analysis/reporting objects*. The distinction is semantic (what is being studied vs. how it is reported) and is enforced to avoid silent shifts of meaning across sections.

Package-locked substrate and declared reporting structures. A target system is denoted by $\mathcal{L}$, but neither its operative state space nor its reachable set is treated as a bare object. A *substrate package* V_Σ locks the system boundary, state-space construction, admissible evolution, and reachability convention, thereby producing a package-locked substrate structure

$$\mathfrak{S}_\Sigma := \mathfrak{S}(\mathcal{L}; V_\Sigma) = (\mathcal{L}, \Omega_\Sigma, \Omega_{\Sigma, \text{reach}}).$$

A task-relative restriction is written $\Omega_{\text{reach}}^{\text{rel}} \subseteq \Omega_{\Sigma, \text{reach}}$ only after the relevant task clause has been declared. Perspective packages and relation packages may use $\mathfrak{S}_\Sigma$ as an operand, but they do not silently redefine it.

A declared package remains an audit bundle of the form $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$. Packages may depend on previously package-locked operands. Such dependency does not add a new top-level slot; it records the relative depth at which the package is declared.

Type discipline and dependency depth. The following rules are normative for every module in this manuscript.

T1. Object type determines slot. Declarations concerning states, carriers, visible domains, or readout structures belong to $\mathcal{I}$; declarations concerning histories or trajectories belong to $\mathcal{C}$; declarations concerning predicates, audited questions, or judgment orientation belong to $\mathcal{Q}$; and declarations concerning reconstruction, transport, or pairing maps belong to the mapping/interface content of $\mathcal{I}$.

T2. Dependency arity determines relative package depth. A package that uses one previously locked structure may be declared over that operand. A package that simultaneously refers to two perspective structures is relationally above both of them. A comparison between already completed relation reports requires a further declared comparison package. No absolute package level is assumed; depth is relative to the declared dependency graph.

T3. No bare carrier. Every state space, reachable set, optimization domain, pairing domain, or relation carrier entering a diagnostic must either be locked by a declared package or be obtained by an explicitly declared canonical construction from package-locked carriers. In particular, neither Ω , Ω_{reach} , nor a relation carrier Z_R may be inserted as an untyped background domain.

T4. Functional role is distinct from descriptive level and dependency depth. The terms *carrier role*, *report role*, and *bridge role* classify typed mathematical components by their function in a completed construction. They do not define descriptive levels and they do not encode package-dependency depth. Packages and declarations are typing/locking specifications rather than members of this role classification; consequently, a single package may declare components belonging to more than one functional role.

Three distinct relation declarations. For a relation diagnostic, the following questions are type-distinct and must not be collapsed:

- (i) *substrate identity*: which package-locked carrier structure is being described;
- (ii) *instance pairing*: which operand instances are declared to correspond;
- (iii) *reconstruction sufficiency*: whether one declared readout can reconstruct the other under an admissible reconstruction class and discrepancy rule.

Identity pairing is a declared special case of instance pairing, not the absence of a pairing clause.

Typing of the two core diagnostics. The readout Ψ is unary at the perspective level: it is evaluated within one package-locked perspective on a locked substrate. The fixed-slice gap $\Delta_{I \rightarrow G}$ is relational: it is produced only after two package-locked perspectives, a package-locked (or canonically derived) pairing structure, and a relation-diagnostic package have been declared. Its direction is fixed by the admissible reconstruction class and by the side on which discrepancy is evaluated. Cross-package comparison between already completed reports remains subject to Normative Rule 2.2.

Partial semantics (well-posedness vs. undefinedness). All reporting-level quantities in this paper are treated as *partial* objects whose meanings are locked only on the domain of declared packages satisfying the relevant clauses. If required clauses are missing (under-specification), if an invoked comparison lacks declared transport/comparability, or if a referenced domain condition is not satisfied, then the corresponding numerical or logical statement is **undefined** relative to the declared regime and must be reported as such. For compactness, a report may attach a short diagnostic status tag (Appendix A) together with a minimal witness list, but the core semantic point is the same: “undefined” is an output state, not an error to be silently repaired by implicit conventions.

Normative Rule 2.1 (Package-local resolution semantics and non-licensing outputs).

- **Package-local predicate.** If a package node $V_\bullet = (\mathcal{I}_\bullet, \varepsilon_\bullet, \mathcal{Q}_\bullet, \mathcal{C}_\bullet)$ produces a nonnegative scalar report x , then

$$x \approx_{\varepsilon_\bullet} 0 \iff x < \varepsilon_\bullet,$$

conditional on well-posedness and on every invoked operability, comparability, and certification gate. The predicate belongs to the package that produces x .

- **Scalar-core scope.** The core predicate uses nonnegative scalar reports and positive scalar thresholds. Typed multicomponent resolution objects require the additional declarations listed in Remark 4.13; their general theory is deferred rather than excluded.
- **No cross-edge inheritance.** Threshold binding is intra-package only. A threshold declared in a substrate, perspective, relation-substrate, relation-diagnostic, or higher comparison package is not inherited by another package node merely because the nodes are connected in the dependency graph.
- **Relation report.** For the fixed-slice relation diagnostic, $\Delta_{I \rightarrow G} \approx_{\varepsilon_R} 0$ means only $\Delta_{I \rightarrow G} < \varepsilon_R$ under V_R . It does not state that either operand structure is identical to the other, nor that endpoint thresholds equal ε_R .
- **Alignment trigger.** A threshold-alignment or conversion clause is required only when one package consumes an already thresholded output from another package, when a readout map itself varies with an endpoint threshold, when a theorem compares thresholds across package nodes, or when already completed reports are numerically compared under reused thresholds.
- **Cross-level equality.** Whenever an identification glues distinct levels, a reported “=” is read through the symmetric resolution-indistinguishability predicate of the package node producing that identification. The predicate need not be transitive unless transitivity is separately established or imposed by declared closure.
- **Ontological non-identity.** For a genuine cross-level relation R , ontological non-identity is recorded separately by the meta-gap axiom $\Delta_R^* > 0$ (Def. 6.1); a finite-resolution report cannot silently replace that semantic function.
- **Refinement behavior.** If a declared or certified structural lower bound $\ell_{I \rightarrow G} \leq \Delta_{I \rightarrow G}$ satisfies $\ell_{I \rightarrow G} \geq \varepsilon_R$, then the report $\Delta_{I \rightarrow G} \approx_{\varepsilon_R} 0$ is unavailable. The licensed output is $\Delta_{I \rightarrow G} \not\approx_{\varepsilon_R} 0$, or **undefined** if a required clause or gate is missing.
- **Undefinedness / non-licensing.** Any numerical statement is licensed only on the domain where the relevant package nodes, operands, pairing, interface, admissibility, and domain clauses are declared and any invoked certificate-relative feasibility gate is satisfied. Comparison between already completed reports additionally requires explicit transport and any required threshold alignment. Otherwise the correct output is **undefined**.

Normative Rule 2.2 (Cross-package comparability and explicit transport).

- **no implicit transport (Normative Rule 2.2).** Numerical reports produced under distinct declared packages V and V' are *not* treated as directly comparable scalars by default (even if they share a symbol name).
- **Explicit transport law.** A cross-package comparison (equality/inequality, limits across packages, or reusing a numerical threshold) is licensed only if a transport/identification rule $\tau_{V \rightarrow V'}$ is explicitly declared together with its domain of validity and compatibility with the relevant representation maps and interface clauses.
- **Non-comparability output.** If $\tau_{V \rightarrow V'}$ is not declared (or does not cover the reported objects), cross-package comparison statements are **undefined**; an optional diagnostic tag is **non-comparable**.
- **Resolution alignment for completed reports.** When already completed reports are compared, their package-local thresholds must be aligned or related by an explicit conversion rule whenever the comparison consumes thresholded outputs or reuses a numerical threshold. Relation formation from declared operands is not itself a comparison between completed reports. Without the required alignment, the comparison is **undefined** (optional tag **RES / resolution-mismatch**).

Normative Rule 2.3 (Carrier/report transport separation).

- **Carrier-side transport acts first on carriers.** A declared map, refinement, composition, or transport between carrier-side structures does not by itself transport a typed report formed from those carriers.
- **No inherited scalar propagation.** In particular, a scalar report attached to one completed construction is not presumed to propagate across package, relation, refinement, or evolutionary edges merely because the underlying carriers can be related.
- **Report propagation requires a bridge.** Any report-to-report propagation law requires a separately declared or derived bridge of the appropriate type together with the certificates required by that bridge. When such a bridge is unavailable or its hypotheses are unverified, the carrier relation may remain meaningful while the requested scalar propagation is not licensed.

Normative Rule 2.4 (Declaration–verification–derivation discipline).

- (i) **Declaration** fixes the typed operands, domains, interfaces, admissible classes, thresholds, transports, and other clauses required to state the requested construction.
- (ii) **Verification** checks that the declared construction is operable and, when a theorem or certificate is invoked, that its additional hypotheses hold on the operands and data actually consumed by that inference.
- (iii) **Derivation** forms only those consequences licensed by the verified declarations and hypotheses.

Declaration does not certify a theorem hypothesis, and verification of hypotheses is not itself the derived conclusion. A completed report may therefore be operable even when a particular theorem-relative propagation or bound remains unverified.

Notation collisions (avoidance rules).

- Within the framework’s diagnostic notation, the symbol Δ is reserved for cross-level residuals, and the bare symbol Δ is not used as an alias for any specific gap object; we always keep subscripts/markers (e.g. $\Delta_{I \rightarrow G}$, $\Delta_{\text{acc}}(\gamma)$, Δ^*) to prevent cross-module collisions.
- Generic scalar placeholders (dummy variables, coordinates, function arguments) are written as $x, y, \dots$, not Δ .
- The symbol A is reserved for *perspectives* (Def. 4.4). Matrix operators used in local linearizations are written as J when local linear models are declared.
- The symbol V is reserved for nodes instantiating the declared four-slot package schema. Semantic subscripts ($V_\Sigma, V_I, V_G, V_{\Sigma_R}, V_R$) indicate package-node function, not absolute depth. The completed dependency record is denoted by $\mathcal{V}$ (Def. 4.12).
- The symbol R is reserved for genuine cross-level relations (Sec. 6.3). Substrate constraint regimes are written as $\mathfrak{R}$ (Def. 4.1) to avoid collision.
- Level-respecting re-expression maps are written as $T_{A_2 \leftarrow A_1}$ (Def. 4.6).

2.2 Equality and resolution-aware approximation

This paper uses “=” as an academic writing convention, but its *reporting semantics* depends on whether a statement stays within a single declared calculus/module or crosses a genuine level boundary.

(i) **Within a declared calculus/module (single-level inference).** When a derivation is explicitly confined to a declared calculus/module (an H-calculus segment in the terminology used later), “=” is read as *equality under the rules of that declared module*: an inference-level identification licensed by the adopted rules and admissibility constraints of the module. This reading is local: it does not by itself assert that a cross-level identification has been made gap-free.

Bare approximation vs. resolution-aware reporting indistinguishability. Within a declared module, the symbol “ $\approx$ ” (without ε) denotes a local analytic/discretization approximation (e.g. a local linearization $v_{k+1} \approx Jv_k$). It does not carry the reporting semantics of $\approx_\varepsilon$. Only $\approx_\varepsilon$ denotes the declared symmetric resolution-indistinguishability predicate under a package; it becomes an equivalence relation only when transitivity is separately established or imposed by declared closure.

(ii) Cross-level identities (limits, completions, quotients, coarse-graining). Whenever “ $=$ ” arises from a construction that *glues distinct levels* (e.g. limiting, completion, quotienting/identification, coarse-graining followed by reconstruction), the default reading is *resolution-aware symmetric indistinguishability under the declared package*:

$$X = Y \quad (\text{in cross-level context}) \quad \rightsquigarrow \quad X \approx_\varepsilon Y,$$

where $\approx_\varepsilon$ denotes the declared symmetric indistinguishability predicate at reporting threshold ε under V . The predicate need not be transitive; a resolution-equivalence relation arises only after transitivity is proved or a declared closure generates equivalence classes.

Normative semantics. All later uses of cross-level equality, resolution-aware approximation, and non-licensing/undefinedness inherit the single authoritative semantics fixed by Normative Rule 2.1. Declaration of a threshold fixes the meaning of the symbol; when a certified fixed-target resolution refinement is invoked, a separate certificate-relative feasibility check determines whether the corresponding ordinary completed package instance is licensed for the declared query clauses (Sec. 4.4).

Two uses of Δ (fixed-slice vs. accumulated). To prevent a common confusion:

- $\Delta_{I \rightarrow G}$ denotes a fixed-slice inter-level residual (compression $\rightarrow$ admissible reconstruction/lifting residual).
- $\Delta_{\text{acc}}(\gamma)$ (when used) denotes a path-updated accumulated report along a declared evolution coordinate γ . Its meaning is package-relative and requires slice-local Ψ_{run} , the declared transport/accumulation clauses, and the inherited accumulated-report type. It is neither interchangeable nor jointly comparable by default with the fixed-slice diagnostic $\Delta_{I \rightarrow G}$; comparison requires the additional bridge of Sec. 7.3.

Throughout the manuscript we do not use the bare symbol Δ as an alias for any specific gap object ($\Delta_{I \rightarrow G}$, $\Delta_{\text{acc}}(\gamma)$, or Δ^*); we always keep subscripts/markers. Generic scalar placeholders (dummy variables, coordinates, function arguments) are written as $x, y, \dots$, not Δ .

2.3 Reader guide and dependency map

This manuscript has a slim core and optional modules. The core consists of the V -structure coupling semantics, the definitions of Ψ and $\Delta_{I \rightarrow G}$, the resolution-aware reporting rules, the fixed-target threshold-feasibility interface, and the minimal dynamical interface. Optional modules are invoked only when their additional objects are declared: probabilistic interfaces for TV/IPM readouts, the full pairwise/partition resolution module, extended dynamical readouts, and geometric-closure modules for model-specific mechanisms. If the required module declarations are absent, the module-specific claim is simply not licensed.

3 Framework Overview

Functional-role decomposition. The framework uses a functional classification of typed mathematical components that is distinct from both descriptive level and package-dependency depth. The classification is applied at the component granularity relevant to a construction: a composite locked tuple may therefore contain components serving different roles. In particular, the complete pairing structure $\mathfrak{P}_R = (Z_R, Z_{R,\text{reach}}, p_I, p_G)$ contains carrier-side data $(Z_R, Z_{R,\text{reach}})$ together with bridge maps (p_I, p_G) .

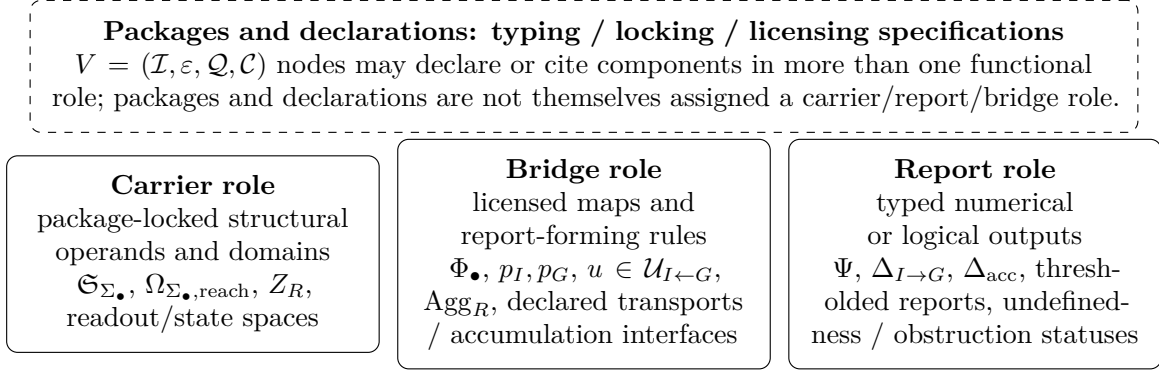


Figure 1: Functional-role decomposition. The three roles classify typed mathematical components by use; they do not define new descriptive levels and do not encode package-dependency depth. No dependency or temporal order is implied by the left-to-right placement.

The role decomposition and the package-dependency graph answer different questions. The former classifies typed components by function; the latter records which declarations and locked operands must be available before a particular construction or report can be formed. For the same reason, packages and declarations are not members of the role classification: they are the specifications under which role-bearing components are typed, locked, derived, or licensed.

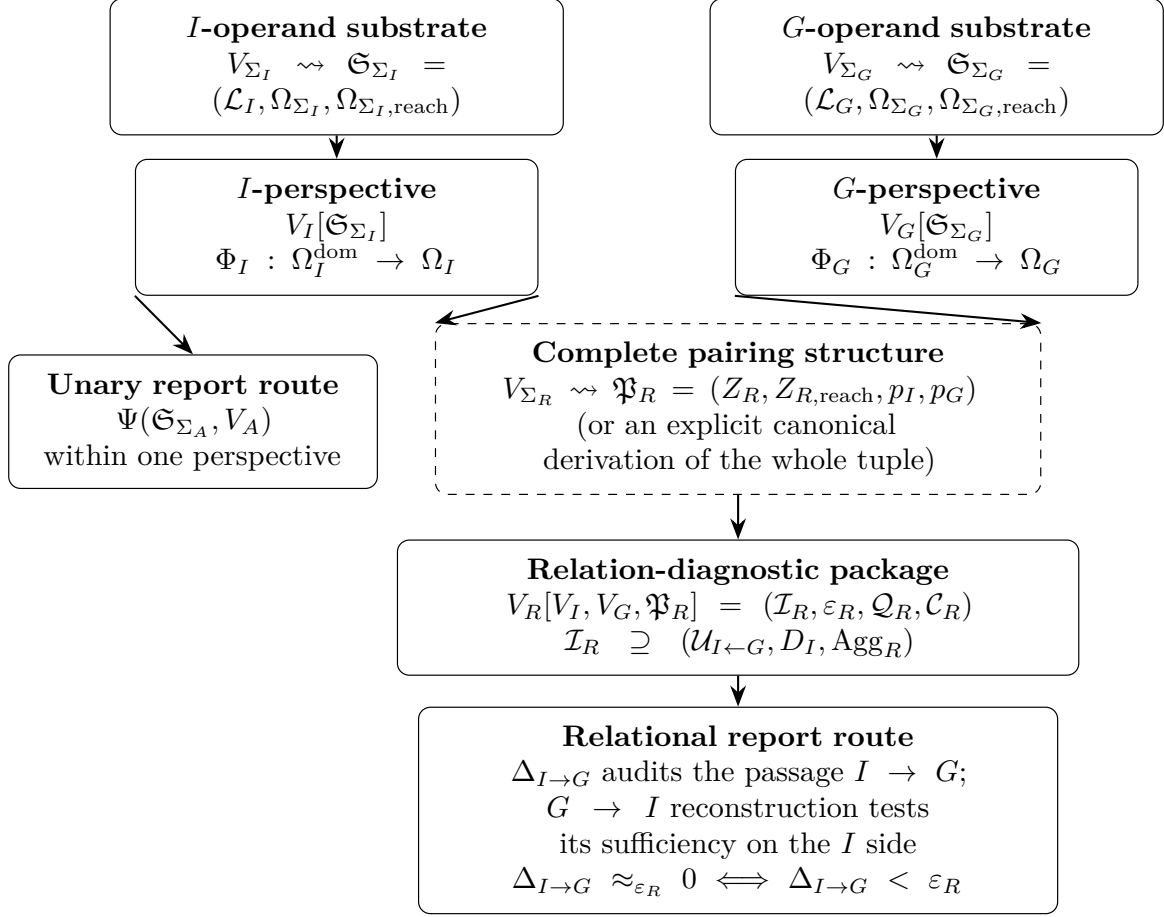


Figure 2: Package-dependency graph. The two operand perspectives may read the same locked substrate or distinct locked substrates. The relation-substrate package locks the complete pairing structure, not only its carrier, unless the whole tuple is canonically derived. Pairing fixes correspondence but does not select a reconstruction direction. The subscript $I \rightarrow G$ names the passage under audit; admissible reverse reconstruction $G \rightarrow I$ tests its sufficiency. The arrows display declaration/operand dependency, not temporal evolution or functional-role membership.

Dependency map. Figure 2 is a dependency map rather than a temporal pipeline. Each arrow records an operand dependency. It may therefore cut across the functional roles of Figure 1: substrate, perspective, and pairing declarations produce or lock carrier- and bridge-side components, while the completed Ψ and $\Delta_{I \rightarrow G}$ objects belong to the report role. No package node may redefine the carrier or readout structure supplied by an upstream node, and later dynamics or closure statements require further package-locked operands.

3.1 Running minimal example: compression, hidden fibers, and future read-out

The following finite-state example is used only to display the type structure of the framework. Let

$$X = \{a, b, c, d\}, \quad \pi(a) = \pi(b) = \pi(c) = 0, \quad \pi(d) = 1,$$

and declare an evolution

$$T(a) = c, \quad T(b) = d, \quad T(c) = c, \quad T(d) = d.$$

Although a and b share the same current macro-report, their one-step macro-readouts separate:

$$\pi(Ta) = 0, \quad \pi(Tb) = 1.$$

Thus endpoint agreement does not by itself license future-readout sufficiency. In the shared-substrate core, a numerical claim requires total operand readouts, identity pairing on the locked reachable substrate, a directional reconstruction class, an I -side discrepancy, aggregation, and the relation threshold ε_R . With those clauses, within-fiber variation can witness a non-negligible $\Delta_{I \rightarrow G}$; without them, the sufficiency claim is undefined.

3.2 Dependency-aware notation quick reference

For quick reference, we list the symbols required by the recursive package-dependency structure. All meanings are package-relative. A threshold symbol carries the subscript of the package node whose output it classifies; the unsubscripted legacy symbols are retained only in explicitly flattened specializations.

Table 1: Minimal notation quick reference for the dependency-aware package structure.

Symbol	Meaning (as used in this paper)
$V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$	Generic four-slot package schema. The same schema is used recursively at substrate, perspective, relation-substrate, relation-diagnostic, and higher comparison nodes; dependency does not add a fifth slot.
$V_{\Sigma_A}, V_{\Sigma_I}, V_{\Sigma_G}$	Substrate packages locking the system boundary, operative state-space construction, admissible evolution, and reachability convention for the unary, I -operand, and G -operand structures. These nodes may coincide.
$\mathfrak{S}_{\Sigma_\bullet}$ ($\mathcal{L}_\bullet, \Omega_{\Sigma_\bullet}, \Omega_{\Sigma_\bullet, \text{reach}}$)	= Package-locked substrate structure produced by $V_{\Sigma_\bullet}$. Neither its carrier nor its reachable domain is a bare background object.
$V_A[\mathfrak{S}_{\Sigma_A}]$	Perspective package reading a locked substrate through $\Phi_A : \Omega_A^{\text{dom}} \rightarrow \Omega_A$. The designated relation operands are $V_I[\mathfrak{S}_{\Sigma_I}]$ and $V_G[\mathfrak{S}_{\Sigma_G}]$.
V_{Σ_R}	Relation-substrate package locking the complete pairing structure $\mathfrak{P}_R$, including the relation carrier, its admissible/reachable domain, and both projections, when that whole tuple is not canonically derived. It may coincide with the operand substrate package in the shared-substrate identity-pairing specialization.
$\mathfrak{P}_R$ ($Z_R, Z_{R, \text{reach}}, p_I, p_G$)	= Package-locked or canonically derived pairing structure, with partial projections $p_I : Z_R \rightarrow \Omega_{\Sigma_I}$ and $p_G : Z_R \rightarrow \Omega_{\Sigma_G}$. Each licensed z selects paired operand instances. Identity pairing is a declared specialization.
$V_R[V_I, V_G, \mathfrak{P}_R]$	Relation-diagnostic package. Its interface content declares $\mathcal{U}_{I \leftarrow G}$, D_I , and Agg_R ; its own slots also supply $\varepsilon_R, \mathcal{Q}_R, \mathcal{C}_R$.
Z_R^{task}	Effective relation-task domain obtained by intersecting the relation-state restriction declared in $\mathcal{I}_R$ with query-induced and history/trajecory-induced restrictions from $\mathcal{Q}_R$ and $\mathcal{C}_R$.
Z_R^{struct}	Structural pairing domain obtained from $Z_{R, \text{reach}}$ after requiring both partial projections to be defined. It tests whether the declared relation carrier and projection legs provide any admissible pairing instance before task restriction.
Z_R^{pair}	Task-licensed pairing domain $Z_R^{\text{struct}} \cap Z_R^{\text{task}}$. Its emptiness is distinguished from structural pairing emptiness because the repair target is the task declaration rather than the pairing structure.
Z_R^{operand}	Operand-reachable pairing domain obtained by further requiring the two projected instances to lie in $\Omega_{\Sigma_I, \text{reach}}$ and $\Omega_{\Sigma_G, \text{reach}}$.
Z_R^{eval}	Final relation-evaluation domain obtained from Z_R^{operand} after requiring both perspective readouts to be defined. The first empty stage identifies the operative undefinedness subtype.

(continued)

Symbol	Meaning (as used in this paper)
Agg_R	Once Z_R^{eval} is fixed, a normalized isotone aggregation map $\text{Agg}_R : [0, \infty]^{Z_R^{\text{eval}}} \rightarrow [0, \infty]$. A stronger zero-detection class may be declared separately. In the core minimax specialization, $\text{Agg}_R = \text{sup}$; the outer infimum over admissible reconstructions remains part of the definition of $\Delta_{I \rightarrow G}$.
Ψ	Unary intra-level structural tension readout evaluated within one package-locked perspective.
$\Delta_{I \rightarrow G}$	Ordered relational reconstruction gap auditing the declared passage $I \rightarrow G$. The pairing fixes correspondence but does not select reconstruction direction; admissible reverse reconstruction $G \rightarrow I$ and the I -side discrepancy test sufficiency. The legacy notation $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ denotes the shared-substrate identity-pairing minimax flattened specialization.
$\Delta_{\text{acc}}(\gamma)$	Accumulated residual scalar along a declared evolution coordinate γ ; distinct from the fixed-slice relation diagnostic.
$\varepsilon_\bullet$	Threshold slot belonging to package node $V_\bullet$. It becomes an active reporting clause only when that node produces or consumes a thresholded report. Binding is intra-package only; distinct node thresholds are independent unless an explicit alignment clause is invoked. The present core treats nonnegative scalar reports and positive scalar thresholds; typed multicomponent resolution objects are explicitly deferred.
$\approx_{\varepsilon_\bullet}$	Package-local symmetric resolution-indistinguishability predicate. For a nonnegative report x , $x \approx_{\varepsilon_\bullet} 0$ means $x < \varepsilon_\bullet$, conditional on report licensing.
Δ_R^*	Meta-gap schema for a genuine cross-level relation R . It is not a reporting tolerance and does not replace the operational $\Delta_{I \rightarrow G}$.
$\Omega, \Omega_{\text{reach}}, V$	Legacy flattened shorthand used in retained theorems and examples. In the shared-substrate core, $\Omega = \Omega_\Sigma$, $\Omega_{\text{reach}} = \Omega_{\Sigma, \text{reach}}$, and V abbreviates the completed dependency record declared by the relevant package card.

4 Formal Objects: Regime, Perspectives, Packages, and Package-Locked Reporting Structures

This section fixes the primitives used throughout the manuscript. The central convention is:

A reportable analysis context is a dependency record of package-locked carriers, perspectives, pairings, and diagnostic clauses; no carrier enters a report as an untyped background space.

The generic package schema remains $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$ at every node. The package-locked reporting structure $\mathcal{V}$ is therefore not determined by a bare pair $(\mathcal{L}, V)$ in the general case; it is determined by the declared dependency record appropriate to the report. The earlier pair notation remains available only as flattened shorthand (Def. 4.12).

4.1 Package-locked substrate regime and fine-grained domain

Definition 4.1 (Constraint regime). A *constraint regime* is a collection $\mathfrak{R}$ of local constraint rules. Each rule $r \in \mathfrak{R}$ involves only finitely many degrees of freedom and is viewed as a relation/predicate on those degrees of freedom.

Definition 4.2 (Substrate package and package-locked substrate). Fix a target system $\mathcal{L}$ and a declared substrate package V_Σ . The package V_Σ locks the system boundary, the operative degrees of freedom, the state-space construction, the admissible evolution, and the reachability convention. It thereby produces the package-locked substrate structure

$$\mathfrak{S}_\Sigma := \mathfrak{S}(\mathcal{L}; V_\Sigma) = (\mathcal{L}, \Omega_\Sigma, \Omega_{\Sigma, \text{reach}}).$$

If $\text{Var}(\mathfrak{R}; V_\Sigma)$ denotes the degrees of freedom retained under the locked boundary convention, then Ω_Σ is the set of assignments admitted by the state-space construction declared in V_Σ . Neither Ω_Σ nor $\Omega_{\Sigma, \text{reach}}$ is treated as package-free.

Remark 4.1 (No global satisfiability assumption). No assumption is made that $\mathfrak{R}$ is jointly satisfiable, nor that the package-locked substrate occupies only states satisfying all constraints. Partial violations are permitted and treated as structurally relevant whenever the declared substrate package admits them.

Definition 4.3 (Package-locked reachability). The reachable set

$$\Omega_{\Sigma, \text{reach}} \subseteq \Omega_\Sigma$$

is the set selected by the admissible evolution and reachability convention locked in V_Σ . A task-relative domain is a further declared restriction

$$\Omega_{\text{reach}}^{\text{rel}} \subseteq \Omega_{\Sigma, \text{reach}}.$$

Unless stated otherwise, a supremum in the shared-substrate core is taken over the relevant declared evaluation subset of $\Omega_{\Sigma, \text{reach}}$.

Remark 4.2 (Recursive carrier locking). The no-bare-carrier rule applies recursively. If a later module introduces a new relation carrier Z_R that is not canonically derived from $\mathfrak{S}_\Sigma$, that carrier must itself be locked by a relation-substrate package before it can enter an optimization or reporting domain.

4.2 Package-locked perspectives, pairing structures, and relation interfaces

Definition 4.4 (Perspective package and representation map). Let $\mathfrak{S}_\Sigma = (\mathcal{L}, \Omega_\Sigma, \Omega_{\Sigma, \text{reach}})$ be a package-locked substrate. A *perspective package* $V_A[\mathfrak{S}_\Sigma]$ declares a visible factor family F_A , an encoding rule Enc_A , a readout domain $\Omega_A^{\text{dom}} \subseteq \Omega_\Sigma$, and the resulting representation map

$$\Phi_A : \Omega_A^{\text{dom}} \rightarrow \Omega_A.$$

The perspective package reads the locked substrate; it does not silently reconstruct the substrate boundary, state space, or reachability convention. Unless a module explicitly invokes partial readout semantics, the current core retains the totality convention $\Omega_A^{\text{dom}} = \Omega_\Sigma$.

Definition 4.5 (Level specification). A *level* is specified operationally by a *level specification*

$$\Lambda := (\Pi_\Lambda, \mathcal{A}_\Lambda),$$

where Π_Λ fixes the retained/suppressed degrees of freedom and $\mathcal{A}_\Lambda$ is the declared admissible family of level-respecting perspective packages. A level specification does not own a reporting threshold. Any thresholded report made at that level uses the active threshold clause of the package node producing that report; comparisons across such reports require the applicable transport and threshold-alignment clauses.

Remark 4.3 (“Whole” versus “individual” is boundary-induced). Within this formalization, “whole” and “individual” are not intrinsic labels of an object; they are induced by the package-locked substrate boundary, the declared level specification, and the corresponding perspective package.

Definition 4.6 (Level-respecting re-expression). For two perspective packages V_{A_1} and V_{A_2} over the same package-locked substrate and within the same admissible family $\mathcal{A}_\Lambda$, a *level-respecting re-expression* is an invertible map

$$T_{A_2 \leftarrow A_1} : \Omega_{A_1} \rightarrow \Omega_{A_2}$$

such that, on their declared common readout domain,

$$\Phi_{A_2}(S) = T_{A_2 \leftarrow A_1}(\Phi_{A_1}(S)).$$

When such a map exists, the two perspectives are treated as equivalent encodings at the same declared level, subject to the package clauses under which the common domain is licensed.

Definition 4.7 (Designated operand perspectives). Fix two package-locked substrate structures

$$\mathfrak{S}_{\Sigma_I} = (\mathcal{L}_I, \Omega_{\Sigma_I}, \Omega_{\Sigma_I, \text{reach}}), \quad \mathfrak{S}_{\Sigma_G} = (\mathcal{L}_G, \Omega_{\Sigma_G}, \Omega_{\Sigma_G, \text{reach}}),$$

and two perspective packages

$$V_I[\mathfrak{S}_{\Sigma_I}], \quad V_G[\mathfrak{S}_{\Sigma_G}],$$

with readout domains and maps

$$\begin{aligned} \Omega_I^{\text{dom}} &\subseteq \Omega_{\Sigma_I}, & \Phi_I : \Omega_I^{\text{dom}} &\rightarrow \Omega_I, \\ \Omega_G^{\text{dom}} &\subseteq \Omega_{\Sigma_G}, & \Phi_G : \Omega_G^{\text{dom}} &\rightarrow \Omega_G. \end{aligned}$$

The two substrate structures may coincide, but equality is not presumed. These maps are components of their respective perspective packages. A relation package may cite them as operands but does not redeclare them as its own internal clauses.

Definition 4.8 (Package-locked pairing structure). A relation diagnostic between V_I and V_G requires a complete pairing structure

$$\mathfrak{P}_R = (Z_R, Z_{R, \text{reach}}, p_I, p_G),$$

where Z_R is a relation carrier, $Z_{R, \text{reach}} \subseteq Z_R$ is its package-locked reachable/admissible domain, and

$$p_I : Z_R \rightarrow \Omega_{\Sigma_I}, \quad p_G : Z_R \rightarrow \Omega_{\Sigma_G}$$

are declared partial projections. The complete tuple $\mathfrak{P}_R$, not only Z_R , must either be locked by a relation-substrate package V_{Σ_R} or be obtained by an explicitly declared canonical construction from already package-locked structures. Thus the ownership or derivation of p_I and p_G is part of the pairing declaration.

In the shared-substrate identity-pairing specialization,

$$\mathfrak{S}_{\Sigma_I} = \mathfrak{S}_{\Sigma_G} = \mathfrak{S}_{\Sigma}, \quad V_{\Sigma_R} = V_{\Sigma},$$

$$Z_R \subseteq \Omega_{\Sigma}, \quad p_I = p_G = \text{id}_{Z_R}.$$

Identity pairing is therefore a declared specialization, not the absence of a pairing clause. A function-type transport $\tau : X_s \rightarrow X_t$ is another specialization, obtained by locking or deriving $Z_R = \text{Dom}(\tau)$, $p_I = \text{id}_{Z_R}$, and $p_G = \tau$. General partial, multi-valued, and many-to-many relations remain representable by the two-projection form.

Definition 4.9 (Directional relation interface). Given operand perspectives V_I, V_G and a complete pairing structure $\mathfrak{P}_R$, a relation-diagnostic package $V_R[V_I, V_G, \mathfrak{P}_R]$ declares the reconstruction and discrepancy mechanism used to audit the passage written $I \rightarrow G$. Its interface content includes

$$\mathcal{U}_{I \leftarrow G} \subseteq \{u : \Omega_G \rightarrow \Omega_I\}, \quad D_I : \Omega_I \times \Omega_I \rightarrow [0, \infty],$$

and an aggregation rule. Once a licensed nonempty evaluation domain Z_R^{eval} has been fixed, the aggregation rule has type

$$\text{Agg}_R : [0, \infty]^{Z_R^{\text{eval}}} \rightarrow [0, \infty].$$

Let $0_{Z_R^{\text{eval}}}$ denote the all-zero residual family. Every admissible aggregation rule is required to be normalized and isotone:

$$\begin{aligned} \text{Agg}_R(0_{Z_R^{\text{eval}}}) &= 0, \\ f(z) \leq g(z) \text{ for all } z \in Z_R^{\text{eval}} &\implies \text{Agg}_R(f) \leq \text{Agg}_R(g). \end{aligned}$$

These are minimum aggregation semantics, not a pointwise zero-detection theorem. A relation package may additionally declare a zero class $\mathcal{N}_R \subseteq [0, \infty]^{Z_R^{\text{eval}}}$ satisfying $\text{Agg}_R(f) = 0 \iff f \in \mathcal{N}_R$. The pairing structure fixes which operand instances correspond but does not select a reconstruction direction. The subscript $I \rightarrow G$ names the compression or reporting passage under audit. Its sufficiency is tested by admissible reverse reconstructions $G \rightarrow I$ and by evaluating D_I in the I -side readout space.

If admissible reconstructions are called *sections*, the relation package must additionally state the relevant consistency clause. Otherwise u is only an admissible reconstruction candidate.

Remark 4.4 (Aggregation semantics and zero detection). Normalization and isotonicity guarantee that the all-zero residual family aggregates to zero and that pointwise enlargement of a residual family cannot reduce its aggregate. They do not imply that $\text{Agg}_R(f) = 0$ forces $f(z) = 0$ at every index. Such a conclusion requires the optional zero-detection clause of Definition 4.9. For the supremum aggregator, $\mathcal{N}_R = \{0_{Z_R^{\text{eval}}}\}$; for an integral or average under a declared measure μ_R , the natural zero class is equality to zero μ_R -almost everywhere.

Proposition 5.1 gives the forward floor statement: an admissible reconstruction with identically zero residual reaches $\Delta_{I \rightarrow G} = 0$. The converse does not hold in general: $\Delta_{I \rightarrow G} = 0$ need not produce an exact reconstruction witness because the outer infimum may be unattained, and the aggregation rule may detect zero only modulo its declared zero class. In contrast, $\Delta_{I \rightarrow G} > 0$ rules out every admissible reconstruction whose residual family is identically zero. Pointwise and fiber-witness interpretations used later are restricted to the declared minimax specialization $\text{Agg}_R = \sup$.

Remark 4.5 (Why the supremum is the core minimax default). The general relation interface keeps Agg_R open, but the core minimax specialization uses the supremum for a structural reason in addition to its worst-case semantics. Let Z_t and Z_s be declared nonempty evaluation domains, let $\rho : Z_t \rightarrow Z_s$ be any map, and let $f : Z_s \rightarrow [0, \infty]$. Then

$$\sup_{z \in Z_t} f(\rho(z)) = \sup_{x \in \rho(Z_t)} f(x) \leq \sup_{x \in Z_s} f(x).$$

Thus arbitrary pullback does not amplify a worst-case aggregate beyond its source value. No auxiliary measure, density, weighting, normalization convention, or mass-ratio correction is required for this inequality. This makes the supremum a canonical declaration-light default for the core minimax report. More general aggregation rules remain licensed extensions and may require additional transport declarations or finite transport certificates when reports are propagated across relation domains. This is a canonicity/design rationale, not a uniqueness theorem for supremum aggregation.

Remark 4.6 (Discrepancy monotonicity under a fixed reporting scaffold). Fix the same operand packages, pairing structure, evaluation domain, admissible reconstruction class, and aggregation rule, and compare two licensed discrepancy rules D_I, D'_I on the same I -side readout space. If

$$D_I(y, y') \leq D'_I(y, y') \quad \text{for all } (y, y') \in \Omega_I \times \Omega_I,$$

then the induced residual families satisfy $r_u^{D_I}(z) \leq r_u^{D'_I}(z)$ for every admissible u and every $z \in Z_R^{\text{eval}}$. Isotonicity of Agg_R , followed by the infimum over the same reconstruction class, gives

$$\Delta_{I \rightarrow G}^{D_I} \leq \Delta_{I \rightarrow G}^{D'_I}.$$

Thus replacing the discrepancy by a pointwise larger licensed discrepancy cannot decrease the reported gap. This is a comparison inside one fixed reporting scaffold; it does not license comparison between independently changed packages.

Remark 4.7 (Deferred common aggregation schema). The normalized/isotone requirements above are currently local to relation diagnostics because $\Delta_{I \rightarrow G}$ exposes a declared aggregation rule, whereas the core definition of Ψ uses a fixed infimum over its licensed evaluation domain. If a later unary or other diagnostic replaces its fixed optimization rule by a package-declared aggregation map, that map must belong to the interface slot of the package producing the report and instantiate a common admissible-aggregation schema, including normalization, isotonicity, and any invoked zero-detection clause. Formal abstraction of that shared schema beyond the present relation interface is deferred.

Remark 4.8 (Operational degrees of freedom must be locked). A fixed-slice relation diagnostic is not well-posed until the following are locked or canonically derived from locked operands:

$$(\mathfrak{S}_{\Sigma_I}, \mathfrak{S}_{\Sigma_G}, V_I, V_G, \mathfrak{P}_R, \mathcal{U}_{I \leftarrow G}, D_I, \text{Agg}_R, \varepsilon_R).$$

Substrate identity, instance pairing, and reconstruction sufficiency are distinct declarations. The first fixes what is being described, the second fixes which instances correspond, and the third fixes the directional residual to be reported.

Remark 4.9 (Staged compatibility with the flattened package notation). Retained theorem statements, calculations, and optional modules still use a single symbol V in several places. Such occurrences are to be read as a *flattened shorthand* only when the surrounding package card supplies the substrate, perspective, pairing, and relation-diagnostic dependency record. This compatibility convention preserves the existing results while preventing the shorthand from being treated as the primitive type structure.

4.3 Declared package V as a coupling specification

A declared package V is an audit bundle because it first functions as a coupling specification: it specifies how objective structure is made reportable through finite access, admissible compression, reconstruction, discrepancy, comparability, and resolution clauses. The package therefore does not stand outside the analysis as a subjective annotation; it is part of the reportability conditions under which any structural quantity in this manuscript is typed and compared. It is not, however, itself assigned a carrier/report/bridge role. Its slots may lock carrier-side domains, declare bridge-side maps or report-forming rules, and activate report-side thresholds within one typed audit specification.

Definition 4.10 (Declared package). A *declared package* is an audit bundle

$$V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C}),$$

where $\mathcal{I}$ is the typed interface content of that package node, $\varepsilon > 0$ is its threshold slot, $\mathcal{Q}$ is its declared query/predicate content, and $\mathcal{C}$ is its declared history or trajectory class when such objects are invoked. The threshold slot becomes an active reporting clause only when that node produces or consumes a thresholded report; otherwise it is present in the schema but is not semantically invoked. The four-slot schema is unchanged across the dependency graph, but the content of $\mathcal{I}$ is module-relative:

- a substrate package V_Σ locks carrier construction, admissible evolution, and reachability;
- a perspective package $V_A[\mathfrak{S}_\Sigma]$ locks a readout domain and representation map Φ_A ;
- a relation-substrate package V_{Σ_R} locks the complete pairing structure $(Z_R, Z_{R,\text{reach}}, p_I, p_G)$ when that tuple is not canonically derived;

- a relation-diagnostic package $V_R[V_I, V_G, \mathfrak{P}_R]$ cites its operand structures and has interface content including $\mathcal{U}_{I \leftarrow G}$, D_I , and Agg_R .

An operand cited by a package is not thereby redeclared as an internal component. In particular, V_R cites Φ_I, Φ_G, p_I, p_G through its locked operands but does not copy them into $\mathcal{I}_R$.

Remark 4.10 (Package completeness is module-relative). Package completeness is relative to the declared report module. A package may be complete for one report while simultaneously serving as the inherited base for a distinct module-specific refinement. The refinement supplies the task-specific operands and clauses required by the second module; it does not retroactively render the original package incomplete. In the RG application [18], the nine-component package V is complete for the Route-I endpoint-divergence module in the flattened sense of Remark 4.9, and simultaneously serves as the common inherited base from which the strict-path and geometric packages are refined.

Remark 4.11 (Typed internal refinements preserve the package schema). The tuple

$$V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$$

remains the authoritative top-level package schema throughout the paper. Optional modules may refine the internal content of an existing component without adding a new top-level slot. In particular, the fixed-target resolution module first declares a raw realization clause $\mathcal{R}_\lambda^{\text{raw}} = (\mathcal{N}_\lambda, D^*, \hat{D}_\lambda)$, then extends it by the certification domain, one-sided envelopes, and certificate object to obtain $\mathcal{R}_\lambda^{\text{cert}}$, with

$$\mathcal{I}_\lambda = \mathcal{I}^{\text{base}} \oplus \mathcal{R}_\lambda^{\text{cert}}.$$

The module refines $\mathcal{Q}$ first by task clauses (ordered null/separation pairs and directional pairwise or partition semantics) and only then, when invoked, by a separate policy-floor clause. The scope under which certificates are reused in scans or adaptive paths, and the adaptive control law itself, remain protocol declarations associated with the package card. None replaces the package definition. Here $\oplus$ denotes typed extension of an existing component, not a new algebraic primitive.

Remark 4.12 (Node-local threshold binding). A package node always has a threshold slot in the four-slot schema, but the slot binds a report only when it is explicitly invoked as an active reporting clause. Thus $\varepsilon_{\Sigma_I}, \varepsilon_{\Sigma_G}, \varepsilon_I, \varepsilon_G, \varepsilon_{\Sigma_R}$, and ε_R , when active, are independent by default and are not inherited along dependency edges. For the relation diagnostic, the default threshold report is

$$\Delta_{I \rightarrow G} \approx_{\varepsilon_R} 0 \quad \Longleftrightarrow \quad \Delta_{I \rightarrow G} < \varepsilon_R.$$

A cross-node threshold-alignment clause is required only when a package consumes thresholded operand reports, when a readout map depends on an endpoint threshold, when a theorem compares resolutions at different nodes, or when one numerical threshold is reused across completed reports. Otherwise no conversion between node-local thresholds is presumed.

Remark 4.13 (Scalar-threshold scope and deferred typed extension). The present core formalizes nonnegative scalar reports with positive scalar thresholds. It does not exclude typed multicomponent resolution objects. A multicomponent extension must additionally declare a resolution space $\mathcal{E}_\bullet$, the report space on which negligibility is tested, a comparison preorder or acceptance region, a componentwise or coupled negligibility predicate, and any transport required across package nodes. Schematically one may write

$$\varepsilon_\bullet \in \mathcal{E}_\bullet, \quad r \approx_{\varepsilon_\bullet} 0 \quad \Longleftrightarrow \quad r \in \mathcal{N}_{\varepsilon_\bullet},$$

but no particular product order, norm, or acceptance geometry is imposed here. The formal multicomponent theory is deferred rather than excluded.

Definition 4.11 (Resolution-aware indistinguishability). Let a package node $V_\bullet$ produce a nonnegative scalar report x and declare threshold $\varepsilon_\bullet > 0$. Conditional on package well-posedness and on every additional operability, comparability, and certification gate invoked by the report,

$$x \approx_{\varepsilon_\bullet} 0 \iff x < \varepsilon_\bullet.$$

The inequality fixes the semantics of that package-local threshold predicate; it does not establish full report licensing and does not align $\varepsilon_\bullet$ with thresholds belonging to other package nodes. When a discrepancy rule is used, the same convention applies only after a symmetric report rule is declared. A directional threshold predicate such as $E_{\lambda,\varepsilon}$ in Sec. 4.4 is not, by itself, an instance of the symmetric relation $\approx_\varepsilon$; an explicit symmetric rule or certified symmetrization is required. Typed multicomponent thresholds are outside the scalar core and are deferred as stated in Remark 4.13.

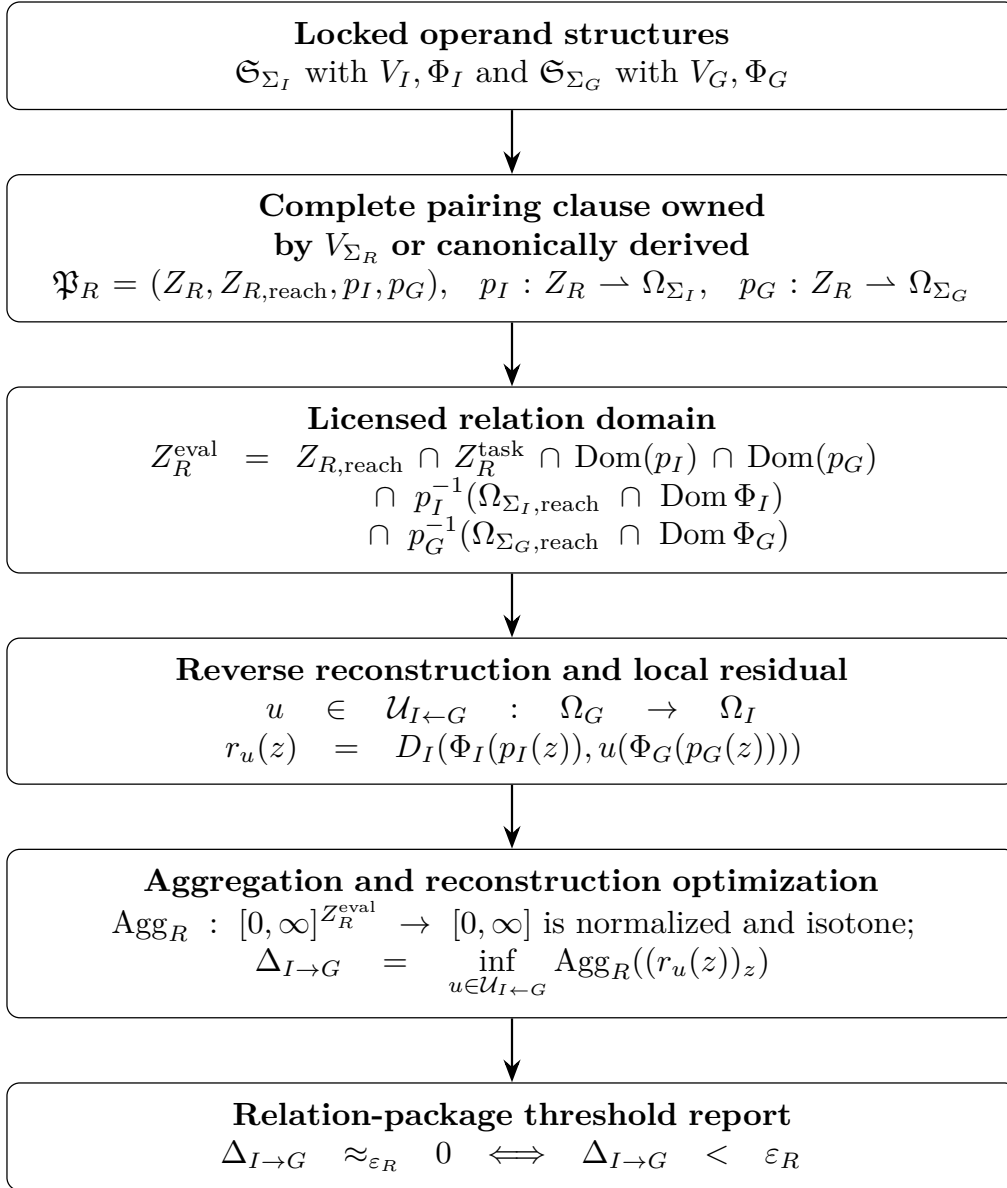


Figure 3: Typed construction of the relation diagnostic. Pairing fixes correspondence but not reconstruction direction. The subscript $I \rightarrow G$ names the passage under audit; the reverse reconstruction $G \rightarrow I$ tests whether the G -side report is sufficient to recover the I -side report. Aggregation produces the scalar before the relation threshold acts.

Remark 4.14 (The reporting threshold is declared but may also require feasibility certification). The framework does not supply a universal, task-independent threshold. Each threshold belongs to the package node whose output it classifies. For the fixed-slice relation diagnostic, ε_R belongs to V_R and is independent by default from thresholds belonging to the substrate, perspective, or pairing nodes. Declaration alone is not always sufficient: when a certified numerical-realization refinement of the relation interface is invoked, the completed relation package must lie in the corresponding certificate-relative set of Sec. 4.4. Measurement resolution, numerical precision, truncation analysis, sampling certification, task loss, and policy tolerance enter through typed clauses rather than as an undifferentiated list of threshold sources. Failure under a current certificate does not prove impossibility under every sharper certificate or improved mechanism.

Protocol 4.1 (Minimal dependency-aware reporting card). A completed report must identify the package nodes and derived structures actually invoked. For a unary Ψ report, items 1–2 and 5 are sufficient together with the conflict-functional clause. For a relational $\Delta_{I \rightarrow G}$ report, the card records at least:

1. **Substrate blocks:** $V_{\Sigma_I}, V_{\Sigma_G}$ and the resulting operand substrate structures, including boundary, carrier construction, admissible evolution, and reachability. State explicitly when the two substrate nodes coincide. For Ψ , record the unary substrate V_{Σ_A} .
2. **Perspective blocks:** V_I, V_G , their readout domains and maps Φ_I, Φ_G ; for Ψ , the relevant V_A, Φ_A, E_A .
3. **Pairing block:** the owner or canonical derivation of the complete tuple

$$\mathfrak{P}_R = (Z_R, Z_{R,\text{reach}}, p_I, p_G),$$

including the domains of partial projections. Identity pairing must be stated explicitly.

4. **Relation-diagnostic block:** $V_R, \mathcal{U}_{I \leftarrow G}, D_I$, normalized/isotone Agg_R , any invoked zero-detection class, the task-relative relation domain, and any non-degeneracy or consistency clauses.
5. **Threshold block:** identify which threshold slots are active and record the threshold of each package node whose thresholded output is actually produced or consumed. For the relation report, state ε_R and the meaning of $\approx_{\varepsilon_R}$. Record a cross-node alignment rule only when one of the triggers in Remark 4.12 is invoked.
6. **Resolution-certification block, when invoked:** the raw and certified refinement family, fixed target discrepancy, reported discrepancy, certification domain, one-sided envelopes, query margins, null/separation clauses, reporting semantics, policy floor if any, and certificate scope.
7. **Comparison block, when invoked:** transport and domain/threshold reconciliation for comparisons between already completed reports. Relation formation from declared operands is not entered here as cross-package comparison.

Definition 4.12 (Package-locked reporting structure (dependency record)). A *package-locked reporting structure* is the typed dependency record sufficient to evaluate a declared report. For a unary perspective report, one may write

$$\mathcal{V}_A := (\mathfrak{S}_{\Sigma_A}, V_A, E_A, [\varepsilon_A], \mathcal{Q}_A, \mathcal{C}_A, \models_{V_A}),$$

where brackets indicate that the threshold slot is recorded as active only when a thresholded unary report is invoked.

For the ordered relation report auditing $I \rightarrow G$, write

$$\mathcal{V}_R := (\mathfrak{S}_{\Sigma_I}, \mathfrak{S}_{\Sigma_G}, V_I, V_G, V_{\Sigma_R} \text{ or a canonical derivation certificate, } \mathfrak{P}_R, V_R, \mathcal{U}_{I \leftarrow G}, D_I, \text{Agg}_R, \varepsilon_R, \mathcal{Q}_R, \mathcal{C}_R, \models_{V_R}).$$

Here $\mathfrak{P}_R = (Z_R, Z_{R,\text{reach}}, p_I, p_G)$ is locked as a complete tuple or canonically derived as a complete tuple. The relation package cites, rather than redeclares, the operand substrates, readouts, and pairing maps. The symbols $\models_{V_A}$ and $\models_{V_R}$ denote evaluation under the corresponding package-locked encodings, admissibility clauses, domains, and reporting rules.

The legacy notation $\mathcal{V}$ or “under V ” may suppress this dependency record only when a package card states the flattening. Quantities reported in this manuscript are package-locked typed readouts or obstructions of the relevant dependency record; no general automorphism-invariance claim is implied by the word “locked.”

Example 4.1 (Dependency-aware Package Card: XOR 2-bit, flattened ID `Pkg_toy_xor`). The card has the following declared blocks.

- **ID / scope.** Reports unary Ψ and the relation gap $\Delta_{I \rightarrow G}$. The flattened identifier is

$$\text{Pkg_toy_xor.}$$

- **Substrate block.** V_{Σ}^{toy} locks the 2-bit boundary with no environment variables and produces

$$\mathfrak{S}_{\Sigma}^{\text{toy}} = (\mathcal{L}, \Omega_{\Sigma}, \Omega_{\Sigma,\text{reach}}), \quad \Omega_{\Sigma} = \Omega_{\Sigma,\text{reach}} = \{00, 01, 10, 11\}.$$

- **Perspective blocks.** V_I^{toy} declares $\Phi_I = \text{id} : \Omega_{\Sigma} \rightarrow \Omega_I (= \Omega_{\Sigma})$. V_G^{toy} declares $\Phi_G(x_1 x_2) = x_1 \oplus x_2 : \Omega_{\Sigma} \rightarrow \Omega_G (= \{0, 1\})$. For the unary report route, $V_A^{\text{toy}} = V_I^{\text{toy}}$ and $\Phi_A = \text{id}$.
- **Pairing block.** The relation carrier is canonically derived from the substrate:

$$Z_R = Z_{R,\text{reach}} = \Omega_{\Sigma,\text{reach}}, \quad p_I = p_G = \text{id}_{Z_R}.$$

Thus this is the shared-substrate identity-pairing specialization and no separate non-derived carrier is introduced.

- **Unary conflict clause.** $E_A(x_1 x_2) = \mathbf{1}\{x_1 = x_2 = 1\}$, giving $\Psi = 0$ in the infimum reading. The alternative $E_A \equiv 1$ would define a different unary package and is not part of this card.
- **Relation-diagnostic block.** V_R^{toy} declares D_I as Hamming distance on 2-bit strings,

$$\mathcal{U}_{I \leftarrow G} = \{u : \{0, 1\} \rightarrow \Omega_{\Sigma} : \Phi_G(u(g)) = g\},$$

and the aggregation rule $\text{Agg}_R = \sup_{z \in Z_R^{\text{eval}}}$ with $Z_R^{\text{eval}} = Z_R$. The outer infimum over u belongs to the definition of $\Delta_{I \rightarrow G}$ rather than to Agg_R .

- **Threshold block.** Only the relation threshold is invoked: $\varepsilon_R = 2$, with $\Delta_{I \rightarrow G} \approx_{\varepsilon_R} 0$ iff $\Delta_{I \rightarrow G} < 2$. No endpoint thresholded report is consumed, so no cross-node threshold alignment is required.
- **Comparison block.** No transport law between this completed report and another package report is declared; such cross-report comparisons are **undefined** under Normative Rule 2.2.

Example 4.2 (Minimal end-to-end pipeline report under V_{toy}). Consider the dependency-aware package card in Example 4.1. The symbol V_{toy} used below is its declared flattened ID. The unary and relational reports are well-defined because the substrate, perspective, identity-pairing, relation-interface, aggregation, and relation-threshold blocks are explicitly locked.

Computing $\Delta_{I \rightarrow G}$ (fixed-slice). For the XOR macro, the fibers are $\Phi_G^{-1}(0) = \{00, 11\}$ and $\Phi_G^{-1}(1) = \{01, 10\}$. Fix any admissible lifting $u : \{0, 1\} \rightarrow \Omega$, i.e. a macro-consistent section satisfying $\Phi_G(u(g)) = g$. On the fiber $\Phi_G^{-1}(0)$, macro-consistency forces $u(0) \in \{00, 11\}$; the best admissible choice still has worst-case Hamming residual 2 (because $D_I(00, 11) = 2$). Likewise, on $\Phi_G^{-1}(1)$, macro-consistency forces $u(1) \in \{01, 10\}$, and the worst-case residual is 2. Therefore

$$\sup_{S \in \Omega_{\Sigma, \text{reach}}} D_I(\Phi_I(S), u(\Phi_G(S))) = 2 \quad \text{for every admissible } u,$$

and hence

$$\Delta_{I \rightarrow G}(\mathcal{L}; V_{\text{toy}}) = \inf_{u \in \mathcal{U}_{I \leftarrow G}} 2 = 2.$$

Because the relation package V_R^{toy} locks $\varepsilon_R = 2$, the strict relation-threshold convention gives $\Delta_{I \rightarrow G} = 2 \not\leq 2$. The licensed output is therefore “not resolution-negligible” under the flattened report ID V_{toy} .

Computing Ψ and the induced H/M regime label. The conflict functional locked by V_A^{toy} yields $\Psi = 0$ in the infimum reading, and the infimum is realized in this toy domain. Under the level-locked H/M classification (Definition 5.5), this unary perspective package yields an H-type regime. Replacing E_A by $E_A \equiv 1$ would define a distinct unary package and report.

Undefinedness check. If D_I , $\mathcal{U}_{I \leftarrow G}$, the pairing block, or the relation evaluation domain is absent, then the flattened expression $\Delta_{I \rightarrow G}(\mathcal{L}; V_{\text{toy}})$ is undefined (under-specification; optional tag U-spec).

Cross-package boundary. Changing D_I , ε_R , $\mathcal{U}_{I \leftarrow G}$, or the pairing structure defines a different relation report. Comparing already completed reports requires explicit transport and any triggered threshold alignment; otherwise the comparison is undefined (Normative Rule 2.2).

4.4 Resolution-controlled refinements and certified threshold feasibility

Declaring a reporting threshold does not by itself establish that the threshold is supportable by the invoked resolution mechanism or fine enough for the declared task. This subsection records a minimal fixed-target certification result. It must be attached to the completed package node whose discrepancy report is being realized; that active node is identified in the package card. It introduces no new top-level component of the package $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$. Instead, when this optional module is invoked, it adds typed subclauses inside the existing interface and query components. Its scope is deliberately narrow: a scalar target discrepancy is held fixed, a parameterized mechanism family reports estimates of that same discrepancy, and the audited task declares ordered-pair separation requirements and, when a full two-sided classifier is invoked, ordered-pair null-preservation requirements. Partition output is treated only after a symmetric threshold relation or a declared certified symmetrization is supplied. The full pairwise/partition module is given in Appendix E.

Remark 4.15 (Declared threshold conversion versus theorem-derived sufficient budget). A declared threshold alignment or conversion rule is part of the package specification that types a comparison or downstream report. A theorem may instead derive a sufficient outer threshold condition from separately verified hypotheses. The latter is a theorem-induced budget: it is a derived conclusion, not part of the original declaration, and it need not be a necessary relation among the realized thresholds of a particular instance. These roles must not be identified. The distinction is an instance of Normative Rule 2.4; the companion composition example is recorded only in Sec. 9.

Definition 4.13 (Fixed-target raw resolution refinement). Fix base package clauses

$$\mathcal{I}^{\text{base}}, \quad \mathcal{Q}^{\text{base}}, \quad \mathcal{C}^{\text{base}},$$

where $\mathcal{I}^{\text{base}}$ retains the original structural interface, including the core clauses required by Definition 4.10. Let X_{adm} be a package-locked report domain, or an explicitly derived domain cited by the active report package, and let

$$D^* : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

be a fixed target discrepancy. The target D^* may be the base discrepancy D itself or a declared scalar report induced by the base interface; it does not replace or redefine the underlying structural interface and need not be symmetric.

A *raw fixed-target resolution refinement* is a declared family

$$\mathfrak{R}_{\text{res}}^{\text{raw}} := \{\mathcal{R}_{\lambda}^{\text{raw}}\}_{\lambda \in \Lambda_{\text{res}}}, \quad \mathcal{R}_{\lambda}^{\text{raw}} := (\mathcal{N}_{\lambda}, D^*, \hat{D}_{\lambda}),$$

where $\mathcal{N}_{\lambda}$ is the resolution or numerical-realization mechanism and

$$\hat{D}_{\lambda} : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

is its report of the same fixed target discrepancy. The control parameter λ may represent, for example, a step size, mesh width, sampling rate, quadrature order, or truncation order; it is not the report threshold ε .

Definition 4.14 (Uniform certification and certified interface refinement). Let $\mathcal{D}_{\text{cert}} \subseteq X_{\text{adm}} \times X_{\text{adm}}$ be the declared certification domain. A two-sided certification scheme consists of one-sided envelopes $\rho_+, \rho_- : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}$ and a declared certificate object Cert such that, deterministically or on a declared uniform confidence event,

$$\hat{D}_{\lambda}(x, y) - D^*(x, y) < \rho_+(\lambda), \quad D^*(x, y) - \hat{D}_{\lambda}(x, y) < \rho_-(\lambda)$$

for all $(x, y) \in \mathcal{D}_{\text{cert}}$. Define the certified refinement subclause

$$\mathcal{R}_{\lambda}^{\text{cert}} := (\mathcal{R}_{\lambda}^{\text{raw}}, \mathcal{D}_{\text{cert}}, \rho_+(\lambda), \rho_-(\lambda), \text{Cert}),$$

and the refined interface

$$\mathcal{I}_{\lambda} := \mathcal{I}^{\text{base}} \oplus \mathcal{R}_{\lambda}^{\text{cert}}.$$

Thus the certificate data are part of the typed internal refinement of $\mathcal{I}$, not a new top-level package slot.

Under the strict-envelope convention above, an exact or non-strict error certificate is represented by adding a declared positive slack. A zero envelope is valid only when the corresponding signed report error is uniformly strictly negative; in particular, symmetric zero envelopes do not represent an exact mechanism under the strict convention.

Definition 4.15 (True and certified query margins). Let $Q_{\text{sep}} \subseteq \mathcal{D}_{\text{cert}}$ be a nonempty, predeclared set of ordered pairs whose declared directional distinctions must remain reportably non-negligible, with $x \neq y$ for every $(x, y) \in Q_{\text{sep}}$. Define the ideal pairwise margin

$$\delta_Q^{\text{pair}} := \inf_{(x, y) \in Q_{\text{sep}}} D^*(x, y).$$

Operational use requires a declared certified lower-margin family

$$\delta_Q^{\text{pair}} : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}, \quad 0 \leq \delta_Q^{\text{pair}}(\lambda) \leq \delta_Q^{\text{pair}},$$

together with the pointwise, simultaneous, or sequential scope under which it remains valid. A family-uniform certificate is the special case in which $\underline{\delta}_Q^{\text{pair}}(\lambda) \equiv \underline{\delta}_Q^{\text{pair}}$ on the invoked mechanism family. When the ideal margin is exactly available, one may take $\underline{\delta}_Q^{\text{pair}}(\lambda) = \delta_Q^{\text{pair}}$ for every invoked λ .

When a full two-sided threshold classifier is invoked, also declare a nonempty ordered-pair null set $Q_0 \subseteq \mathcal{D}_{\text{cert}}$ satisfying

$$D^*(x, y) = 0 \quad \text{for every } (x, y) \in Q_0,$$

whose pairs must be reported negligible. Let Sem specify directional pairwise output or partition output, and let $p \geq 0$ be any declared policy floor.

Definition 4.16 (Task-indexed query refinements and completed packages). Let

$$\sigma \in \{\text{pair}, \text{part}\}, \quad \tau \in \{\text{sep}, \text{class}\}$$

index the report semantics and the task mode. Define the certificate-level query refinements

$$\mathcal{Q}^{\sigma, \text{sep}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, \text{Sem} = \sigma),$$

and, when the declared null set Q_0 is present,

$$\mathcal{Q}^{\sigma, \text{class}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, Q_0, \text{Sem} = \sigma).$$

If a policy floor is invoked, define

$$\mathcal{Q}^{\sigma, \tau, p} := \mathcal{Q}^{\sigma, \tau} \oplus p.$$

For each $(\lambda, \varepsilon) \in \Lambda_{\text{res}} \times \mathbb{R}_{>0}$, the corresponding ordinary completed packages under the unchanged schema are

$$V_{\lambda, \varepsilon}^{\sigma, \tau} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau}, \mathcal{C}^{\text{base}}),$$

and, when policy is invoked,

$$V_{\lambda, \varepsilon}^{\sigma, \tau, p} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau, p}, \mathcal{C}^{\text{base}}).$$

Thus directional-pairwise versus partition semantics, separation-only versus full classification, and certificate-only versus policy-restricted use are distinct internal query refinements of the same top-level package schema. The ρ_+ floor belongs only to the $\tau = \text{class}$ mode.

Definition 4.17 (Task-indexed package-admissible threshold sets). For each fixed mechanism parameter λ and task index (σ, τ) , define

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau}(\lambda) := \left\{ \varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau} \text{ is admissible} \right\},$$

and, when a policy clause is invoked,

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau, p}(\lambda) := \left\{ \varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau, p} \text{ is admissible} \right\}.$$

These task-indexed sets collect the package-admissibility restrictions not already expressed by the algebraic certificate inequalities. The certificate inequalities below generate algebraic threshold slices; the actually licensed slices are their intersections with the corresponding $\mathcal{E}_{\text{adm}}^{\sigma, \tau}$ sets. No endpoint criterion may silently replace that intersection unless the required algebraic slice is known to lie inside the relevant task-indexed package-admissible threshold set.

Envelope	Error controlled	Operational roles
ρ_+	over-report allowance: $\widehat{D}_{\lambda} - D^* < \rho_+$	sets the null-preservation floor for a declared Q_0 ; converts an observed report into a certified lower bound $D^* > \widehat{D}_{\lambda} - \rho_+$; may therefore also enter the construction of a certified query margin
ρ_-	under-report allowance: $D^* - \widehat{D}_{\lambda} < \rho_-$	erodes the certified separation-query ceiling and converts a certified lower bound on D^* into the display guarantee $\widehat{D}_{\lambda} > D^* - \rho_-$

The subscripts refer to the sign of the report error, not to whether the resulting bound on D^* is upper or lower. In particular, ρ_+ is the envelope used when an observed positive report is converted into a lower bound on the target discrepancy.

Definition 4.18 (Certified directional pairwise feasibility sets). Under the directional threshold convention

$$\widehat{D}_\lambda(x, y) < \varepsilon \implies (x, y) \text{ is reported negligible in the declared direction,}$$

define the algebraic slices

$$I_\lambda^{\text{sep, cert}} := \left(0, \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right],$$

and, when Q_0 is declared,

$$I_\lambda^{\text{class, cert}} := (0, \infty) \cap \left[\rho_+(\lambda), \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right].$$

The package-conditioned certified sets are

$$\mathcal{A}_{Q_{\text{sep}}}^{\text{sep, cert}} := \left\{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair, sep}}(\lambda) \cap I_\lambda^{\text{sep, cert}}\right\},$$

and

$$\mathcal{A}_{Q_0, Q_{\text{sep}}}^{\text{class, cert}} := \left\{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap I_\lambda^{\text{class, cert}}\right\}.$$

The first protects the declared directional distinctions; the second additionally guarantees negligible reports on the declared ordered null pairs.

The policy-restricted slices are

$$I_{\lambda, p}^{\text{sep, cert}} := I_\lambda^{\text{sep, cert}} \cap [p, \infty), \quad I_{\lambda, p}^{\text{class, cert}} := I_\lambda^{\text{class, cert}} \cap [p, \infty),$$

and the corresponding licensed sets use $\mathcal{E}_{\text{adm}}^{\text{pair, sep}, p}(\lambda)$ and $\mathcal{E}_{\text{adm}}^{\text{pair, class}, p}(\lambda)$, respectively.

Proposition 4.1 (Fixed-mechanism certified feasibility). *Fix λ and write*

$$\underline{b}_\lambda := \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda).$$

The exact fixed-mechanism licensed slices are

$$\mathcal{E}_{\text{adm}}^{\text{pair, sep}}(\lambda) \cap (0, \underline{b}_\lambda],$$

for separation-only reporting, and

$$\mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap (0, \infty) \cap [\rho_+(\lambda), \underline{b}_\lambda]$$

for two-sided classification. Policy replaces the separation and classification admissibility sets by $\mathcal{E}_{\text{adm}}^{\text{pair, sep}, p}$ and $\mathcal{E}_{\text{adm}}^{\text{pair, class}, p}$, respectively, and intersects the algebraic slice with $[p, \infty)$. Consequently, each licensed slice is nonempty if and only if the displayed intersection is nonempty.

Corollary 4.1 (Endpoint criteria under admissibility-neutrality). *Suppose the relevant algebraic slice is contained in the corresponding package-admissible threshold set. Then the separation-only slice is nonempty exactly when $\underline{b}_\lambda > 0$, and the two-sided classification slice is nonempty exactly when*

$$\rho_+(\lambda) \leq \underline{b}_\lambda \quad \text{and} \quad \underline{b}_\lambda > 0.$$

The policy-restricted endpoint conditions are, respectively,

$$p \leq \underline{b}_\lambda, \quad \underline{b}_\lambda > 0,$$

and

$$\max\{p, \rho_+(\lambda)\} \leq \underline{b}_\lambda, \quad \underline{b}_\lambda > 0.$$

If $\rho_+ = \rho_- = \rho$, the full-classification algebraic slice is nonempty exactly when

$$2\rho(\lambda) \leq \underline{\delta}_Q^{\text{pair}}(\lambda) \quad \text{and} \quad \underline{\delta}_Q^{\text{pair}}(\lambda) > 0.$$

These endpoint tests certify package-conditioned feasibility only under the stated admissibility containment.

Remark 4.16 (Ideal and certified margins). Replacing $\underline{\delta}_Q^{\text{pair}}(\lambda)$ by the ideal margin δ_Q^{pair} defines the corresponding ideal algebraic geometry. The operational certificate slices are conservative because $\underline{\delta}_Q^{\text{pair}}(\lambda) \leq \delta_Q^{\text{pair}}$. Failure of a package-conditioned certified set may arise from an empty algebraic certificate slice, from package inadmissibility on that slice, or from both; it need not imply failure of the ideal geometry.

Remark 4.17 (Certificate-relative, package-conditioned output). The framework does not derive a universal, task-independent threshold. It derives certificate-relative, package- and query-conditioned sets once a fixed target discrepancy, a typed resolution refinement of the interface, typed task and policy refinements, and external physical, numerical, or statistical certificates have been declared. Failure to enter one of these sets does not, by itself, prove that no sharper margin certificate, sharper mechanism certificate, or improved mechanism could license the report.

Remark 4.18 (Multiple audited queries). If one completed package uses one threshold for several audited query bundles, the configuration must lie in the intersection of their certified sets. Replacing the bundles by unions of their Q_{sep} and Q_0 clauses is equivalent only when all bundles share the same D^* , raw and certified refinement family, certification domain, envelopes, margin-certification construction, certificate scope, report semantics, and policy clauses. If policy floors differ, the joint policy clause uses $p_{\text{joint}} := \sup_i p_i$. If directional pairwise and partition semantics or certificate scopes differ, the intersection formulation must be retained. A threshold certified for only one member of $\mathcal{Q}$ is not thereby certified for the whole package.

Remark 4.19 (Pairwise scope and partition boundary). Proposition 4.1 uses an ordered-pair threshold-report predicate and does not require a symmetric discrepancy. This directional predicate is not, by itself, the framework’s symmetric resolution-indistinguishability predicate $\approx_\varepsilon$; it may instantiate that relation only after a symmetric rule or a declared certified symmetrization is supplied. If the deliverable must be an equivalence partition containing all threshold-level identifications, direct pairwise separation is insufficient because finite-resolution indistinguishability need not be transitive. Appendix E introduces the corresponding bottleneck-chain margin and identifies transitive closure as the precise contact point with finite-resolution repair non-uniqueness.

4.5 Comparability across packages

Definition 4.19 (Comparability and transport law for completed reports). Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be two completed package-locked reporting structures producing reports R_1 and R_2 . Cross-report comparison is defined only if a comparison package declares a transport law covering the report objects actually compared. The declaration includes, as required:

- maps transporting the relevant report carriers or representation spaces into a common typed comparison space, together with their domains of validity;
- transformation rules for any discrepancy, aggregation, ordering, or admissible class consumed by the comparison;
- reconciliation of the operand or evaluation domains when the comparison depends on them;

- threshold alignment or conversion only when thresholded outputs are consumed, a numerical threshold is reused, or the comparison itself is thresholded.

A common codomain or common scalar notation is not by itself a transport declaration. If the required transport does not cover both completed reports, the comparison is **undefined**; see Normative Rule 2.2. Relation formation from declared operands is not a comparison between completed reports and is therefore not governed by this definition.

Remark 4.20 (Construction bridges and transport bridges). A transport law compares, identifies, or re-expresses report objects that are already well-defined under their respective completed packages. When an invoked comparison requires an object that has not yet been instantiated for one requested module, transport is not yet applicable. The declaration schema of that module must first supply the task-specific operands and construction clauses required to define the object and its completed package. Only after each requested module yields a well-posed report object may transport, resolution alignment, or scalar comparison be invoked. Thus construction may precede transport when module-specific packages have different operand types or arities. This distinction introduces no new framework primitive: the construction remains part of the corresponding module declaration.

Absence of a required construction is under-specification of the requested module report, whereas absence of transport between two already-defined reports is non-comparability.

5 Core Quantities: Ψ and the Fixed-Slice Gap $\Delta_{I \rightarrow G}$

This section defines the two core quantities of the framework. Both are package-locked, but their dependency types differ. A unary Ψ report requires a locked substrate and perspective record; a relational $\Delta_{I \rightarrow G}$ report additionally requires a complete pairing structure and a relation-diagnostic package (Sec. 4).

Main-line convention. The authoritative fixed-slice type is $\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R)$. Retained downstream theorems and examples may use $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ only as the declared shared-substrate flattened shorthand. Accumulated quantities and ontological guardrails are written explicitly as $\Delta_{\text{acc}}(\gamma)$ and Δ_R^* .

5.1 Intra-level structural tension readout Ψ

Definition 5.1 (Conflict functional (declared)). Fix a package-locked substrate $\mathfrak{S}_{\Sigma_A}$ and a perspective package $V_A[\mathfrak{S}_{\Sigma_A}]$ at a fixed level specification. Write

$$\mathfrak{S}_{\Sigma_A} = (\mathcal{L}_A, \Omega_{\Sigma_A}, \Omega_{\Sigma_A, \text{reach}}),$$

and let the perspective readout be

$$\Phi_A : \Omega_A^{\text{dom}} \rightarrow \Omega_A.$$

A *conflict functional* under A is a declared map

$$E_A : \Omega_A \rightarrow \mathbb{R}_{\geq 0},$$

whose audited tension source, zero condition, and discrepancy meaning are fixed by the interface content of V_A or by an explicitly cited unary-report refinement. The conflict functional does not redefine the substrate reachability convention.

Definition 5.2 (Ψ (unary intra-level structural tension readout)). Fix a package-locked substrate $\mathfrak{S}_{\Sigma_A}$ and a perspective package $V_A[\mathfrak{S}_{\Sigma_A}]$ declaring Φ_A and an admissible conflict functional E_A . Let $\Omega_A^{\text{rel}} \subseteq \Omega_{\Sigma_A}$ be any declared unary state-domain restriction. Its source slot in V_A must be stated; query- or history-induced restrictions are recorded through $\mathcal{Q}_A$ or $\mathcal{C}_A$ before the effective domain is formed. Set $\Omega_A^{\text{rel}} := \Omega_{\Sigma_A}$ when no additional restriction is invoked. Define

$$\Omega_A^{\text{eval}} := \Omega_{\Sigma_A, \text{reach}} \cap \Omega_A^{\text{rel}} \cap \text{Dom}(\Phi_A).$$

The unary intra-level structural tension readout is

$$\Psi(\mathfrak{S}_{\Sigma_A}, V_A) := \inf_{S \in \Omega_A^{\text{eval}}} E_A(\Phi_A(S)).$$

Non-vacuity clause. The variables, audited tension source, zero condition, discrepancy meaning, and evaluation-domain construction of E_A must be explicitly tied to the locked unary dependency record. Otherwise the report is **undefined** rather than freely assigned.

Operability gate. If $\Omega_A^{\text{eval}} = \emptyset$, numerical reporting is **undefined** under Protocol 8.4.

Flattened shorthand. When Φ_A is total on the locked reachable substrate, no further unary task restriction is invoked, and the substrate and perspective declarations are flattened into one audit bundle V , the legacy notation

$$\Psi(\mathcal{L}, A; V) := \inf_{S \in \Omega_{\text{reach}}} E_A(\Phi_A(S))$$

is retained as an exact shorthand. When the unary threshold slot is active, the package-local report is $\Psi \approx_{\varepsilon_A} 0$ iff $\Psi < \varepsilon_A$.

Remark 5.1 (Allowed values and interpretation). By construction, $\Psi(\mathfrak{S}_{\Sigma_A}, V_A) \in [0, \infty]$ and in particular Ψ may be equal to 0.

Neutral reading. $\Psi(\mathfrak{S}_{\Sigma_A}, V_A) = 0$ means that, within the declared reachability and within the chosen conflict functional, there exist reachable states whose residual conflict can be made arbitrarily small under A . $\Psi(\mathfrak{S}_{\Sigma_A}, V_A) > 0$ means that a strictly positive lower bound persists under the same declared regime. No cross-level claims are implied by Ψ alone.

Regime dependence. Changing any of the locked clauses (e.g. reachable set, admissibility, or the choice of E_A) generally changes the operational meaning and value of Ψ . Therefore, any reported Ψ must be accompanied by the relevant package elements (Sec. 4.3, Protocol 4.1).

5.2 Inter-level structural gap $\Delta_{I \rightarrow G}$

Definition 5.3 ($\Delta_{I \rightarrow G}$ (package-locked relational reconstruction gap)). Fix:

- (i) package-locked operand substrates

$$\mathfrak{S}_{\Sigma_I} = (\mathcal{L}_I, \Omega_{\Sigma_I}, \Omega_{\Sigma_I, \text{reach}}), \quad \mathfrak{S}_{\Sigma_G} = (\mathcal{L}_G, \Omega_{\Sigma_G}, \Omega_{\Sigma_G, \text{reach}});$$

- (ii) perspective packages $V_I[\mathfrak{S}_{\Sigma_I}]$ and $V_G[\mathfrak{S}_{\Sigma_G}]$ with maps

$$\Phi_I : \Omega_I^{\text{dom}} \rightarrow \Omega_I, \quad \Phi_G : \Omega_G^{\text{dom}} \rightarrow \Omega_G;$$

- (iii) a complete package-locked or canonically derived pairing structure

$$\mathfrak{P}_R = (Z_R, Z_{R, \text{reach}}, p_I, p_G), \quad p_I : Z_R \rightharpoonup \Omega_{\Sigma_I}, \quad p_G : Z_R \rightharpoonup \Omega_{\Sigma_G};$$

- (iv) a relation-diagnostic package $V_R[V_I, V_G, \mathfrak{P}_R]$ declaring $\mathcal{U}_{I \leftarrow G}$, D_I , Agg_R , and the relation threshold ε_R when a thresholded relation report is invoked.

If the relation interface directly restricts relation states, let $Z_{R,I}^{\text{rel}} \subseteq Z_R$ denote that $\mathcal{I}_R$ -clause, with $Z_{R,I}^{\text{rel}} := Z_R$ when absent. Query predicates in $\mathcal{Q}_R$ and history/trajectory clauses in $\mathcal{C}_R$ may induce further licensed subdomains, denoted $Z_{R,Q}^{\text{rel}}$ and $Z_{R,C}^{\text{rel}}$. Define

$$Z_R^{\text{task}} := Z_{R,I}^{\text{rel}} \cap Z_{R,Q}^{\text{rel}} \cap Z_{R,C}^{\text{rel}}.$$

This is the effective relation-task domain; the source slot of every restriction must be stated.

The relation domain is resolved in four stages. First define the structural pairing domain

$$Z_R^{\text{struct}} := Z_{R,\text{reach}} \cap \text{Dom}(p_I) \cap \text{Dom}(p_G).$$

Then impose the effective task restriction:

$$Z_R^{\text{pair}} := Z_R^{\text{struct}} \cap Z_R^{\text{task}}.$$

Next impose both operand-reachability gates:

$$Z_R^{\text{operand}} := Z_R^{\text{pair}} \cap p_I^{-1}(\Omega_{\Sigma_I,\text{reach}}) \cap p_G^{-1}(\Omega_{\Sigma_G,\text{reach}}).$$

Finally impose both perspective-readout domain gates:

$$Z_R^{\text{eval}} := Z_R^{\text{operand}} \cap p_I^{-1}(\text{Dom } \Phi_I) \cap p_G^{-1}(\text{Dom } \Phi_G).$$

Thus

$$Z_R^{\text{struct}} \supseteq Z_R^{\text{pair}} \supseteq Z_R^{\text{operand}} \supseteq Z_R^{\text{eval}}.$$

Here inverse images under the partial projections are interpreted on their declared domains, and an absent Q - or C -induced restriction is interpreted as Z_R . Thus a relation package may further narrow the operand reachable domains but may not reopen states declared unreachable by either operand substrate. Auditing such counterfactual states requires redeclaring the relevant substrate package. Once the final domain is fixed,

$$\text{Agg}_R : [0, \infty]^{Z_R^{\text{eval}}} \rightarrow [0, \infty]$$

is normalized and isotone as required by Definition 4.9. For $u \in \mathcal{U}_{I \leftarrow G}$ define

$$r_u(z) := D_I(\Phi_I(p_I(z)), u(\Phi_G(p_G(z)))) , \quad z \in Z_R^{\text{eval}}.$$

The general relation gap is

$$\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R) := \inf_{u \in \mathcal{U}_{I \leftarrow G}} \text{Agg}_R((r_u(z))_{z \in Z_R^{\text{eval}}}).$$

For the core minimax specialization, $\text{Agg}_R = \sup$ on the licensed residual family, so

$$\Delta_{I \rightarrow G} = \inf_{u \in \mathcal{U}_{I \leftarrow G}} \sup_{z \in Z_R^{\text{eval}}} D_I(\Phi_I(p_I(z)), u(\Phi_G(p_G(z)))) .$$

The pairing structure fixes correspondence but does not select reconstruction direction. The subscript $I \rightarrow G$ names the compression or reporting passage under audit. Its sufficiency is tested through admissible reverse reconstructions $G \rightarrow I$ and discrepancy in the I -side readout space. The reverse audit $\Delta_{G \rightarrow I}$ may reuse the same substrates and pairing after declaring $\mathcal{U}_{G \leftarrow I}$, D_G , and the corresponding relation package clauses.

Shared-substrate identity-pairing specialization and flattened shorthand. The original notation $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ is retained when

$$\mathfrak{S}_{\Sigma_I} = \mathfrak{S}_{\Sigma_G} = \mathfrak{S}_{\Sigma}, \quad V_{\Sigma_R} = V_{\Sigma},$$

$$Z_R = Z_{R,\text{reach}} = \Omega_{\Sigma,\text{reach}}, \quad p_I = p_G = \text{id}_{\Omega_{\Sigma,\text{reach}}},$$

there are no additional relation-domain restrictions, $\text{Agg}_R = \text{sup}$, and both readouts are total on the reachable substrate:

$$\boxed{\Omega_{\Sigma,\text{reach}} \subseteq \text{Dom}(\Phi_I) \cap \text{Dom}(\Phi_G)}.$$

Under these conditions the recursively declared structure may be flattened into one audit bundle V , and the definition reduces exactly to

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) := \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D(\Phi_I(S), u(\Phi_G(S))).$$

Threshold binding. When the relation threshold slot is active, $\Delta_{I \rightarrow G} \approx_{\varepsilon_R} 0$ means $\Delta_{I \rightarrow G} < \varepsilon_R$. Thresholds belonging to operand, substrate, or pairing packages are independent by default. Alignment is required only under the triggers stated in Remark 4.12.

Operability gate. The numerical report is licensed only when $Z_R^{\text{eval}} \neq \emptyset$ and $\mathcal{U}_{I \leftarrow G} \neq \emptyset$. Empty-domain causes are recorded by the first failed stage:

- $Z_R^{\text{struct}} = \emptyset$: the declared relation carrier and projection domains provide no structural pairing instance (optional subtype ADM/pairing-structure-empty);
- $Z_R^{\text{struct}} \neq \emptyset$ but $Z_R^{\text{pair}} = \emptyset$: structural pairing instances exist, but the effective task restriction selects none of them (optional subtype ADM/task-domain-empty);
- $Z_R^{\text{pair}} \neq \emptyset$ but $Z_R^{\text{operand}} = \emptyset$: task-licensed pairing instances exist, but none is simultaneously reachable under both operand substrate declarations (optional subtype DOM/reachability-incompatible);
- $Z_R^{\text{operand}} \neq \emptyset$ but $Z_R^{\text{eval}} = \emptyset$: operand-reachable pairs exist, but the two perspective readout domains have no jointly evaluable instance (optional subtype DOM/readout-domain-empty).

If $\mathcal{U}_{I \leftarrow G} = \emptyset$, the report is likewise **undefined** (optional subtype ADM/reconstruction-class-empty). These subtypes refine the existing status vocabulary and do not introduce new top-level output states. If the infimum is attained, the attaining map may be called a best admissible reconstruction; otherwise only η -optimal admissible reconstructions are licensed.

Proposition 5.1 (Exact zero residual reaches the gap floor). *Suppose the relation report of Definition 5.3 is operable. If there exists $u_0 \in \mathcal{U}_{I \leftarrow G}$ such that*

$$r_{u_0} = 0_{Z_R^{\text{eval}}},$$

then

$$\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R) = 0.$$

Proof. Normalization gives

$$\text{Agg}_R(r_{u_0}) = \text{Agg}_R(0_{Z_R^{\text{eval}}}) = 0,$$

so the outer infimum satisfies $\Delta_{I \rightarrow G} \leq 0$. On the other hand, Agg_R has codomain $[0, \infty]$, hence every value entering the infimum is nonnegative and $\Delta_{I \rightarrow G} \geq 0$. Therefore $\Delta_{I \rightarrow G} = 0$. $\square$

Remark 5.2 (Arity-shifted diagnostic duality). The Ψ - $\Delta_{I \rightarrow G}$ pairing is a diagnostic duality across different arities, not an algebraic duality or an equality of package depth. The readout Ψ asks for irreducible tension within one package-locked perspective; $\Delta_{I \rightarrow G}$ asks for irreducible reconstruction residual between two package-locked perspectives under a declared pairing.

As an independence check, suppose $V_I = V_G = V_A$, the pairing is identity, $\text{id}_{\Omega_A} \in \mathcal{U}_{A \leftarrow A}$, and $D_A(y, y) = 0$ for every $y \in \Omega_A$. By the standing normalization of Agg_R , the resulting residual family aggregates to zero, so $\Delta_{A \rightarrow A} = 0$, while $\Psi(\mathfrak{S}_{\Sigma_A}, V_A)$ may still be strictly positive. Hence Ψ is not a degenerate instance of the relation-gap formula.

Standing specialization for the fiber bound. From Definition 5.4 through Corollary 5.1, assume the shared-substrate, total-readout, identity-pairing, minimax specialization of Definition 5.3. Thus $Z_R^{\text{eval}} = \Omega_{\Sigma, \text{reach}}$, $p_I = p_G = \text{id}$, $\text{Agg}_R = \text{sup}$, and the legacy symbols $\Omega_{\text{reach}}, \mathcal{U}, D$ abbreviate $\Omega_{\Sigma, \text{reach}}, \mathcal{U}_{I \leftarrow G}, D_I$. The active threshold is ε_R .

Definition 5.4 (Macro fibers and interface diameter). Under the standing shared-substrate specialization, for any macro value $g \in \Phi_G(\Omega_{\Sigma, \text{reach}})$ define the *macro fiber*

$$F_g := \{S \in \Omega_{\Sigma, \text{reach}} : \Phi_G(S) = g\}.$$

Its interface image is $\Phi_I(F_g) \subseteq \Omega_I$. If D_I is a pseudometric on Ω_I , define

$$\text{diam}_{D_I}(g) := \sup_{S, S' \in F_g} D_I(\Phi_I(S), \Phi_I(S')) \in [0, \infty].$$

See Section 5.3 for the corresponding geometric reading.

Assumption 5.1 (Pseudometric discrepancy for the fiber-diameter bound). The declared discrepancy rule D_I is a pseudometric on Ω_I : for all $x, y, z \in \Omega_I$,

$$D_I(x, x) = 0, \quad D_I(x, y) = D_I(y, x), \quad D_I(x, z) \leq D_I(x, y) + D_I(y, z).$$

This assumption is satisfied by common metric discrepancies used in auditing. When D_I is directed, the lower-bound mechanism below is not available in this form and requires a separately declared directed or two-sided variant.

5.3 Geometric intuition: projection and fibers

It is helpful to read the cross-level interface as a many-to-one projection. The map $\Phi_G : \Omega_{\Sigma, \text{reach}} \rightarrow \Phi_G(\Omega_{\Sigma, \text{reach}})$ collapses multiple micro states into a single macro report. For any $g \in \Phi_G(\Omega_{\Sigma, \text{reach}})$, the fiber

$$F_g := \{S \in \Omega_{\Sigma, \text{reach}} : \Phi_G(S) = g\}$$

collects the micro states that are indistinguishable at the reporting level.

A lifting $u \in \mathcal{U}_{I \leftarrow G}$ selects a representative $u(g) \in \Omega_I$ for each macro value g . Given the interface pseudometric D_I on the internal-side images, the worst-case representative radius at g is

$$r_u(g) := \sup_{T \in F_g} D_I(\Phi_I(T), u(g)).$$

By Assumption 5.1, any single representative controls pairwise variation on the same fiber: for all $S, S' \in F_g$,

$$D_I(\Phi_I(S), \Phi_I(S')) \leq D_I(\Phi_I(S), u(g)) + D_I(u(g), \Phi_I(S')) \leq 2r_u(g).$$

Taking the supremum over $S, S' \in F_g$ yields the structural bound

$$\text{diam}_{D_I}(g) \leq 2r_u(g).$$

Consequently, whenever a fiber has nonzero interface-image diameter, no lifting can be lossless in worst case: the residual is enforced by the fiber thickness under D_I , rather than by estimation noise. In the diagram below, the left panel displays the interface-image fiber $\Phi_I(F_g) \subseteq \Omega_I$, not the state fiber $F_g \subseteq \Omega_{\Sigma, \text{reach}}$ itself; this keeps $u(g)$ and the compared points in the same type space.

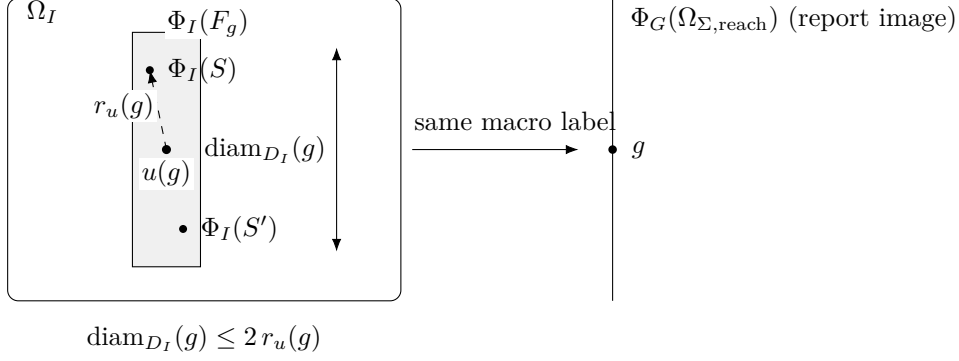


Figure 4: Fiber geometry in the shared-substrate identity-pairing minimax specialization. Here $Z_R^{\text{eval}} = \Omega_{\Sigma, \text{reach}}$ and $p_I = p_G = \text{id}$, so Φ_G collects substrate instances into a report fiber F_g and $\Phi_I(F_g) \subseteq \Omega_I$ lies in the reconstruction codomain. The representative $u(g)$ and both compared points therefore have the same type. The interface-image thickness controls the unavoidable worst-case representative radius through $\text{diam}_{D_I}(g) \leq 2r_u(g)$ (Assumption 5.1). This figure illustrates the flattened theorem specialization; it does not replace the general pairing structure of Definition 4.8.

Finally, a thresholded relation report uses the relation-package threshold ε_R . A distinct audit or realization package may carry its own threshold, but that threshold is not inherited by V_R unless an explicit alignment clause is invoked. Statements such as “reportable as 0” refer only to the package-local threshold predicate for the relevant completed residual and do not express ontological equality across levels.

Theorem 5.1 (Fiber-diameter lower bound for $\Delta_{I \rightarrow G}$). *Assume the standing shared-substrate specialization makes $\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R)$ well-defined, with $\Omega_{\Sigma, \text{reach}} \neq \emptyset$ and $\mathcal{U}_{I \leftarrow G} \neq \emptyset$, and assume Assumption 5.1. Then*

$$\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R) \geq \frac{1}{2} \sup_{g \in \Phi_G(\Omega_{\Sigma, \text{reach}})} \text{diam}_{D_I}(g).$$

In particular, if some g satisfies $\text{diam}_{D_I}(g) > 0$, then the relation gap is strictly positive.

Proof. Fix $u \in \mathcal{U}_{I \leftarrow G}$ and $g \in \Phi_G(\Omega_{\Sigma, \text{reach}})$. For any $S, S' \in F_g$, Assumption 5.1 gives

$$\begin{aligned} D_I(\Phi_I(S), \Phi_I(S')) &\leq D_I(\Phi_I(S), u(g)) + D_I(u(g), \Phi_I(S')) \\ &\leq 2 \sup_{T \in F_g} D_I(\Phi_I(T), u(g)). \end{aligned}$$

Hence

$$\text{diam}_{D_I}(g) \leq 2 \sup_{T \in \Omega_{\Sigma, \text{reach}}} D_I(\Phi_I(T), u(\Phi_G(T))).$$

Take the infimum over $u \in \mathcal{U}_{I \leftarrow G}$ and then the supremum over g . □

Corollary 5.1 (Resolution witness for a non-negligible relation report). *Under the standing specialization and Assumption 5.1, suppose there exist $S, S' \in \Omega_{\Sigma, \text{reach}}$ such that*

$$\Phi_G(S) = \Phi_G(S'), \quad D_I(\Phi_I(S), \Phi_I(S')) \geq 2\varepsilon_R.$$

Then

$$\Delta_{I \rightarrow G}(V_I, V_G, \mathfrak{P}_R \mid V_R) \geq \varepsilon_R,$$

and therefore $\Delta_{I \rightarrow G} \not\approx_{\varepsilon_R} 0$ under the active relation-package threshold.

Remark 5.3 (Finite-precision licensing of a $\Delta_{I \rightarrow G}$ witness). If $\Delta_{I \rightarrow G}(\mathcal{L}; V) = +\infty$, then it is non-negligible at every finite threshold and no finite residual-axis estimation construction is required. Assume henceforth that

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) < \infty.$$

The finite-precision mechanisms below operate on the standing flattened specialization. They leave the structural clauses $(\Phi_I, \Phi_G, D_I, \mathcal{U}_{I \leftarrow G}, \Omega_{\Sigma, \text{reach}})$, and hence the fixed relation target, unchanged while certifying how that target or its witnesses are numerically reported. For every completed refinement package used below, declare the identity transport on the shared residual report axis. In this remark the symbol ε denotes the active relation threshold ε_R of the completed refinement package, and λ indexes only its realization mechanism. Under this transport,

$$\tau_{\lambda, \varepsilon}^\Delta = \text{id}_{\mathbb{R}_{\geq 0}}, \quad \Delta_{I \rightarrow G}(\mathcal{L}; V_{\lambda, \varepsilon}^{\text{pair, class}}) \equiv_{\tau_{\lambda, \varepsilon}^\Delta} \Delta_{I \rightarrow G}(\mathcal{L}; V).$$

The equality of target values is therefore licensed by the shared structural projection and the declared identity transport, not inferred merely from reuse of the symbol. For this fixed system-package instance, take the residual report domain

$$X_\Delta := \{0, \Delta_{I \rightarrow G}(\mathcal{L}; V)\} \subseteq \mathbb{R}_{\geq 0}, \quad D^*(r, s) := |r - s|,$$

with

$$Q_0^\Delta := \{(0, 0)\}, \quad Q_{\text{sep}}^\Delta := \{(\Delta_{I \rightarrow G}(\mathcal{L}; V), 0)\}.$$

Its pairwise separation margin is $\delta_Q^{\text{pair}} = \Delta_{I \rightarrow G}(\mathcal{L}; V)$. Write

$$L := \frac{1}{2} \sup_{g \in \Phi_G(\Omega_{\text{reach}})} \text{diam}_{D_I}(g),$$

so that Theorem 5.1 gives $L \leq \Delta_{I \rightarrow G}(\mathcal{L}; V)$. When L is not exactly available, let $0 \leq \underline{L} \leq L$ be any declared certified lower bound; taking $\underline{L} = L$ recovers the exact case. Thus $\underline{L}$ is a certified lower bound on the residual-axis separation margin, not the margin itself, and any set constructed from it is conservative.

The minimal licensing rule for a positive witness is one-sided: any certified quantity $\underline{\Delta}_\lambda$ satisfying

$$\underline{\Delta}_\lambda \leq \Delta_{I \rightarrow G}(\mathcal{L}; V), \quad \underline{\Delta}_\lambda \geq \varepsilon,$$

licenses $\Delta_{I \rightarrow G}(\mathcal{L}; V) \not\approx_\varepsilon 0$. Three common sources of such a lower bound should be kept distinct.

1. **Structural-bound route.** The theorem-generated certificate itself gives $\underline{\Delta}_\lambda := \underline{L}$; hence $\varepsilon \leq \underline{L}$ is sufficient, with no additional envelope for a direct estimate of $\Delta_{I \rightarrow G}$.
2. **Witness-pair route.** Suppose a finite-precision mechanism reports a fiber-pair distance $\hat{d}_\lambda$ for states $S, S' \in \Omega_{\text{reach}}$ with $\Phi_G(S) = \Phi_G(S')$, and supplies a one-sided *over-report allowance* $\eta_+(\lambda)$ such that

$$\hat{d}_\lambda - D_I(\Phi_I(S), \Phi_I(S')) < \eta_+(\lambda).$$

Then

$$\underline{\Delta}_\lambda^{\text{fib}} := \frac{1}{2}(\hat{d}_\lambda - \eta_+(\lambda)) \leq \Delta_{I \rightarrow G}(\mathcal{L}; V).$$

In particular, $\hat{d}_\lambda - \eta_+(\lambda) \geq 2\varepsilon$ licenses the non-negligibility report through Corollary 5.1.

3. **Direct-estimate route.** If a mechanism reports $\hat{\Delta}_\lambda$ with

$$\hat{D}_\lambda(\Delta_{I \rightarrow G}(\mathcal{L}; V), 0) = \hat{\Delta}_\lambda$$

and the over-report envelope of Definition 4.14 gives

$$\widehat{\Delta}_\lambda - \Delta_{I \rightarrow G}(\mathcal{L}; V) < \rho_+(\lambda),$$

then $\underline{\Delta}_\lambda^{\text{dir}} := \widehat{\Delta}_\lambda - \rho_+(\lambda)$ is a certified lower bound. Hence $\widehat{\Delta}_\lambda - \rho_+(\lambda) \geq \varepsilon$ is sufficient for the positive report.

These one-sided witness rules are weaker than the full two-sided classifier of Sec. 4.4. A scalar estimate $\widehat{\Delta}_\lambda$ alone supports only the one-sided positive-witness route. The full two-sided slice additionally requires a certified residual-axis discrepancy report $\widehat{D}_\lambda$ on both the target pair $(\Delta_{I \rightarrow G}, 0)$ and the declared null pair $Q_0^\Delta = \{(0, 0)\}$. If the same threshold must both merge the declared null residual and keep the fixed target residual $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ non-negligible, define the conservative algebraic full-classification slice

$$I_{\lambda, \underline{L}}^{\Delta, \text{class}} := (0, \infty) \cap [\rho_+(\lambda), \underline{L} - \rho_-(\lambda)].$$

The actually licensed slice is

$$\mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap I_{\lambda, \underline{L}}^{\Delta, \text{class}}.$$

The algebraic slice is nonempty exactly when

$$\rho_+(\lambda) \leq \underline{L} - \rho_-(\lambda) \quad \text{and} \quad \underline{L} - \rho_-(\lambda) > 0.$$

In the symmetric case its algebraic nonemptiness is equivalent to

$$2\rho(\lambda) \leq \underline{L} \quad \text{and} \quad \underline{L} > 0.$$

These endpoint conditions imply a licensed package slice only when the algebraic slice intersects the task-indexed package-admissible threshold set. Failure of the conservative algebraic slice does not establish failure of the true algebraic slice because $\underline{L}$ may be strictly below $\Delta_{I \rightarrow G}(\mathcal{L}; V)$; failure of the licensed intersection may additionally arise from package inadmissibility.

The two factors of two appearing in this discussion express two-sided separation geometry but arise through different arguments. In Corollary 5.1, the factor follows from the triangle inequality on a structural fiber. In the symmetric classification criterion, it is the sum of the over-report and under-report allowances. The constants are analogous but are not identified. This remark licenses only fiber-witness-supported or directly certified $\Delta_{I \rightarrow G}$ reports; it does not cover every $\Delta_{I \rightarrow G}$ or Ψ computation.

Remark 5.4 (Fiber witnesses as audit artifacts). A pair (S, S') with $\Phi_G(S) = \Phi_G(S')$ and large interface discrepancy constitutes a *fiber witness*. When a report asserts that $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ is not resolution-negligible at relation threshold ε_R , including such a witness (or a reproducible procedure for generating it under V) provides an explicit, package-locked explanation of why the “ $\approx_{\varepsilon_R} 0$ ” convention is unavailable.

Probabilistic reporting module. A distribution-valued implementation of the same diagnostic, including TV/IPM reporting semantics and event-probability bounds, is recorded in Appendix B. This module is not needed for the core definitions in this section and introduces no new primitive; it only translates package-locked residual statements into a declared distribution-level reporting vocabulary when the relevant interface clauses are supplied.

Remark 5.5 (Probabilistic reporting when single-representative sufficiency fails). The probabilistic reading used in Appendix B should not be treated as a new core primitive of the framework. It is an interpretive consequence of the same V -locked reporting discipline. When an endpoint or compressed report is sufficient under V , the original structure can be represented by a single admissible V -level report. When single-representative sufficiency fails, the original structure is

not thereby invalidated; what fails is the license to report it through one unique representative at the declared resolution.

In this situation, $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ marks the obstruction to such single-representative reporting. It does not by itself determine a probability law. A distribution-valued report is licensed only when the package additionally declares a candidate/fiber space, a fiber measure or prior, and the corresponding reporting interface. Under those clauses, if V is not enlarged by additional complement coordinates, a finer interface, or a different admissible reconstruction class, one available replacement is a distribution-valued report. Informally, the unresolved candidates inside the V -fiber are no longer collapsed into a single representative, but are reported as a conditional distribution or mixture relative to the original structure.

One may write this intuition schematically as follows. This notation is not a new diagnostic module; it only re-labels the currently invoked distribution-valued reporting situation, with e_V serving as the declared compressed report and $\rho_{\mathfrak{S}}$ serving as the original-structure report object whose single-representative recovery is at issue. Let $\mathcal{X}_V$ be the declared candidate/fiber space, let $e_V : \mathcal{X}_V \rightarrow \text{End}_V$ be the declared endpoint map, and let $\rho_{\mathfrak{S}} : \mathcal{X}_V \rightarrow \text{Rep}$ denote the report object induced by the original structure $\mathfrak{S}$, where Rep may itself be a space of probability laws. If $\mu_V(\cdot | e)$ is a declared conditional measure on the fiber $e_V^{-1}(e)$, then the corresponding probabilistic report is the pushforward

$$P_V^{\mathfrak{S}}(\cdot | e) := (\rho_{\mathfrak{S}})_{\#} \mu_V(\cdot | e) \in \mathcal{P}(\text{Rep}).$$

Bayesian conditioning may then be read as the familiar update of this fiber measure after a V -declared evidence event. This reading explains why Bayesian forms become natural once a distribution-valued reporting clause and a declared evidence event are supplied, but it does not make Bayesian probability part of the structural core.

5.4 Optional H/M regime labels induced by Ψ

The H/M vocabulary is an optional shorthand for a fixed-level Ψ -report. The naming is mnemonic: H refers to a homogeneous-cluster closure regime, where some admissible perspective removes the declared intra-level tension; M refers to a mixed heterogeneous package, where heterogeneous contributions remain mixed under every admissible perspective. The formal test is only the Ψ -criterion below.

Definition 5.5 (Optional H/M label at a fixed level). Fix a level specification $\Lambda = (\Pi_{\Lambda}, \mathcal{A}_{\Lambda})$ over a package-locked substrate and, for every $A \in \mathcal{A}_{\Lambda}$, a completed unary report $\Psi_A := \Psi(\mathfrak{S}_{\Sigma_A}, V_A)$.

- H -type at Λ : there exists $A \in \mathcal{A}_{\Lambda}$ such that $\Psi_A = 0$.
- M -type at Λ : $\Psi_A > 0$ for every $A \in \mathcal{A}_{\Lambda}$.

For a finite-resolution H/M label, a classification package V_{HM} must cite the completed unary reports, declare a common comparison axis or transports to one, and activate its own threshold ε_{HM} . The finite-resolution predicates are then $\Psi_A < \varepsilon_{HM}$ for an H witness and $\Psi_A \geq \varepsilon_{HM}$ for every A in the M report, subject to the declared transports and threshold alignments. If the level, admissible family, conflict functional, evaluation domains, comparison clauses, or active classification threshold is absent, the label is undefined. The label is level-locked and is not identified with $\Delta_{I \rightarrow G}$ or Δ_{acc} .

6 Structural Fracture Law and Diagnostic Translation

6.1 Structural fracture law (canonical statement)

Structural fracture law. When a logical system simultaneously contains opposing elements, and these elements produce feedback on the system, if one attempts to

connect, across levels, the system's whole (macroscopic determinacy) and the system's internal factors (microscopic behavioral logic) by a single continuous-logic principle within a single perspective, then a judgment discontinuity is structurally unavoidable. In short:

$$\begin{aligned} & \text{opposing elements} + \text{system self-feedback} \\ & + \text{cross-level continuous logic} \Rightarrow \text{judgment discontinuity.} \end{aligned}$$

6.2 Diagnostic translation into Ψ - $\Delta_{I \rightarrow G}$ language

The fracture law is used here as orientation, not as a derivation engine. The formulas in the cross-level paragraph below use the shared-substrate, total-readout, identity-pairing, minimax specialization of Definition 5.3; the general pairing form remains available through that definition.

Cross-level gluing. A continuity-biased gluing claim asks for an admissible lift $u \in \mathcal{U}$ such that

$$\sup_{S \in \Omega_{\text{reach}}} D_I(\Phi_I(S), u(\Phi_G(S))) < \varepsilon_R.$$

The corresponding diagnostic object is

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D_I(\Phi_I(S), u(\Phi_G(S))).$$

If this value is not resolution-negligible under the declared package, the single-valued cross-level report is not licensed at that resolution.

Intra-level oppositionality. At a fixed level, opposing contributions are read through the conflict functional used to define $\Psi(\mathfrak{S}_{\Sigma_A}, V_A)$ (or its declared flattened shorthand). If no admissible perspective at the level drives the readout below the declared threshold, the situation receives the optional M -type label of Definition 5.5.

Audit/fiber formulation. For audited decidability or extendability claims, the same pressure appears as fiber non-constancy: two admissible histories share the declared endpoint/readout but differ on a declared query. This gives a concrete witness that the queried judgment is not single-valued under the locked viewpoint. Any closure-restoring response requires a separate complement or extension package; it is not built into the present core.

6.3 Meta-gap Δ^* (cross-level non-identity postulate)

Definition 6.1 (Δ_R^* (meta-gap schema)). The meta-gap Δ_R^* is a semantic non-identity marker attached to a genuine cross-level relation R . It records that a genuine cross-level identification is not an ontological equality. It is not an operational inf-sup diagnostic and is not itself compared with ε ; only a separately declared package-level realization may enter such a comparison. The framework adopts the postulate schema

$$(\text{genuine cross-level relation } R) \implies \Delta_R^* > 0.$$

Here “ > 0 ” is schematic notation for semantic non-identity, not a numerical order comparison on an operational carrier. When R is unambiguous from context, we may write Δ^* for Δ_R^* .

Remark 6.1 (Visibility is resolution-dependent; ontology is not). The meta-gap Δ_R^* itself is neither assigned an operational magnitude nor compared with ε . If a package separately declares an operational realization $r_V(R) \in \mathbb{R}_{\geq 0}$ of the relation, that realized residual may satisfy $r_V(R) < \varepsilon$ under the active realization-package threshold and therefore be reported as resolution-negligible, while $\Delta_R^* > 0$ remains unchanged as a semantic non-identity postulate. Accordingly, *diagnostic* threshold comparisons are carried out using the package-relative gap $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ (Sec. 5.2), not Δ_R^* .

6.4 From static fracture to declared dynamics

The structural fracture law supplies the static orientation for the framework: opposing elements, system self-feedback, and a single cross-level continuous-logic principle mark the regime in which a cross-level judgment discontinuity is expected under the locked viewpoint. The first move of the present framework is to translate that discontinuity into a package-conditioned residual report: the obstruction is no longer left as an informal statement, but is read through declared interfaces, admissible reconstructions, discrepancy rules, and resolution clauses.

The second move is dynamical. When the same package-locked obstruction persists along a declared evolution, refinement, hierarchy, or scale parameter, the framework needs a minimal way to report persistence without importing a domain-specific dynamics. The purpose of the next section is exactly this: it supplies a type-preserving accumulation update in which a slice-local running readout $\Psi_{\text{run}}(\gamma)$ contributes compatible increments to an already typed accumulated report $\Delta_{\text{acc}}(\gamma)$ under declared transport across γ .

The dynamical interface does not add a new fracture law. It supplies the minimal reporting structure required when the same static obstruction is followed along a declared evolution parameter.

7 Minimal Dynamical Interface

Module summary.

Scope. Provide the minimal core interface by which a static package-locked obstruction is followed along a declared evolution parameter γ , without adding new core primitives or silently identifying path-accumulated and fixed-slice relational reports.

Dependencies. A declared evolution parameter γ ; a path $\mathfrak{P}_\gamma$ of completed unary-report records; slice-local formation of $\Psi_{\text{run}}(\gamma)$; transports to a common report axis; and an accumulated-report package V_{acc} fixing the initial accumulated-report type, compatible increments, and any type/pairing transport required along the path.

Outputs. A reproducible type-preserving accumulation update for $\Delta_{\text{acc}}(\gamma)$, together with package-local accumulated reports. Any comparison with a separately completed fixed-slice endpoint report remains a distinct bridge request.

Non-invocation rule. If the path, slice-local running-readout declaration, transports, accumulated-package clauses, or active threshold required by a claim are absent, the accumulated report is undefined under the reporting protocol. If an endpoint–path comparison bridge is absent, equality, order, or difference expressions between the two report types are not licensed.

7.1 Declared evolution path and slice-local running readout

This section is the point at which the static fracture diagnostic becomes a minimal dynamics of declared residual readout. The minimal interface has two logically distinct steps: local report formation on each declared γ -slice, followed by a type-preserving accumulation update. Neither step supplies a fixed-slice endpoint relation report by itself.

Source and status of the dynamical interface. The running quantity $\Psi_{\text{run}}(\gamma)$ is not introduced by differentiating a preselected accumulated curve. It is formed from the package-locked unary report available on the γ -slice and then transported to the common report axis. Only after that local report is independently typed does the accumulated-report package consume it as an increment. The relation

$$\frac{d}{d\gamma}\Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma) \quad \text{a.e.}$$

is therefore a consequence of the declared accumulation update under local integrability, not the definition by which the running readout is manufactured.

Definition 7.1 (Declared unary-report path and running readout). Let $\gamma \in [\gamma_0, \gamma_1]$ be a declared evolution parameter. A *declared unary-report path* is a γ -indexed family of completed dependency records

$$\mathfrak{P}_\gamma := (\mathfrak{S}_{\Sigma_A, \gamma}, V_{A, \gamma}, E_{A, \gamma}, \models_{V_{A, \gamma}})$$

for which $\Psi(\mathfrak{S}_{\Sigma_A, \gamma}, V_{A, \gamma})$ is well-posed at every γ . The path declaration includes transports of the completed unary report objects to a common nonnegative report axis, together with the parameter-measure convention. A common codomain without those transports is insufficient.

Define the transported running readout

$$\Psi_{\text{run}}(\gamma) := \tau_{\gamma \rightarrow \gamma_0}^\Psi(\Psi(\mathfrak{S}_{\Sigma_A, \gamma}, V_{A, \gamma})).$$

Its declaration is *slice-local*: its operands are the package-locked carrier/readout data of the γ -slice together with the declared transport needed to place that report on the common axis. The accumulated object Δ_{acc} is not an operand of this declaration. Consequently, a retrospective assignment $\Psi_{\text{run}} := d\Delta_{\text{acc}}/d\gamma$ does not, by itself, instantiate the forward dynamical interface.

This dependency clause is the dynamical specialization of Normative Rule 2.4: the dependency graph points from the package-locked γ -slice to $\Psi_{\text{run}}(\gamma)$ and from the running readout into the accumulated update; it contains no reverse operand edge from Δ_{acc} into the declaration of Ψ_{run} . This prevents the minimal interface from becoming vacuous by circularly selecting the local driver from the accumulated report it is supposed to update.

7.2 Type-preserving accumulation update

Definition 7.2 (Type-preserving accumulated structural quantity). Let $\mathfrak{P}_\gamma$ and Ψ_{run} be as in Definition 7.1. An accumulated-report package $V_{\text{acc}}[\mathfrak{P}_\gamma]$ declares the initial accumulated-report type and value, the transported accumulated carrier/report family along γ , the rule by which transported running increments act on that family, its discrepancy, and an active threshold ε_{acc} whenever a thresholded accumulated report is made. In the scalar specialization used in this manuscript, the compatible update is

$$\Delta_{\text{acc}}(\gamma) := \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(s) ds,$$

whenever the integral is well-defined.

The accumulated-report type at γ is inherited from the declared initial type through the transport clauses of V_{acc} ; it is not created by the integral. Where that inherited report type contains a pairing or correspondence clause, preservation or declared transport of that clause is part of the accumulated-package specification. Thus the primary invariant of the update is *type preservation*. Pairing preservation is a conditional obligation only when the inherited accumulated-report type itself contains a pairing or correspondence clause. In the scalar specialization used in this manuscript, the accumulated scalar carries no independent pairing clause, so that condition is inactive here. This terminology does not identify Δ_{acc} with the fixed-slice relational report $\Delta_{I \rightarrow G}$.

If $\Psi_{\text{run}} \in L^1_{\text{loc}}$, then Δ_{acc} is absolutely continuous and

$$\frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma) \quad \text{for almost every } \gamma.$$

Under the default accumulated-carrier discrepancy $d_{\text{acc}}(a, b) = |a - b|$, the package-local threshold predicate is

$$a \approx_{\varepsilon_{\text{acc}}} b \iff |a - b| < \varepsilon_{\text{acc}}.$$

Under reparameterization, Ψ_{run} must be transported as a density; otherwise cross-axis comparison is **undefined**.

The role of the explicit transport law is therefore stronger than parameter bookkeeping. It keeps the running increments and the inherited accumulated-report type inside one licensed update family. Carrier-side evolution alone does not provide this report transport; the requirement is an instance of Normative Rule 2.3.

7.3 Accumulated versus fixed-slice endpoint reports

The accumulated object $\Delta_{\text{acc}}(\gamma)$ and a separately completed fixed-slice endpoint report $\Delta_{I \rightarrow G}$ answer different typed questions. Sharing the symbol Δ does not make them commensurable. Before an endpoint–path comparison bridge is declared, schematic expressions such as

$$\Delta_{\text{acc}} \stackrel{?}{=} \Delta_{I \rightarrow G}, \quad \Delta_{\text{acc}} \stackrel{?}{\leq} \Delta_{I \rightarrow G}, \quad \Delta_{\text{acc}} - \Delta_{I \rightarrow G}$$

are not licensed comparisons: equality, order, and subtraction already presuppose that the two completed report structures have been transported into a common comparison type.

Remark 7.1 (Two independent duties of an endpoint–path bridge). A bridge between an accumulated path report and a fixed-slice endpoint report can have two logically independent duties.

- (i) **Comparability / commensurability.** The bridge declares or derives the common comparison structure on which the two completed reports may be jointly consumed, including any algebraic operation needed for a proposed difference report. Without this first duty, equality, inequality, difference, or limit expressions across the two report types are **undefined/non-licensed**.
- (ii) **Equality or order certification.** After comparability has been established, an additional verified theorem or certificate may license a particular equality, inequality, or extremal relation. If subtraction is already well-typed under the comparison structure, such a certificate may further justify interpreting the resulting difference as a nonnegative excess. Comparability alone does not supply any such equality or order statement.

This separation is a direct application of Normative Rule 2.4: declaring a comparison structure is not the same act as verifying the hypotheses of a theorem that orders the compared reports.

Remark 7.2 (RG geometry as a worked realization, not a framework law). The RG companion paper [18] provides a concrete realization of this two-stage logic in its geometric module. Once the declared construction supplies a common positive-definite information metric, the path length L_{II} and endpoint distance d_{end} become commensurable in the same induced geometry and their difference is algebraically well-typed; the curve-length/distance theorem then licenses $L_{II} \geq d_{\text{end}}$, justifying the nonnegative geodesic-excess report $\Gamma_{\text{geo}} := L_{II} - d_{\text{end}}$. Neither the inequality nor the subtraction is a generic consequence of the present framework, and neither L_{II} nor d_{end} is identified here with Δ_{acc} or $\Delta_{I \rightarrow G}$ by default. The example illustrates the bridge discipline rather than supplying a universal endpoint–path law.

Symbol	Role in this manuscript	Gating / notes
$\Delta_{I \rightarrow G}(\mathcal{L}; V)$	Flattened shared-substrate, identity-pairing, min-max specialization of the ordered relational reconstruction gap.	Requires a locked substrate, total operand readouts on the evaluated reachable domain, identity pairing, $\mathcal{U}_{I \leftarrow G}$, D_I , aggregation, and the relation threshold ε_R ; threshold semantics follow Normative Rule 2.1.
$\Delta_{\text{acc}}(\gamma)$	Path-updated accumulated report along a declared evolution parameter γ (integral / discrete accumulation interface).	Requires a declared path $\mathfrak{P}_\gamma$, slice-local Ψ_{run} , transport to a common running-report axis, and an accumulated-report package V_{acc} preserving the inherited report type and any carried pairing clause. Not comparable with $\Delta_{I \rightarrow G}$ by default.
Δ_R^*	Meta-gap schema (cross-level non-identity postulate) guarding against ontological equality readings across genuine cross-level relations.	Semantic guardrail, not an operational inf-sup diagnostic; never used as a numerical threshold criterion.

Table 2: Disambiguation of Δ objects used in this paper. Numerical reporting is always package-relative; a shared symbol does not imply default commensurability. Ontological statements use Δ_R^* (with R suppressed when unambiguous) only as a semantic axiom (Sec. 6.3).

Remark 7.3 (Domain of validity and reporting convention). The interface in Definitions 7.1–7.2 is operational and path-dependent: it requires a declared evolution parameter, a declared path $\mathfrak{P}_\gamma$ of completed report records, slice-local formation of Ψ_{run} , explicit transports to a common report axis, and an accumulated-report package V_{acc} .

Discrete variant. If γ indexes a discrete sequence (e.g. $\gamma = n \in \mathbb{N}$), the forward update may be written

$$\Delta_{\text{acc}}(n+1) = \Delta_{\text{acc}}(n) + \Psi_{\text{run}}(n),$$

where $\Psi_{\text{run}}(n)$ is independently formed from the n -slice under Definition 7.1. Equivalently,

$$\Delta_{\text{acc}}(n) = \Delta_{\text{acc}}(n_0) + \sum_{k=n_0}^{n-1} \Psi_{\text{run}}(k),$$

under the same transported-report and accumulated-package requirements.

Resolution. Any statement of the form “ $\Delta_{\text{acc}}(\gamma) \approx_{\varepsilon_{\text{acc}}} 0$ ” (if used) is evaluated under the declared accumulated-carrier discrepancy; under the default rule above it means $|\Delta_{\text{acc}}(\gamma)| < \varepsilon_{\text{acc}}$ and does not assert an ontological identity. Geometric closures and self-similar solutions are deferred to appendices (or later chapters) to keep the main line slim.

Remark 7.4 (Dynamical analogies deferred). Analogies with familiar dynamical lexicons such as force, motion, energy, action, driver, or accumulated response are not part of the slim core. They may be used only as optional dynamical analogies under an explicit package and are deferred to the extended dynamics module when needed. The main text uses only the minimal type-preserving accumulation interface of Definitions 7.1–7.2; pairing preservation is additionally required only for accumulated-report types that actually carry a declared pairing or correspondence clause.

8 Well-posed Reporting and Licensed Comparisons

This section states the minimal declaration conditions needed for reproducible reporting, and specifies when cross-report comparisons are licensed. In this framework, Ψ and $\Delta_{I \rightarrow G}$ (and any other Δ -variant when explicitly invoked) are not treated as context-free invariants: their meanings are uniquely locked only after the relevant clauses of a declared package have been stated. When the required clauses are not declared (or when cross-report comparisons invoke undeclared transport/domain/resolution conversions), the corresponding numerical statement is *undefined relative to the declared regime*. For compactness, we optionally attach short diagnostic status tags (Appendix A), but the core semantics is simply partiality (well-posed vs. undefined) and licensed vs. unlicensed comparisons.

8.1 Package card template (fillable)

The minimal reporting card (Protocol 4.1) specifies what must be *declared*. For long-term reuse, it is often convenient to record these clauses in a single “package card” block with a short identifier. The following template is a fillable form; it introduces no new primitives.

Table 3: Dependency-aware package card template. Fill only the blocks invoked by the report.

Block / field	Declaration
Report ID, target quantity, and any flat-tened shorthand	_____
Operand substrate node(s) $V_{\Sigma_I}, V_{\Sigma_G}$: systems, boundaries, carriers, evolution, reachability; state any coincidence	_____
Unary perspective V_A : domain, Φ_A , E_A (only if Ψ is invoked)	_____
Operand perspectives V_I, V_G : readout domains and maps	_____
Complete pairing origin: V_{Σ_R} or canonical derivation	_____
Complete pairing $\mathfrak{P}_R = (Z_R, Z_{R,\text{reach}}, p_I, p_G)$; owner/derivation and identity status	_____
Empty-domain diagnosis stage: pairing structure, task domain, operand reachability, or readout domain	_____
Relation package V_R : $\mathcal{U}_{I \leftarrow G}$, D_I , normalized/isotone Agg_R , any zero-detection class, domain and consistency clauses	_____
Package-local thresholds actually invoked	_____

Block / field	Declaration
Cross-node threshold alignment, only if triggered	_____
Resolution refinement and certification, if invoked	_____
Queries, histories/trajectories, policy, and certificate scope	_____
Transport for comparison between completed reports, if invoked	_____
Optional module-specific operands and constructions	_____

If a probabilistic/Bayesian clause is invoked, any additional candidate or fiber carrier must satisfy the no-bare-carrier rule; the card must record its locking or canonical derivation, measure or prior, evidence event, likelihood/readout kernel, and normalization domain.

8.2 What must be declared to make Ψ and $\Delta_{I \rightarrow G}$ well-defined

Protocol 8.1 (Minimal declaration requirements). A report is well-defined only if every carrier and mechanism entering its formula is package-locked or explicitly derived from package-locked operands.

1. **For Ψ :** declare V_Σ , the relevant locked reachable domain, a perspective package V_A with readout domain and map Φ_A , the conflict functional E_A , and the threshold of the package node if a thresholded unary report is made.
2. **For $\Delta_{I \rightarrow G}$:** declare $V_{\Sigma_I}, V_{\Sigma_G}$, state whether they coincide, and declare V_I, V_G with their readout domains and maps.
3. **Pairing:** declare the owner or canonical derivation of the complete tuple $\mathfrak{P}_R$, including $Z_{R,\text{reach}}$, the partial projection domains, and p_I, p_G . Identity pairing must be stated. Resolve the domain successively as $Z_R^{\text{struct}}, Z_R^{\text{pair}}, Z_R^{\text{operand}}$, and Z_R^{eval} , so that structural projection availability, task selection, both operand-reachability gates, and both readout-domain gates are separately auditable.
4. **Directional relation mechanism:** declare $V_R, \mathcal{U}_{I \leftarrow G}, D_I$, a normalized and isotone Agg_R , any stronger zero-detection class that is invoked, the task-relative relation domain, and any consistency or non-degeneracy constraints.
5. **Threshold semantics:** declare ε_R for a thresholded relation report. Other node-local thresholds are declared only when their thresholded outputs are invoked. Add cross-node alignment only under the triggers of Remark 4.12.
6. **Optional certification and comparison:** declare every invoked resolution-certification clause. Comparison between already completed reports additionally requires transport, domain reconciliation, and any required threshold alignment.

Operability. The actual evaluation domain must respect the reachable domains of both operand substrates, and it and every admissible family used by an infimum, supremum, or other aggregation must be nonempty. When an evaluated relation domain is empty, record the first empty stage: structural pairing domain, task-licensed pairing domain, operand-reachable domain, or final readout-evaluation domain. Empty or undeclared operands yield **undefined**, not a surrogate value induced by empty-set conventions. If any required declaration is absent, the symbol denotes an under-specified family rather than one well-posed report.

Remark 8.1 (Operability versus theorem-relative verification). Operability asks whether a declared report or bridge can be formed as a well-typed object on a nonempty licensed domain. Theorem-relative verification asks a different question: whether additional hypotheses required for a particular derived statement hold on the data consumed by that theorem. Failure of the latter blocks that theorem-derived conclusion but does not, by itself, make the underlying completed report undefined. This is the protocol-level instance of Normative Rule 2.4; no additional theorem-failure status tag is introduced.

8.3 Resolution robustness and locked adaptation

When a fixed-target resolution refinement is invoked, the active completed report-package node and its transported report axis must be identified. The certified sets of Sec. 4.4 determine which ordinary completed task packages, and which further policy-refined packages when a policy floor is invoked, are licensed under the declared report semantics. Reporting across more than one configuration requires an additional comparability and certification layer.

Definition 8.1 (Resolution-control robustness). Fix a report semantics and a corresponding certified configuration set $\mathcal{A}$. Suppose the family declares either:

1. a common report space $\mathcal{Y}$ whose identity maps are explicitly adopted as the transports across the invoked configurations; or
2. separately declared transports into a common report space $\mathcal{Y}$.

Let

$$R : \mathcal{A} \rightarrow \mathcal{Y}$$

be the transported diagnostic output. The report is:

- *configuration-robust* if R is constant on all of $\mathcal{A}$;
- *component-robust* if R is constant on a declared connected or ordered component of $\mathcal{A}$;
- *configuration-sensitive* otherwise.

A numerical report may replace literal equality by a separately declared equivalence in $\mathcal{Y}$, but the reporting threshold ε may not silently be reused for a second semantic role. A common codomain alone is not a transport declaration.

Protocol 8.2 (Certification scope for scans and adaptive paths). A certificate must match the way the completed resolution packages are used.

1. A report at one predeclared configuration (λ, ε) may use a pointwise certificate.
2. A finite scan over several predeclared configurations requires a simultaneous or multiplicity-corrected certificate over the scanned set.
3. A continuous scan requires a uniform certificate over its declared configuration domain.
4. An adaptive path whose next configuration or stopping time depends on previous observations requires a sequentially valid or anytime-valid certificate under the declared control and stopping rules.

Pointwise confidence guarantees may not be reinterpreted as a joint guarantee for a scan or adaptive path.

Definition 8.2 (Adaptive resolution meta-protocol). An adaptive resolution procedure is governed by the meta-protocol

$$\bar{V} := (\mathfrak{R}_{\text{res}}^{\text{cert}}, \mathcal{Q}^{\sigma, \tau}, [p], \hat{e}, \text{Cert}, \mathsf{T}, \text{Ctrl}, \text{Accept}, \text{Reject}, \text{Stop}),$$

where

$$\mathfrak{R}_{\text{res}}^{\text{cert}} := \{\mathcal{R}_{\lambda}^{\text{cert}}\}_{\lambda \in \Lambda_{\text{res}}}$$

is the declared certified interface-refinement family, $\mathcal{Q}^{\sigma, \tau}$ contains the task-indexed query clauses and report semantics, $[p]$ is the optional separate policy refinement, $\hat{e}$ is the declared error estimator, Cert is valid for the adaptive use, T declares the transports between visited package reports (including any explicitly adopted identity transports), Ctrl is the update law, and the remaining entries are the acceptance, rejection, and stopping rules. Writing the full configuration as

$$c_n := (\lambda_n, \varepsilon_n),$$

the controller may evolve by

$$c_{n+1} = \text{Ctrl}(c_n, \hat{e}_n),$$

with every visited c_n required to lie in the applicable package-conditioned certified set and, when policy is invoked, in its policy-restricted subset. For partition semantics, the sequential certificate must also cover every visited threshold-edge domain.

The object $\bar{V}$ is a higher-level protocol governing a declared path through ordinary packages

$$V_{\lambda_0, \varepsilon_0}^{\sigma, \tau}, V_{\lambda_1, \varepsilon_1}^{\sigma, \tau}, \dots$$

or through their policy-refined counterparts. It does not replace or redefine the top-level package schema of Definition 4.10. Fixed-threshold adaptation is the special case $\varepsilon_n \equiv \varepsilon_0$.

Protocol 8.3 (Lock the package path, not necessarily a constant value). Adaptive variation defines a declared path through completed packages. It is therefore package variation, but it is not *undeclared* viewpoint drift when the certified refinement family, task clauses, optional policy clause, report semantics, margin certificates, sequential mechanism certificate, transports, update law, acceptance/rejection rules, and stopping rule are locked before the substantive diagnostic. What is locked is $\bar{V}$ and the admissible package path, not necessarily constant numerical values of λ or ε . A post-hoc change of any of these clauses in response to the observed conclusion defines a new meta-protocol and must be reported as such.

Remark 8.2 (Sensitive reports). If R is configuration-sensitive, a report should provide a certified scan, certified transition boundaries between stable regions, or a declared selection objective. Certified feasibility alone does not select a unique completed package.

8.4 Partiality and licensed comparisons (when numerical reporting is undefined)

The framework distinguishes two outcomes: (i) a numerical report (when the quantity is uniquely locked and well-defined under a declared package and domain), and (ii) an explicit *undefinedness* or *non-licensing* statement (when the attempted report or comparison is not meaningful under the declared clauses). The latter is a semantic classification, not a value judgment.

Definition 8.3 (Undefinedness / non-licensing modes (minimal set)). The following modes cover the minimal reasons used in this paper for treating a numerical report or comparison as undefined under the declared regime:

1. **Under-specification:** one or more required clauses of Protocol 8.1, or one or more operand/construction clauses required by the declaration schema of a requested module, are missing, so the quantity is not uniquely locked.

2. **Admissibility/operability violation:** the attempted computation invokes an element (perspective, discrepancy, lifting, encoding, or interface sub-clause) outside the declared admissible family, or the declared structural-pairing, task-licensed pairing, or reconstruction class is empty after its own admissibility gates; the report is undefined relative to the declared regime.
3. **Domain mismatch or incompatibility:** the audited domain (reachable set, history class, or query set) differs across reports without an explicit reconciliation rule, or a declared relation has pairing instances but no jointly operand-reachable or jointly readable instance; comparisons or aggregations across domains are undefined.
4. **Non-comparability (missing transport):** two completed report objects are produced under distinct reporting structures without a declared transport/comparability law in the sense of Def. 4.19; equality/inequality comparisons are not licensed.
5. **Resolution mismatch:** reporting thresholds or audit resolutions differ without a declared conversion rule; cross-report comparisons are not licensed under the declared resolution-aware semantics.

Status	Meaning under the declared regime	Minimal witness (optional)
ok	Quantity is well-posed and the numerical report is licensed (package clauses locked, admissible instantiation, domain matches; and transport/resolution clauses declared if a cross-report comparison is performed).	Package card (Protocol 4.1) + reproduction notes.
U-spec	Under-specification: at least one clause needed to instantiate and lock the requested module report is missing.	List missing package items or module-specific operand/construction clauses.
ADM	Admissibility/operability violation: an invoked component is inadmissible, or the structural-pairing, task-licensed pairing, or reconstruction class is empty after its own gates.	Identify the offending component or empty admissible stage and the declared family.
DOM	Domain mismatch/incompatibility: domains lack reconciliation, or pairing instances have no jointly operand-reachable or jointly readable instance.	Specify the first empty domain stage or the missing reconciliation clause.
NC	Non-comparability: two already-defined reports are compared without a declared transport law.	Specify the two completed reporting structures and the missing transport/comparability clause.
RES	Resolution mismatch: cross-report comparison is requested with different ε (or audit resolutions) and no declared conversion rule.	Specify the resolutions and the missing conversion rule.

Table 4: Status-style reporting (optional). These labels are short names for the undefinedness / non-licensing modes (Def. 8.3) and inherit the domain-gating protocol (Protocol 8.4).

Remark 8.3 (Node scope of status and obstruction reports). Status and obstruction outputs belong to the report role: each is a typed report about a specified construction or comparison node and the operands consumed there. Its scope is therefore inherited from that report type rather than imposed as an additional global convention. A status assigned to one node does not retroactively invalidate already completed upstream or sibling reports. It may, however, block any downstream construction whose declared operands require the failed or non-licensed node. Thus node-local status does not imply downstream irrelevance.

Protocol 8.4 (Licensing rule for reporting and cross-report comparison). When a numerical report (or a cross-report comparison) is requested, the following checks determine whether the request is licensed under the declared regime:

1. **Locking check:** verify that the package clauses required by Protocol 8.1 have been declared and, for each requested module, that the operand and construction clauses required by its declaration schema have been supplied. If not, the request is **undefined** by under-specification (optional tag: U-spec, Appendix A).
2. **Admissibility check:** verify that instantiated components lie in the declared admissible family. If not, the request is **undefined** by admissibility violation (optional tag: ADM, Appendix A).
3. **Domain check:** verify that the audited domain matches the declared one, or that an explicit reconciliation rule is given. If not, the request is **undefined** by domain mismatch (optional tag: DOM, Appendix A).
4. **Operability check:** verify that the declared audited domain and every admissible family invoked by the requested definition are nonempty. For $\Delta_{I \rightarrow G}$, inspect the staged domains in order: Z_R^{struct} , then Z_R^{pair} , then Z_R^{operand} , then Z_R^{eval} , and finally verify $\mathcal{U}_{I \leftarrow G} \neq \emptyset$. Report the first failure using the optional subtype ADM/pairing-structure-empty, ADM/task-domain-empty, DOM/reachability-incompatible, DOM/readout-domain-empty, or ADM/reconstruction-class-empty. The request remains undefined; the subtype only localizes its cause.
5. **Resolution-feasibility check (when a certified mechanism family is invoked):** verify that the realized completed package $V_{\lambda, \varepsilon}$ belongs to the invoked certificate-relative set and that the certificate scope matches the use (fixed, scanned, or adaptive). If not, the object-level threshold report is not licensed; report current-certificate infeasibility, policy-induced infeasibility, chain-separation obstruction, or certificate-scope failure as applicable (Appendix E).
6. **Comparability check (cross-report only):** when comparing across packages, verify that a transport law is declared (Def. 4.19). If not, the comparison is not licensed (optional tag: NC, Appendix A).
7. **Resolution check (cross-report only):** verify that thresholds are identical or related by a declared conversion rule. If not, the comparison is not licensed (optional tag: RES, Appendix A).
8. **Otherwise:** report the numerical value together with the minimal package card (Protocol 4.1) and any additional clauses needed for reproducibility.

Remark 8.4 (Reporting protocol versus theorem-specific inference). Protocol 8.4 licenses formation of the requested report or cross-report comparison. It does not silently verify hypotheses belonging to a separate theorem that may later consume that report. Any such inference follows Normative Rule 2.4: first lock the theorem’s typed inputs and certificates, then verify its hypotheses on the consumed data, and only then derive the theorem-specific conclusion.

Remark 8.5 (Neutrality of undefinedness diagnoses). An undefinedness/non-licensing statement indicates that a numerical report or comparison is not meaningful under the *declared* package clauses, domain, and resolution semantics. It does not evaluate the system under study; it specifies which declaration or reconciliation clause would be required to make the requested numerical statement well-posed.

8.5 Canonical audited example: Markov coarse-graining and fiber non-constancy

This subsection records a short, non-toy instantiation showing how the protocol-level outputs (ok vs. undefined) arise from an explicit witness.

Setup (finite Markov chain under a locked endpoint). Let X be a finite state space with Markov kernel $K(x, \cdot)$ [9]. Fix a declared coarse map $\phi : X \rightarrow Y$ and take the endpoint operator for one-step histories $h = (x_0, x_1)$ to be $\text{End}(h) := \phi(x_0) \in Y$. Lock a discrete endpoint discrepancy and declare any $\varepsilon < 1$, so endpoint indistinguishability reduces to equality of macro labels.

Audited predictive queries. For each event $A \subseteq Y$, consider the one-step macro predictive query

$$q_A(h) := \mathbb{P}(\phi(X_1) \in A \mid X_0 = x_0). \quad (1)$$

This is a natural audited family when the intended statement treats macro-level predictions as the reportable content.

Witness for non-single-valuedness. If there exist micro states $x, x' \in X$ with $\phi(x) = \phi(x')$ such that the induced macro predictive laws differ, i.e. $\bar{K}_x(\cdot) \neq \bar{K}_{x'}(\cdot)$, then there exists an event A with $q_A(h_1) \neq q_A(h_2)$ for histories h_1, h_2 sharing the same endpoint label. Hence any endpoint-based report that requires these audited predictive queries to be single-valued on endpoint fibers is not licensed under the declared package.

Protocol consequence. The witness shows the mechanism: the coarse endpoint identifies microstates inside one macro fiber, while the audited predictive content varies within that fiber. Therefore any claim requiring single-valued endpoint-based prediction is not licensed under the locked package. With a declared discrepancy this becomes a non-negligibility witness; without the required clauses it is reported as **undefined** rather than replaced by an endpoint-matching surrogate.

9 Discussion

The philosophical point of the coupling specification V is not that structural gaps are failures of description, nor that reality contains a primitive viewpoint-independent discontinuity. A successful formalism may fit its target structure with high precision inside a declared regime; quantum mechanics is a paradigmatic example of such a deep alignment between formal description and objective behavior. The boundary appears when a valid descriptive structure is extended across levels, scales, or reporting interfaces while being asked to preserve structural content without loss. In this sense, the gap is neither merely linguistic nor a standalone ontological discontinuity. It is a report, conditioned by V , of an objective cross-level structural relation: the package does not create the underlying structure, but it determines the form in which cross-level residuals, admissible equivalences, and non-licensing states become reportable.

Under this coupling, Ψ and $\Delta_{I \rightarrow G}$ separate two different questions. Ψ asks whether residual tension remains inside a declared level and perspective. $\Delta_{I \rightarrow G}$ asks whether lower-level structure

can be compressed into an upper-level report and reconstructed within the declared admissible class. Their separation prevents within-level tension, cross-level residuality, approximation, and undefinedness from collapsing into a single informal notion of error.

The remaining protocol discipline follows from this distinction and from Normative Rule 2.4. When a certified fixed-target resolution refinement is invoked, declarations first lock the mechanism family, certificate objects, task clauses, and candidate threshold semantics; verification then checks the applicable certificate-relative feasibility conditions; only then is the corresponding thresholded report derived as licensed. When the required clauses or certificates are absent, the corresponding report is undefined or the requested derivation is not licensed. Macro-closure, residual complements, and closure restoration remain downstream mechanisms because they require additional future-readout and enlarged-fiber declarations.

A companion composition module makes the carrier/report separation concrete [16]. Relation carriers may admit a licensed composite assembly while their scalar gap reports still possess no inherited propagation law. Under separately verified assembly, reconstruction, discrepancy, and aggregation certificates, one obtains the explicit bound

$$\Delta_{I \rightarrow G} \leq C_{IM} \Delta_{I \rightarrow M} + LC_{MG} \Delta_{M \rightarrow G},$$

where C_{IM} and C_{MG} are the relevant aggregation pullback costs and L is the certified middle-to-outer reconstruction distortion. The associated outer threshold condition

$$\varepsilon_{IG} \geq C_{IM} \varepsilon_{IM} + LC_{MG} \varepsilon_{MG}$$

is a sufficient theorem-induced uniform budget, not a constitutive threshold declaration and not a necessary relation for every realized chain. This is an explicit realization of Normative Rules 2.3 and 2.4: scalar propagation is licensed and certified rather than inherited. In the same module, supremum aggregation is especially declaration-light because arbitrary pullback does not amplify the worst-case aggregate, whereas more general aggregation rules can require additional transport control.

10 Conclusion

The framework's main output is not a universal scalar formula for every obstruction. It is a disciplined report space: under a declared package V and finite reporting resolution ε , cross-level sufficiency, residuality, approximation, comparability, and undefinedness become typed and reproducible outputs. Within that report space, V functions as the coupling specification between descriptive access and objective structure; Ψ records intra-level residual tension; $\Delta_{I \rightarrow G}$ records cross-level residuality under declared compression and reconstruction; and Δ_{acc} provides a type-preserving accumulated readout when a declared evolution axis and accumulated-report transport are supplied. It is not comparable with a fixed-slice endpoint report by default; endpoint-path comparability and any subsequent equality/order statement are separately licensed. The resolution-feasibility module adds one further distinction: a threshold can be declared without being feasible, and when a certified fixed-target resolution refinement is invoked its licensed values are constrained jointly by the mechanism certificate and the declared query.

The framework therefore treats V -structure coupling as the semantic basis of cross-level reportability. It distinguishes what is structurally carried, what is reported, and what requires a separately licensed bridge before one report may be propagated into another; declaration, verification, and derivation remain distinct steps in that process.

A Diagnostic status tags and optional witnesses

Module summary.

Scope. Provide optional short tags and optional minimal witnesses for the undefinedness / non-licensing modes used in the main text.

Dependencies. None beyond the core reporting protocol (Sec. 8) and Def. 8.3.

Outputs. A lightweight status vocabulary (ok, U-spec, ADM, DOM, NC, RES) and suggested minimal witness fields.

Non-invocation rule. If omitted, the manuscript may report undefinedness in full prose; the core semantics is unchanged.

This appendix provides an *optional* compact labeling scheme for the undefinedness/non-licensing modes used in the main text (Def. 8.3). These tags are not part of the core semantics; they serve as short names for common regimes of under-specification or unlicensed comparison.

Status-based presentation (optional). A report may be presented in the lightweight form

$$\text{Report} := (\text{status}, \text{payload}, \text{witness}),$$

where $\text{status} \in \{\text{ok}, \text{U-spec}, \text{ADM}, \text{DOM}, \text{NC}, \text{RES}\}$, payload is the numerical value when **ok** (and omitted otherwise), and witness is an optional minimal record that makes the relevant missing clause or mismatch explicit.

Definition A.1 (Diagnostic status tags). The diagnostic tags used in this paper are:

1. **U-spec (under-specification):** one or more clauses required to lock the quantity, including any operand/construction clauses required by the declaration schema of the requested module, are missing.
2. **ADM (admissibility/operability violation):** the attempted computation invokes an element outside the declared admissible family, or an admissible structural-pairing, task-licensed pairing, or reconstruction class is empty after its own gates. Optional subtypes used here are ADM/pairing-structure-empty, ADM/task-domain-empty, and ADM/reconstruction-class-empty.
3. **DOM (domain mismatch/incompatibility):** the audited domain differs across reports without an explicit reconciliation rule, or a relation has pairing instances but no jointly operand-reachable or jointly readable instance. Optional subtypes used here are DOM/reachability-incompatible and DOM/readout-domain-empty.
4. **NC (non-comparability):** two completed reports are produced under distinct reporting structures without a declared transport/comparability law (Def. 4.19); equality/inequality comparisons are not licensed.
5. **RES (resolution mismatch):** thresholds/resolutions differ without a declared conversion rule; cross-report comparisons are not licensed under the declared resolution-aware semantics.

Remark A.1 (ADM/DOM classification principle). For the staged relation-domain failures above, ADM records emptiness or inadmissibility generated within a package node’s own declared clauses, whereas DOM records failure of otherwise admissible package-locked domains to align across dependency edges. Accordingly, structural-pairing emptiness, task-induced emptiness, and an empty reconstruction class are ADM subtypes; operand-reachability incompatibility and joint readout-domain failure are DOM subtypes. This principle guides assignment of optional future subtypes and does not replace the full definitions of ADM and DOM.

Remark A.2 (Scope of the short status vocabulary). The short tags above cover the core undefinedness and cross-package mismatch modes. They are not intended to exhaust the certificate-relative non-licensing states introduced by Appendix E, such as current-certificate infeasibility, policy-induced infeasibility, insufficient certificate scope, or chain-separation obstruction. Those states are reported in full typed prose unless a later application explicitly declares an extended status vocabulary.

Remark A.3 (Status tags are node-scoped). The short status tags inherit the scope of the typed report to which they are attached. They are node-scoped, not globally contagious: a tag on one construction or comparison does not retroactively change already completed upstream or sibling reports, although it blocks any downstream request whose declared operands require that non-licensed node. This is the compact-tag form of Remark 8.3.

Remark A.4 (Minimal witnesses (optional)). When a diagnostic tag is used, the following witness data are sufficient (but not required) to make the mode explicit:

- **U-spec**: list the missing declaration items relative to Protocol 8.1, the declaration schema of the requested module, or the minimal package card (Protocol 4.1); identify missing operands/construction clauses explicitly when applicable.
- **ADM**: specify the offending component and declared admissible family, or record the first empty admissible stage: $Z_R^{\text{struct}} = \emptyset$, $Z_R^{\text{struct}} \neq \emptyset$ with $Z_R^{\text{pair}} = \emptyset$, or $\mathcal{U}_{I \leftarrow G} = \emptyset$. For ADM/task-domain-empty, identify which $\mathcal{I}_R$, $\mathcal{Q}_R$, or $\mathcal{C}_R$ restriction induced the empty task intersection when available.
- **DOM**: specify the two audited domains and missing reconciliation clause, or record the staged witness $Z_R^{\text{pair}} \neq \emptyset$ with $Z_R^{\text{operand}} = \emptyset$, or $Z_R^{\text{operand}} \neq \emptyset$ with $Z_R^{\text{eval}} = \emptyset$.
- **NC**: specify the two completed reporting structures, the two already-defined reports, and the missing transport/comparability declaration.
- **RES**: specify the compared thresholds/resolutions and the missing conversion rule.

Worked example (strict endpoint-sufficiency residual and independent geodesic-excess diagnostic in RG flows). Explicit instantiations of the strict and geometric diagnostics under their respective module-specific packages, including typed status reports and operational witnesses, are given in [18].

B Extended probabilistic instantiation of $\Delta_{I \rightarrow G}$

Module summary.

Scope. Clarify how the package-locked residual $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ functions as a certificate-level structural limit and how distribution-valued interface clauses translate it into TV/IPM reporting readouts without replacing the slim-core witness mechanisms.

Dependencies. The shared-substrate, total-readout, identity-pairing, minimax specialization of Definition 5.3, with a completed relation package fixing $\mathcal{U}_{I \leftarrow G}$, D_I , $\text{Agg}_R = \sup$, and active threshold ε_R . For the probabilistic module, additionally a declared observation space $(\mathcal{Y}, \Sigma)$ and distribution-valued interface clauses; for the Bayesian reading, a declared candidate/fiber space $\mathcal{X}_V$, fiber measure or prior, evidence event, likelihood/readout kernel, and normalization domain.

Outputs. (i) Certificate lower bounds via fiber witnesses / fiber diameter and via audit single-valuedness; (ii) probabilistic reporting readouts: uniform event-gap control under TV and uniform audit predicate / advantage-error control under an IPM; (iii) an optional Bayesian intuition for conditionalized distribution-valued reports.

Non-invocation rule. If the interface is not distribution-valued (or the required clauses are absent), the probabilistic module is not invoked; the slim core remains unaffected. If the Bayesian clauses are absent, the Bayesian reading is simply not licensed.

B.1 Minimax residual as a certificate-level structural limit

This appendix begins in the flattened shared-substrate specialization. The slim core then treats $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ as the shorthand for the completed relation residual:

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D_I(\Phi_I(S), u(\Phi_G(S))),$$

where $\Omega_{\text{reach}} = \Omega_{\Sigma, \text{reach}}$, $\mathcal{U} = \mathcal{U}_{I \leftarrow G}$, and D_I are supplied by the completed flattened dependency record. The primary function of $\Delta_{I \rightarrow G}$ in this module is certificate-level: explicit witnesses can force resolution-non-negligibility under the authoritative reporting semantics (Normative Rule 2.1).

A canonical certificate is provided by the fiber-diameter mechanism. Under the pseudometric assumption on D_I , Theorem 5.1 gives the witness lower bound

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) \geq \frac{1}{2} \sup_{g \in \Phi_G(\Omega_{\text{reach}})} \text{diam}_{D_I}(g),$$

and Corollary 5.1 upgrades this to a resolution-aware non-negligibility certificate: if there exist $S, S' \in \Omega_{\text{reach}}$ with $\Phi_G(S) = \Phi_G(S')$ and $D_I(\Phi_I(S), \Phi_I(S')) \geq 2\varepsilon_R$, then $\Delta_{I \rightarrow G}(\mathcal{L}; V) \geq \varepsilon_R$ and hence $\Delta_{I \rightarrow G}(\mathcal{L}; V) \not\approx_{\varepsilon_R} 0$ under the declared reporting convention. In this sense, a fiber witness (Remark 5.4) functions as an audit artifact: it explains, in package-locked terms, why the “ $\approx_{\varepsilon_R} 0$ ” convention is unavailable at the stated resolution.

B.2 Alignment with audit semantics: single-valued decidability and fiber non-constancy

When a claim is explicitly about decidability/extendability under a locked audit package $V^{\text{aud}} = (\mathcal{I}^{\text{aud}}, \varepsilon, \mathcal{Q}, \mathcal{C})$, whose interface $\mathcal{I}^{\text{aud}}$ locks the endpoint operator **End** and the corresponding endpoint discrepancy, the relevant local predicate is single-valuedness on endpoint fibers: for every declared query $q \in \mathcal{Q}$, admissible histories that share the same endpoint/readout at resolution ε must satisfy $q(h_1) = q(h_2)$. If some q varies within an endpoint fiber at the locked audit resolution, the audited claim is not licensed as a single-valued endpoint report. Any restoration then requires a separately declared extension/complement workflow rather than an implicit repair.

This audit reading is consistent with the slim-core witness logic above: a non-closure witness supplies an explicit obstruction to single-valued audit decidability under the declared viewpoint, and the correct output is either an extension workflow or a non-licensing report. Throughout, “reportable as 0” is used only as a resolution-aware reporting convention at finite ε , not as an ontological identity.

B.3 Probabilistic instantiation as reporting interface (TV/IPM)

Setup (shared-substrate flattened specialization). This subsection instantiates the total-readout, identity-pairing, minimax specialization of Definition 5.3. The symbols $\Omega_{\text{reach}}, \mathcal{U}, D$ abbreviate $\Omega_{\Sigma, \text{reach}}, \mathcal{U}_{I \leftarrow G}, D_I$, and any thresholded conclusion uses the active relation threshold ε_R . Fix an observation space $(\mathcal{Y}, \Sigma)$. Under the declared interface, each audited state $S \in \Omega_{\text{reach}}$ induces an outcome distribution $P_S := \Phi_I(S) \in \mathcal{P}(\mathcal{Y})$. Given the coarse description $\Phi_G(S)$ and an admissible lifting $u \in \mathcal{U}$, the reconstructed outcome distribution is $Q_S^u := u(\Phi_G(S)) \in \mathcal{P}(\mathcal{Y})$. The diagnostic retains its minimax form:

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} D_I(P_S, Q_S^u),$$

with D_I declared as the relation-package discrepancy on distributions.

Remark B.1 (interpretation boundary; no sampling semantics). In this module, P_S and Q_S^u are declared distribution-valued readouts. TV/IPM consequences are deterministic reporting bounds for the minimax residual, not sampling-based statistical inference. Estimation, significance, or average-case semantics requires an additional package clause.

(i) Event-probability gaps via total variation. If V declares total variation distance $D_{I, \text{TV}}(P, Q) := \sup_{B \in \Sigma} |P(B) - Q(B)|$ [3], then

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} \sup_{B \in \Sigma} |P_S(B) - Q_S^u(B)|.$$

Accordingly, a report $\Delta_{I \rightarrow G}(\mathcal{L}; V) < \varepsilon_R$ licenses the existence of an admissible $u_{\varepsilon_R} \in \mathcal{U}$ such that

$$\sup_{S \in \Omega_{\text{reach}}} \sup_{B \in \Sigma} |P_S(B) - Q_S^{u_{\varepsilon_R}}(B)| < \varepsilon_R,$$

i.e. all event-probability gaps are ε_R -bounded at the declared reporting resolution.

(ii) Audit predicates and advantage/error control via an IPM. Let V declare an admissible audit predicate family $\mathcal{T} \subseteq \{f : \mathcal{Y} \rightarrow [0, 1]\}$, and define the induced IPM [8, 3]

$$d_{\mathcal{T}}(P, Q) := \sup_{f \in \mathcal{T}} |\mathbb{E}_P[f] - \mathbb{E}_Q[f]|.$$

If V sets $D_I := d_{\mathcal{T}}$, then

$$\Delta_{I \rightarrow G}(\mathcal{L}; V) = \inf_{u \in \mathcal{U}} \sup_{S \in \Omega_{\text{reach}}} d_{\mathcal{T}}(P_S, Q_S^u),$$

and a report $\Delta_{I \rightarrow G}(\mathcal{L}; V) < \varepsilon_R$ licenses the existence of $u_{\varepsilon_R} \in \mathcal{U}$ such that

$$\sup_{S \in \Omega_{\text{reach}}} \sup_{f \in \mathcal{T}} |\mathbb{E}_{P_S}[f] - \mathbb{E}_{Q_S^{u_{\varepsilon_R}}}[f]| < \varepsilon_R.$$

To translate $d_{\mathcal{T}}$ into an audit advantage/error readout, consider the equal-prior one-shot decision task induced by the declared predicate family $\mathcal{T}$ [2] with $Y \sim P$ under $H = 0$ and $Y \sim Q$ under $H = 1$. This is a decision-theoretic interpretation of the IPM under the declared test family, not

an additional statistical testing protocol. Any $f \in \mathcal{T}$ induces a randomized decision rule that outputs 1 with probability $f(Y)$. Its equal-prior error probability is

$$P_e(f; P, Q) = \frac{1}{2}(\mathbb{E}_P[f] + \mathbb{E}_Q[1 - f]) = \frac{1}{2}(1 + \mathbb{E}_P[f] - \mathbb{E}_Q[f]).$$

Define the achievable audit advantage within $\mathcal{T}$ as $\text{Adv}_{\mathcal{T}}(P, Q) := \frac{1}{2} - \inf_{f \in \mathcal{T}} P_e(f; P, Q)$. Then

$$\text{Adv}_{\mathcal{T}}(P, Q) \leq \frac{1}{2} d_{\mathcal{T}}(P, Q),$$

with equality if $\mathcal{T}$ is closed under complement ($f \in \mathcal{T} \Rightarrow 1 - f \in \mathcal{T}$). Under the declared instantiation $D = d_{\mathcal{T}}$, a report $\Delta_{I \rightarrow G}(\mathcal{L}; V) < \varepsilon_R$ therefore bounds the worst-case audit advantage (after admissible reconstruction) by $\varepsilon/2$ at the declared resolution, and yields the corresponding equal-prior audit error lower bound.

Interpretation. The probabilistic module is a reporting interface: it converts package-locked residual statements into declared distribution-level readouts (event-gap bounds; predicate-gap and advantage/error bounds) once the relevant clauses are declared. When a fiber witness or audit non-closure witness is available, the certificate lower bounds above already lock the corresponding non-licensing or extension conclusion within the same package.

B.4 Bayesian intuition under declared evidence

Bayesian conditionalization is not used as a primitive in the framework. It is an optional interpretation available only when the probabilistic module has already declared the needed clauses: a candidate/fiber space $\mathcal{X}_V$, a prior or fiber measure, an evidence event, a likelihood or readout kernel, and a normalization domain. This schematic notation presupposes the existence of the relevant conditional measure; otherwise the probabilistic report is **undefined**.

Let $\mathcal{X}_V$ be the declared candidate/fiber space, let $e_V : \mathcal{X}_V \rightarrow \text{End}_V$ be the declared endpoint map, let $e \in \text{End}_V$ be the reported endpoint, and let $\mu_V(\cdot \mid e)$ denote a declared conditional measure on the fiber $e_V^{-1}(e)$. If B is a V -declared evidence event with likelihood/readout kernel $L_V(B \mid x)$ and the normalizing denominator is positive and finite, the updated fiber measure is schematically

$$d\mu_V(x \mid e, B) = \frac{L_V(B \mid x) d\mu_V(x \mid e)}{\int_{e_V^{-1}(e)} L_V(B \mid x') d\mu_V(x' \mid e)}.$$

For a report object $\rho_{\mathfrak{S}} : \mathcal{X}_V \rightarrow \text{Rep}$ induced by the original structure $\mathfrak{S}$, the updated distribution-valued report is then

$$\text{P}_V^{\mathfrak{S}}(\cdot \mid e, B) := (\rho_{\mathfrak{S}})_{\#} \mu_V(\cdot \mid e, B).$$

This subsection is only a Bayesian intuition for conditionalized distribution-valued reporting. It introduces no new gap object and does not change the meanings of Ψ , $\Delta_{I \rightarrow G}$, or Δ_{acc} . If the prior/fiber measure, evidence event, likelihood/readout kernel, or normalization condition is not declared, the Bayesian reading is **undefined** under the same non-licensing semantics used throughout the framework.

C Extended Ψ - $\Delta_{I \rightarrow G}$ Dynamics (short optional module)

Module summary.

Scope. Record a short optional dynamical appendix for Δ -variant separation and simple accumulated-readout templates.

Dependencies. A declared path of completed unary-report structures along γ , slice-local formation of Ψ_{run} , transports to a common report axis, an accumulated-report package V_{acc} , and any additional carrier assumptions used by the chosen template.

Outputs. Variant-safe statements about $\Delta_{\text{acc}}(\gamma)$, together with optional proportional and conservation templates that respect the dependency direction of the core accumulation interface.

Non-invocation rule. If the required declarations are not supplied, this module is not invoked. A fixed-slice endpoint report is not jointly consumed with Δ_{acc} without an additional comparison bridge.

This appendix is a compact support module. It does not introduce a new dynamical theory; it records how the minimal type-preserving accumulation update may be reused without confusing fixed-slice, accumulated, and meta-gap objects or reversing the dependency from the local running report to the accumulated report. Pairing preservation is an additional conditional obligation only for accumulated-report types that contain such a clause.

C.1 Notation and Δ -variants

- **Fixed-slice gap:** $\Delta_{I \rightarrow G}(\mathcal{L}_\gamma; V_\gamma)$ is the compression–reconstruction residual defined by a completed relation diagnostic.
- **Accumulated readout:** $\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^\gamma \Psi_{\text{run}}(s) ds$ is meaningful only under a declared path of completed unary reports, slice-local Ψ_{run} , transport to a common report axis, and V_{acc} preserving the inherited accumulated-report type.
- **Endpoint–path comparison:** joint use of a fixed-slice endpoint report and Δ_{acc} requires a separately licensed comparison bridge. Such a bridge may establish commensurability without establishing equality or order.
- **Meta-gap:** Δ_R^* is the cross-level non-identity marker of Definition 6.1; it is not a numerical substitute for either numerical report above.

Lemma C.1 (Variant separation). *The three symbols $\Delta_{I \rightarrow G}$, Δ_{acc} , and Δ_R^* name distinct objects. Before an explicit package-level comparison bridge is declared, equality, order, subtraction, or limit expressions jointly consuming $\Delta_{I \rightarrow G}$ and Δ_{acc} are not licensed. A bridge that establishes a common comparison type may also declare the algebraic operations needed to form a difference, but it does not by itself imply any equality or order relation; those require an additional verified certificate. The meta-gap Δ_R^* remains a semantic guardrail rather than a numerical operand.*

Lemma C.2 (Accumulation interface). *Assume Ψ_{run} is independently instantiated from package-locked data of each declared γ -slice as in Definition 7.1, and let V_{acc} supply the type-preserving update of Definition 7.2. If $\Psi_{\text{run}} \in L_{\text{loc}}^1$ on the declared interval, then*

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^\gamma \Psi_{\text{run}}(s) ds$$

defines an absolutely continuous accumulated readout and

$$\frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma)$$

holds almost everywhere. If $\Psi_{\text{run}} \geq 0$, then Δ_{acc} is non-decreasing on the declared interval.

Remark C.1 (Discrete index). For a discrete hierarchy or grid index k , let Ψ_k be independently formed from the k -slice under the discrete analogue of Definition 7.1. The forward update is

$$\Delta_{\text{acc},k+1} = \Delta_{\text{acc},k} + \Psi_k.$$

Hence $\Delta_{\text{acc},k+1} - \Delta_{\text{acc},k} = \Psi_k$ follows from the update; it is not used to define the local running report retrospectively.

C.2 Optional proportional and conservation templates

Lemma C.3 (Optional proportional specialization). *Assume Ψ_{run} and Δ_{acc} are independently instantiated through the core accumulation interface, with Δ_{acc} absolutely continuous. Suppose that the independently instantiated pair satisfies the additional model relation*

$$\Psi_{\text{run}}(\gamma) = b \Delta_{\text{acc}}(\gamma) \quad a.e.$$

on a declared interval, for a declared constant b . This relation is a hypothesis on the jointly instantiated reports, not a declaration of either report from the other, and therefore introduces no reverse operand edge from Δ_{acc} into the declaration of Ψ_{run} . Since $\Psi_{\text{run}} \geq 0$, the sign of b must be compatible with the sign of the accumulated carrier on that interval. The accumulation interface then gives the linear ODE $\Delta'_{\text{acc}} = b \Delta_{\text{acc}}$ almost everywhere, whose absolutely continuous solution is uniquely

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) e^{b(\gamma - \gamma_0)}$$

on the full declared interval, including the case $\Delta_{\text{acc}}(\gamma_0) = 0$. The specialization is invoked only on a nonempty licensed subdomain of independently instantiated report pairs on which the hypothesis is verified; otherwise no proportional-specialization consequence is derived. The same non-invocation discipline is used by the spiral specialization in Remark D.4. This is an optional model specialization, not a core law.

Lemma C.4 (Regime-relative conservation test). *Fix a declared interval $[\gamma_0, \gamma_1]$, a transport law across γ , an initial value, the default accumulated-carrier discrepancy $d_{\text{acc}}(a, b) = |a - b|$, and an active accumulated-package threshold ε_{acc} . Suppose an audit gives an estimate I of $\int_{\gamma_0}^{\gamma_1} \Psi_{\text{run}}(s) ds$ with error bound $E \geq 0$:*

$$\left| I - \int_{\gamma_0}^{\gamma_1} \Psi_{\text{run}}(s) ds \right| \leq E.$$

If $|I| + E < \varepsilon_{\text{acc}}$, then the report

$$\Delta_{\text{acc}}(\gamma_1) \approx_{\varepsilon_{\text{acc}}} \Delta_{\text{acc}}(\gamma_0)$$

is licensed. Otherwise this certificate alone does not license conservation at accumulated-package threshold ε_{acc} .

Remark C.2 (Required fields). Any use of the conservation template reports the evolution path, transport law, accumulation rule, initial value, estimate/error bound, and threshold. Missing fields make the conservation statement undefined.

D Logarithmic Spiral Closure and Self-Similar Geometry

Module summary.

Scope. Provide a short optional visualization for the accumulated interface $(\gamma, \Delta_{\text{acc}}(\gamma))$ under a declared planar embedding.

Dependencies. An accumulated quantity $\Delta_{\text{acc}}(\gamma)$, embedding constants, and an optional proportional-model hypothesis evaluated on independently instantiated reports.

Outputs. A logarithmic-spiral visualization when the proportional hypothesis is verified on a nonempty licensed regime and the positivity gate holds.

Non-invocation rule. If the embedding is absent or the proportional hypothesis is not verified on a nonempty licensed regime, no spiral claim is invoked.

This appendix records an *optional* geometric visualization for the accumulated interface introduced in Section 7. It turns the scalar pair $(\gamma, \Delta_{\text{acc}}(\gamma))$ into a declared planar curve and isolates the proportional specialization in which the embedding becomes a logarithmic spiral.

D.1 Objects and scope of this appendix

Fix a declared evolution parameter γ and an accumulated quantity $\Delta_{\text{acc}}(\gamma)$ as in Section 7, together with an intra-level rate $\Psi_{\text{run}}(\gamma)$ satisfying the interface

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) + \int_{\gamma_0}^{\gamma} \Psi_{\text{run}}(s) ds, \quad \frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = \Psi_{\text{run}}(\gamma) \quad \text{a.e. when } \Psi_{\text{run}} \in L_{\text{loc}}^1 \quad (2)$$

on any interval where the integral is well-defined. In this appendix, $\Delta_{\text{acc}}(\gamma)$ is an *accumulated readout* along γ . It is not jointly comparable with a fixed-slice inter-level gap $\Delta_{I \rightarrow G}(\mathcal{L}; V)$ unless an endpoint–path comparison bridge first supplies a common comparison type; any subsequent equality or order claim requires an additional verified certificate as in Remark 7.1.

D.2 Declared planar embedding (auxiliary)

Definition D.1 (Planar embedding of $(\gamma, \Delta_{\text{acc}})$). Declare constants $a_\theta \neq 0$, $a_r > 0$, and an offset $\theta_0 \in \mathbb{R}$. On any interval where $a_r \Delta_{\text{acc}}(\gamma) > 0$, define a planar curve in polar coordinates (r, θ) by

$$\theta(\gamma) = a_\theta \gamma + \theta_0, \quad r(\gamma) = a_r \Delta_{\text{acc}}(\gamma), \quad (3)$$

and interpret the geometric trajectory as $\gamma \mapsto (r(\gamma), \theta(\gamma))$. The positivity condition is part of the embedding declaration; if it fails on a region, the polar-spiral report is not licensed there.

Remark D.1 (Status of the embedding). The map (3) is an auxiliary visualization/closure device. It does not assert any claim about the physical phase-space geometry of $\mathcal{L}$. All geometric statements below are statements about the embedded curve defined by the declared map.

D.3 Logarithmic-spiral specialization under a local proportionality hypothesis

Hypothesis D.1 (Local proportionality specialization). On a declared γ -interval I , assume Δ_{acc} is absolutely continuous and that Ψ_{run} and Δ_{acc} have been independently instantiated under the core accumulation interface. Suppose that this independently instantiated pair satisfies the optional model relation

$$\Psi_{\text{run}}(\gamma) = b \cdot \Delta_{\text{acc}}(\gamma) \quad \text{for a.e. } \gamma \in I, \quad (4)$$

where $b \in \mathbb{R}$ is a declared constant (a regime parameter). The relation filters the licensed report pairs admitted to this specialization; it does not define Ψ_{run} from Δ_{acc} . The specialization is invoked only when this filtered licensed regime is nonempty. Because $\Psi_{\text{run}} \geq 0$, it carries the same sign-compatibility gate as Lemma C.3: on any positively embedded interval with

$\Delta_{\text{acc}}(\gamma) > 0$, it requires $b \geq 0$. If a signed accumulated carrier is intended, that carrier must be declared separately.

Combining (2) and (4) yields the linear ODE

$$\frac{d}{d\gamma} \Delta_{\text{acc}}(\gamma) = b \Delta_{\text{acc}}(\gamma) \quad \text{for almost every } \gamma \in I, \quad (5)$$

and uniqueness of the absolutely continuous solution gives, on the full declared interval,

$$\Delta_{\text{acc}}(\gamma) = \Delta_{\text{acc}}(\gamma_0) \exp(b(\gamma - \gamma_0)), \quad (6)$$

including the zero-initial-value case. Under the embedding (3), this becomes a spiral law.

Lemma D.1 (Logarithmic spiral under the proportional specialization). *Assume Definition D.1 and Hypothesis D.1 on an interval I , and suppose $a_r \Delta_{\text{acc}}(\gamma) > 0$ on I . Then the embedded curve satisfies*

$$\frac{dr}{d\theta} = \kappa r, \quad \kappa := \frac{b}{a_\theta}, \quad (7)$$

hence

$$r(\theta) = r(\theta_{\text{ref}}) \exp(\kappa(\theta - \theta_{\text{ref}})). \quad (8)$$

If $\kappa \neq 0$, the trajectory is a non-degenerate logarithmic spiral in the (r, θ) plane. If $\kappa = 0$, it is the degenerate constant-radius case.

Remark D.2 (Inward vs. outward winding). If $\kappa > 0$, then $r(\theta)$ is increasing in θ on the interval. If $\kappa < 0$, then $r(\theta)$ is decreasing in θ on the interval and approaches 0 as $\theta \rightarrow +\infty$ along that regime (if the interval extends to $+\infty$ in θ). If $\kappa = 0$, the embedded curve has constant radius. These are geometric properties of the embedded curve under the declared closure.

D.4 Equiangular property and a characterization

A classical geometric characterization of logarithmic spirals is equiangularity: the angle between the tangent direction and the radius vector is constant along the curve.

Lemma D.2 (Equiangular relation in polar form). *Let a C^1 polar curve be given by $r = r(\theta)$ with $r(\theta) > 0$ on an interval and $dr/d\theta \neq 0$ at the point under consideration. For the unsigned geometric angle $\varphi(\theta) \in (0, \pi/2)$ between the tangent direction and the radius vector,*

$$|\tan \varphi(\theta)| = \left| \frac{r(\theta)}{dr/d\theta} \right|. \quad (9)$$

Equivalently, the signed formula $\tan \varphi_{\text{or}} = r/(dr/d\theta)$ may be used when an oriented-angle convention is explicitly declared.

Corollary D.1 (Log spiral implies equiangularity). *If $r(\theta) = r_0 e^{\kappa(\theta - \theta_0)}$ with $\kappa \neq 0$, then $dr/d\theta = \kappa r$ and (9) yields $|\tan \varphi(\theta)| = 1/|\kappa|$, hence $\varphi(\theta)$ is constant under the unsigned angle convention.*

Corollary D.2 (Equiangularity implies log spiral). *If a signed angle convention is declared and $\varphi_{\text{or}}(\theta) \equiv \varphi_0$ is constant with $\tan \varphi_0 \neq 0$, then the signed relation gives $dr/d\theta = \kappa r$ with $\kappa = 1/\tan \varphi_0$, hence $r(\theta) = r_0 e^{\kappa(\theta - \theta_0)}$. For an unsigned-angle statement, the same conclusion requires a fixed orientation/sign gate for $dr/d\theta$ on the declared interval.*

D.5 Self-similarity constraints and a functional characterization

The spiral in Lemma D.1 can also be related to “rotation + scaling” self-similarity constraints. A single declared scaling step is not, by itself, enough to force a pure logarithmic spiral: it permits a shift-invariant modulation on the declared interval. A pure logarithmic spiral follows only after a no-modulation condition or a stronger continuous local one-parameter scaling law is declared.

Definition D.2 (Rotation–scaling self-similarity (single-step shape invariance)). A polar curve $r = r(\theta)$ is said to satisfy a single-step rotation–scaling law if there exist constants (ϕ, λ) with $\phi \neq 0$ and $\lambda > 0$ such that for all θ for which both sides are in the declared interval,

$$r(\theta + \phi) = \lambda r(\theta). \quad (10)$$

Lemma D.3 (Single-step self-similarity and the no-modulation gate). *Assume $r(\theta) > 0$ and r is continuous on a declared interval stable under the shift $\theta \mapsto \theta + \phi$, and suppose (10) holds for fixed (ϕ, λ) . Put*

$$\kappa := \frac{\log \lambda}{\phi}, \quad p(\theta) := e^{-\kappa\theta} r(\theta).$$

Then

$$p(\theta + \phi) = p(\theta), \quad r(\theta) = e^{\kappa\theta} p(\theta).$$

Thus the single-step law licenses an exponential factor plus a ϕ -shift-invariant modulation on the declared interval. A pure logarithmic spiral follows only if the package declares the no-modulation condition that p is constant on the interval, or if it strengthens the symmetry to a continuous local one-parameter scaling law

$$r(\theta + s) = e^{\kappa s} r(\theta)$$

for all sufficiently small admissible s . Under either strengthened declaration, $dr/d\theta = \kappa r$ and the curve is a logarithmic spiral.

Remark D.3 (Role of symmetry declarations). The single-step constraint alone does not eliminate shift-invariant modulation; the no-modulation or local one-parameter scaling clause is an additional declaration.

D.6 Discrete evolution index (hierarchy depth) version

When the evolution parameter is a discrete index (e.g. a hierarchy depth $k \in \mathbb{N}$), the same structure appears in difference form.

Hypothesis D.2 (Discrete proportional model relation). Let Ψ_k be independently instantiated from the declared k -slice and let the discrete accumulated readout satisfy the forward update

$$\Delta_{\text{acc},k+1} = \Delta_{\text{acc},k} + \Psi_k.$$

Suppose that the independently instantiated pair satisfies the optional proportionality

$$\Psi_k = b \Delta_{\text{acc},k}, \quad (11)$$

with declared constant b (regime parameter). This is a hypothesis that filters admissible instantiated pairs; it does not define Ψ_k from $\Delta_{\text{acc},k}$. The specialization is invoked only when the filtered licensed regime is nonempty. On any positive accumulated regime where $\Delta_{\text{acc},k} > 0$ and $\Psi_k \geq 0$, the hypothesis requires $b \geq 0$.

The forward update then gives $\Delta_{\text{acc},k+1} = (1 + b)\Delta_{\text{acc},k}$, hence $\Delta_{\text{acc},k} = \Delta_{\text{acc},0}(1 + b)^k$ on any nonempty licensed regime where the hypothesis is verified. Under the embedding $\theta_k = a_\theta k + \theta_0$ and $r_k = a_r \Delta_{\text{acc},k}$, this yields a discrete sampling of a non-degenerate logarithmic spiral when $b \neq 0$; when $b = 0$, it yields the degenerate constant-radius sampling.

D.7 Reproducible reporting clauses for spiral closure

Any use of the spiral closure in later case studies is regarded as well-posed only if it reports the following declarations explicitly:

1. the Δ -variant used (*accumulated* Δ_{acc} by default); if a fixed-slice $\Delta_{I \rightarrow G}$ is jointly consumed, the endpoint–path comparison bridge and any additional equality/order certificate must be declared separately;
2. the evolution parameter domain (continuous γ or discrete k) and the reference point;
3. the independently instantiated running readout (Ψ_{run} in the continuous case or Ψ_k in the discrete case), the proportional hypothesis (e.g. $\Psi_{\text{run}} = b\Delta_{\text{acc}}$ or $\Psi_k = b\Delta_{\text{acc},k}$), and the nonempty licensed regime on which that hypothesis is verified;
4. the embedding constants $(a_\theta, a_r, \theta_0)$ used to map the trajectory into polar coordinates.

Remark D.4 (Relation to the core framework and non-invocation). Appendix D introduces no new core definition. It only records one optional specialization path: the accumulation interface plus a verified proportional model relation plus a declared planar embedding $\Rightarrow$ logarithmic spiral geometry for the accumulated readout. If the proportional hypothesis is not verified on a nonempty licensed regime of independently instantiated reports, none of the spiral claims are invoked.

E Resolution Feasibility and Control: Pairwise and Partition Semantics

Module summary.

Scope. Provide the full fixed-target scalar resolution-feasibility module underlying Sec. 4.4, including separation-only and two-sided classification tasks under pairwise and partition output semantics.

Dependencies. The unchanged package schema $V = (\mathcal{I}, \varepsilon, \mathcal{Q}, \mathcal{C})$; typed fixed-target refinements of the existing interface and query components; certified one-sided error envelopes; and a positive report threshold.

Outputs. Certificate-relative pairwise and partition feasibility sets; chain-separation conditions; certificate/policy localization; robustness, selection, and adaptive-control clauses.

Non-invocation rule. If the resolution parameter changes the semantic target object rather than estimating a fixed target discrepancy, this appendix is not invoked without an additional transport construction.

E.1 Purpose and restricted scope

This appendix is attached to a declared active report-package node and its fixed target discrepancy. Every domain introduced below must be package-locked or explicitly derived from locked carriers. It addresses a prior licensing question: given typed resolution refinements of an existing declared package and declared threshold-reporting requirements, when does any positive reporting threshold remain certified? The threshold is confined by a certified lower bound on the query margin, the certified error of the mechanism, any invoked null-preservation demand, and any additional policy floor. Certificate-only and policy-restricted sets are kept distinct.

The construction preserves the top-level package tuple and is restricted to a fixed scalar target discrepancy and to mechanisms that estimate or numerically realize that same target. The additional objects below are internal refinements of $\mathcal{I}$ and $\mathcal{Q}$, not new top-level slots. It treats two output semantics:

1. *directional pairwise semantics*, in which each ordered queried pair is judged directly by a thresholded discrepancy and no symmetry is presumed;
2. *partition semantics*, in which a symmetric threshold relation, or a declared certified symmetrization, is closed transitively into equivalence classes.

It also separates two task types:

1. *separation-only licensing*, which requires declared ordered positive pairs to remain reportably non-negligible;
2. *two-sided classification*, which additionally requires declared ordered null pairs to be reported negligible by the same threshold.

Closure, reconstructability, future-readout constancy, and scale families that change the target object itself require different demand functionals and are not reduced to the present module.

E.2 Target discrepancy, interface family, and certification

Definition E.1 (Fixed target discrepancy). Let X_{adm} be a package-locked admissible domain or an explicitly derived domain cited by the active report package. A fixed target discrepancy is a declared map

$$D^* : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

that does not depend on the resolution-control parameter or on the report threshold. It may be a metric, pseudometric, declared symmetrization of a divergence, or another scalar discrepancy whose type is fixed by the enclosing package.

Definition E.2 (Raw and certified resolution refinements of an existing interface). Fix base clauses $\mathcal{I}^{\text{base}}$, $\mathcal{Q}^{\text{base}}$, and $\mathcal{C}^{\text{base}}$ under Definition 4.10. Let $\lambda \in \Lambda_{\text{res}}$ parameterize raw realization subclauses

$$\mathfrak{R}_{\text{res}}^{\text{raw}} := \{\mathcal{R}_{\lambda}^{\text{raw}}\}_{\lambda \in \Lambda_{\text{res}}}, \quad \mathcal{R}_{\lambda}^{\text{raw}} := (\mathcal{N}_{\lambda}, D^*, \hat{D}_{\lambda}),$$

where $\mathcal{N}_{\lambda}$ is the declared measurement, resolution, or numerical-realization mechanism and

$$\hat{D}_{\lambda} : X_{\text{adm}} \times X_{\text{adm}} \rightarrow \mathbb{R}_{\geq 0}$$

reports the same fixed target discrepancy. The base structural interface is retained, and λ is not the report threshold ϵ .

Let $\mathcal{D} \subseteq X_{\text{adm}} \times X_{\text{adm}}$ be the declared certification domain. For each λ , let $\rho_+(\lambda), \rho_-(\lambda) \geq 0$ satisfy, deterministically or on a declared confidence event,

$$\hat{D}_{\lambda}(x, y) - D^*(x, y) < \rho_+(\lambda), \tag{12}$$

$$D^*(x, y) - \hat{D}_{\lambda}(x, y) < \rho_-(\lambda) \tag{13}$$

for every $(x, y) \in \mathcal{D}$. If a non-strict bound or exact equality is the available result, a declared positive slack may be added to obtain the strict envelope used by the threshold rule. A zero envelope is valid only when the corresponding signed report error is uniformly strictly negative; symmetric zero envelopes do not represent an exact mechanism under this strict convention.

With Cert denoting the declared certificate object, define

$$\mathcal{R}_{\lambda}^{\text{cert}} := (\mathcal{R}_{\lambda}^{\text{raw}}, \mathcal{D}, \rho_+(\lambda), \rho_-(\lambda), \text{Cert}), \quad \mathcal{I}_{\lambda} := \mathcal{I}^{\text{base}} \oplus \mathcal{R}_{\lambda}^{\text{cert}}.$$

The symbol $\oplus$ denotes typed internal extension of $\mathcal{I}$, not a new top-level package operation.

Envelope	Error controlled	Operational roles
ρ_+	over-report allowance: $\hat{D}_\lambda - D^* < \rho_+$	null-preservation floor for a declared Q_0 ; converts an observed report into the certified lower bound $D^* > \hat{D}_\lambda - \rho_+$; may also enter the construction of a certified query margin
ρ_-	under-report allowance: $D^* - \hat{D}_\lambda < \rho_-$	certified separation-ceiling erosion; converts a certified lower bound on D^* into the display guarantee $\hat{D}_\lambda > D^* - \rho_-$

The subscripts encode the sign of the report error. They do not name the direction of the resulting bound on D^* . This role map is used throughout the appendix.

Remark E.1 (Exogenous certification). The physical, numerical, or statistical arguments that produce the mechanism envelopes and the certified query-margin bounds are inputs to this module. Instrument specifications, conditioning analysis, rounding-error bounds, concentration inequalities, and independent calibration may contribute to those certificates; none is automatically identical to the final certified quantity. The framework operates downstream of the declared certificates.

Definition E.3 (Pointwise, simultaneous, and sequential certificates). How the declared certificate is used is a protocol-level scope clause. A certificate for one predeclared completed configuration (λ, ε) is *pointwise in the configuration*. For a declared scan set $\mathcal{G}_0 \subseteq \Lambda_{\text{res}} \times \mathbb{R}_{>0}$, a simultaneous certificate satisfies

$$\mathbb{P}\left(\bigcap_{(\lambda, \varepsilon) \in \mathcal{G}_0} \mathcal{C}(\lambda)\right) \geq 1 - \alpha,$$

where $\mathcal{C}(\lambda)$ is the event on which (12)–(13) and every certified query-margin lower bound used at that configuration are valid. A finite scan may use a declared familywise correction; an infinite scan requires a uniform certificate. For an adaptively chosen sequence of completed configurations $c_1, c_2, \dots$, a sequentially valid certificate satisfies

$$\mathbb{P}(\mathcal{C}_n \text{ holds at every visited step } n) \geq 1 - \alpha$$

under the declared data-dependent stopping rule. Pointwise guarantees cannot be reused as a joint guarantee for a scan or adaptive path; time-uniform confidence sequences provide one standard sequential construction [6].

Protocol E.1 (Certification scope must match use). A single predeclared configuration may use a pointwise certificate. A predetermined scan requires simultaneous certification. An adaptive controller requires sequentially valid certification under the declared controller and stopping rule.

E.3 Threshold semantics and typed query clauses

Definition E.4 (Direct threshold relation). For $\varepsilon > 0$, define the ordered relation

$$x E_{\lambda, \varepsilon} y \quad :\Longleftrightarrow \quad \hat{D}_\lambda(x, y) < \varepsilon.$$

Under directional pairwise semantics, an ordered pair in $E_{\lambda, \varepsilon}$ is reported negligible in that direction. Symmetry is not presumed, and the relation need not be transitive. Accordingly, $E_{\lambda, \varepsilon}$ is a directional threshold-report predicate, not the framework’s symmetric resolution-indistinguishability predicate $\approx_\varepsilon$. It may instantiate $\approx_\varepsilon$ only after a symmetric rule or a declared certified symmetrization is supplied. The word *merge* is reserved below for a symmetric relation that is closed into a partition.

Definition E.5 (True and certified pairwise query margins). Let

$$Q_{\text{sep}} \subseteq \mathcal{D}, \quad Q_{\text{sep}} \neq \emptyset,$$

be declared before the main diagnostic, with every ordered pair required to remain reportably non-negligible in its declared direction and with $x \neq y$ for every $(x, y) \in Q_{\text{sep}}$. Define the ideal margin

$$\delta_Q^{\text{pair}} := \inf_{(x,y) \in Q_{\text{sep}}} D^*(x, y).$$

Operational certification requires a declared lower-margin family

$$\underline{\delta}_Q^{\text{pair}} : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}, \quad 0 \leq \underline{\delta}_Q^{\text{pair}}(\lambda) \leq \delta_Q^{\text{pair}},$$

together with the pointwise, simultaneous, or sequential scope under which it remains valid. A family-uniform lower bound is the special case in which $\underline{\delta}_Q^{\text{pair}}(\lambda)$ is constant on the invoked mechanism family. If the ideal margin is exactly available, one may take $\underline{\delta}_Q^{\text{pair}}(\lambda) = \delta_Q^{\text{pair}}$ for every invoked λ .

When a full two-sided classifier is invoked, additionally declare a nonempty ordered-pair set

$$Q_0 \subseteq \mathcal{D}$$

whose pairs must be reported negligible, with

$$D^*(x, y) = 0 \quad \text{for every } (x, y) \in Q_0.$$

The Q_0 clause is not implied by Q_{sep} and must not be introduced silently.

Definition E.6 (Policy floor). A reporting policy may impose $p \geq 0$, forbidding thresholds below p even when the certificate-only task clauses would permit them. Policy does not certify capability and is therefore introduced as a further query refinement rather than built into the pre-policy task package.

Definition E.7 (Task-indexed query refinements and completed packages). Let

$$\sigma \in \{\text{pair}, \text{part}\}, \quad \tau \in \{\text{sep}, \text{class}\}$$

index report semantics and task mode. Define

$$\mathcal{Q}^{\sigma, \text{sep}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, \text{Sem} = \sigma),$$

and, when Q_0 is declared,

$$\mathcal{Q}^{\sigma, \text{class}} := \mathcal{Q}^{\text{base}} \oplus (Q_{\text{sep}}, Q_0, \text{Sem} = \sigma).$$

If policy is invoked, define

$$\mathcal{Q}^{\sigma, \tau, p} := \mathcal{Q}^{\sigma, \tau} \oplus p.$$

For each $(\lambda, \varepsilon) \in \Lambda_{\text{res}} \times \mathbb{R}_{>0}$, define

$$V_{\lambda, \varepsilon}^{\sigma, \tau} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau}, \mathcal{C}^{\text{base}}),$$

and, when the policy clause is invoked,

$$V_{\lambda, \varepsilon}^{\sigma, \tau, p} := (\mathcal{I}_{\lambda}, \varepsilon, \mathcal{Q}^{\sigma, \tau, p}, \mathcal{C}^{\text{base}}).$$

All four combinations are instances of the original package schema and remain distinct because adding Q_0 or changing pairwise to partition semantics may change package admissibility.

Definition E.8 (Task-indexed package-admissible threshold sets). For each fixed λ and task index (σ, τ) , define

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau}(\lambda) := \{\varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau} \text{ is admissible}\},$$

and, when policy is invoked,

$$\mathcal{E}_{\text{adm}}^{\sigma, \tau, p}(\lambda) := \{\varepsilon > 0 : V_{\lambda, \varepsilon}^{\sigma, \tau, p} \text{ is admissible}\}.$$

The certificate inequalities generate algebraic threshold slices. The licensed configuration slices are their intersections with the corresponding task-indexed package-admissible threshold sets. Endpoint inequalities alone are sufficient only when the required algebraic slice is known to lie inside the relevant $\mathcal{E}_{\text{adm}}^{\sigma, \tau}$ set.

Definition E.9 (Collision and vanishing uniform margin). A *representation collision* occurs when some $(x, y) \in Q_{\text{sep}}$ satisfies $D^*(x, y) = 0$. A *vanishing ideal margin* occurs when every required pair has positive discrepancy but $\delta_Q^{\text{pair}} = 0$. A certified lower bound may also vanish even when the ideal margin is positive; that is certificate-relative conservatism, not a representation collision.

On the certification event, every required-distinct ordered pair satisfies

$$\hat{D}_\lambda(x, y) > D^*(x, y) - \rho_-(\lambda) \geq \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda).$$

For each declared null ordered pair, the over-report envelope gives

$$\hat{D}_\lambda(x, y) < \rho_+(\lambda).$$

Thus ρ_- supplies the upper threshold constraint for a declared certified margin. The ρ_+ envelope enters the threshold interval as a lower endpoint only when null preservation is required; separately, it may be used upstream to construct a certified lower bound from observed discrepancy reports.

E.4 Pairwise certified feasibility

Definition E.10 (Directional pairwise certified sets). For fixed λ , define the algebraic slices

$$I_\lambda^{\text{pair, sep, cert}} := \left(0, \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right],$$

and, when Q_0 is declared,

$$I_\lambda^{\text{pair, class, cert}} := (0, \infty) \cap \left[\rho_+(\lambda), \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda)\right].$$

The package-conditioned sets are

$$\mathcal{A}_{Q_{\text{sep}}}^{\text{pair, sep, cert}} := \{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair, sep}}(\lambda) \cap I_\lambda^{\text{pair, sep, cert}}\},$$

and

$$\mathcal{A}_{Q_0, Q_{\text{sep}}}^{\text{pair, class, cert}} := \{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{pair, class}}(\lambda) \cap I_\lambda^{\text{pair, class, cert}}\}.$$

Policy intersects the algebraic slice with $[p, \infty)$ and uses $\mathcal{E}_{\text{adm}}^{\text{pair, sep, } p}(\lambda)$ or $\mathcal{E}_{\text{adm}}^{\text{pair, class, } p}(\lambda)$, respectively.

For fixed λ , write

$$\underline{b}_\lambda^{\text{pair}} := \underline{\delta}_Q^{\text{pair}}(\lambda) - \rho_-(\lambda).$$

Proposition E.1 (Directional pairwise fixed-mechanism slices). *For fixed λ , the exact licensed slices are*

$$\mathcal{E}_{\text{adm}}^{\text{pair,sep}}(\lambda) \cap (0, \underline{b}_{\lambda}^{\text{pair}}]$$

for separation-only reporting and

$$\mathcal{E}_{\text{adm}}^{\text{pair,class}}(\lambda) \cap (0, \infty) \cap [\rho_+(\lambda), \underline{b}_{\lambda}^{\text{pair}}]$$

for full classification. Policy uses $\mathcal{E}_{\text{adm}}^{\text{pair,sep},p}(\lambda)$ or $\mathcal{E}_{\text{adm}}^{\text{pair,class},p}(\lambda)$, respectively, and additionally intersects with $[p, \infty)$. Each licensed slice is nonempty exactly when the corresponding displayed intersection is nonempty.

Corollary E.1 (Pairwise endpoint criteria under admissibility-neutrality). *Assume the relevant algebraic slice is contained in the corresponding package-admissible threshold set. Then the separation-only slice is nonempty exactly when $\underline{b}_{\lambda}^{\text{pair}} > 0$, and the full-classification slice is nonempty exactly when*

$$\rho_+(\lambda) \leq \underline{b}_{\lambda}^{\text{pair}} \quad \text{and} \quad \underline{b}_{\lambda}^{\text{pair}} > 0.$$

Policy replaces the lower endpoint by p or $\max\{p, \rho_+(\lambda)\}$, respectively. If $\rho_+ = \rho_- = \rho$, the full-classification algebraic slice is nonempty exactly when

$$2\rho(\lambda) \leq \underline{\delta}_Q^{\text{pair}}(\lambda) \quad \text{and} \quad \underline{\delta}_Q^{\text{pair}}(\lambda) > 0.$$

These endpoint criteria certify package-conditioned feasibility only under the stated admissibility containment.

Remark E.2 (Ideal and certified margins). Replacing $\underline{\delta}_Q^{\text{pair}}(\lambda)$ by the true margin δ_Q^{pair} gives the corresponding ideal algebraic geometry. The operational certificate slices are conservative. Failure of a package-conditioned certified set may reflect a weak margin certificate, package inadmissibility on the algebraic slice, or both, even when the ideal margin is positive.

Remark E.3 (Classical adjacency). The geometry $\rho_+ + \rho_- \leq \delta$ is the standard requirement that two-sided uncertainty allowances fit inside a required separation margin. Closely related forms appear in margin, packing, and two-point minimax arguments [10]. No novelty is claimed for the inequality in isolation. Its role here is to locate that geometry inside a declared package with explicitly typed null and separation queries and a declared certified margin.

Remark E.4 (Multiple queries and family geometry). If one package uses one threshold for several audited query bundles, its completed configuration must lie in the intersection of their certified sets. Replacing those bundles by unions of their Q_0 and Q_{sep} clauses is equivalent only when they share the same target discrepancy, raw and certified refinement family, certification domain, envelopes, margin-certification construction, certificate scope, report semantics, and policy clauses. If policy floors differ, use $p_{\text{joint}} := \sup_i p_i$. When semantics or certificate scopes differ, retain the intersection formulation.

For fixed λ , a threshold slice may be empty, a singleton, or a nondegenerate interval. The family-level set need not be connected: discrete mechanism families may yield isolated configurations or several components. The word *margin* is therefore reserved for an ordered slice or component, not for every multi-point family-level set.

E.5 Partition semantics and chain separation

Assume for this subsection that the direct threshold relation is symmetric, either because D^* and $\widehat{D}_{\lambda}$ are symmetric or because a symmetrization was explicitly declared and certified.

Definition E.11 (Threshold graph and generated partition). Let $G_{\lambda,\varepsilon}$ be the undirected graph on X_{adm} with edge $\{x, y\}$ when $\widehat{D}_{\lambda}(x, y) < \varepsilon$. The threshold partition $\Pi_{\lambda,\varepsilon}$ is the partition into connected components of $G_{\lambda,\varepsilon}$, equivalently the equivalence relation generated by $E_{\lambda,\varepsilon}$.

Definition E.12 (Chain-edge certification condition). Let

$$\mathcal{D}_{\text{edge}}(\lambda, \varepsilon) := \{(x, y) \in X_{\text{adm}}^2 : \widehat{D}_\lambda(x, y) < \varepsilon\}.$$

Write $\text{EdgeCert}(\lambda, \varepsilon)$ when

$$\mathcal{D}_{\text{edge}}(\lambda, \varepsilon) \subseteq \mathcal{D},$$

and the certified under-report envelope is valid jointly on that edge domain. The stronger condition $\mathcal{D} = X_{\text{adm}}^2$ is sufficient. For a scan, EdgeCert must hold simultaneously on every visited edge domain; for an adaptive path it must hold under the declared sequential certificate.

Definition E.13 (Bottleneck chain distance). Set

$$B^\star(x, x) := 0.$$

For $x \neq y$, define

$$B^\star(x, y) := \inf_{\substack{n \geq 1 \\ x = x_0, \dots, x_n = y}} \max_{1 \leq i \leq n} D^\star(x_{i-1}, x_i),$$

where all chain vertices lie in X_{adm} . Equivalently, one may allow the length-zero chain and assign it bottleneck 0. This is the smallest target-level edge threshold at which x and y can be connected through a chain. The construction is the standard bottleneck geometry underlying single-linkage threshold connectivity [4].

Definition E.14 (True and certified partition query margins). Define the ideal bottleneck margin

$$\delta_Q^{\text{chain}} := \inf_{(x, y) \in Q_{\text{sep}}} B^\star(x, y).$$

Because the one-edge chain is admissible,

$$\delta_Q^{\text{chain}} \leq \delta_Q^{\text{pair}}.$$

Operational use requires a declared certified lower-margin family

$$\underline{\delta}_Q^{\text{chain}} : \Lambda_{\text{res}} \rightarrow \mathbb{R}_{\geq 0}, \quad 0 \leq \underline{\delta}_Q^{\text{chain}}(\lambda) \leq \delta_Q^{\text{chain}},$$

with a certificate scope matching the scan or adaptive use. When pairwise and partition certificate slices are compared, choose compatible lower bounds satisfying

$$\underline{\delta}_Q^{\text{chain}}(\lambda) \leq \underline{\delta}_Q^{\text{pair}}(\lambda)$$

for every invoked λ . Exact margins may be used when available; family-uniform bounds are the constant special case.

Proposition E.2 (Certified chain-separation sufficiency). *Assume $\text{EdgeCert}(\lambda, \varepsilon)$. If*

$$\varepsilon \leq \underline{\delta}_Q^{\text{chain}}(\lambda) - \rho_-(\lambda),$$

then no pair in Q_{sep} lies in the same block of $\Pi_{\lambda, \varepsilon}$.

Proof. Suppose a required pair (x, y) is connected by a reported threshold chain $x = x_0, \dots, x_n = y$. For every edge,

$$\widehat{D}_\lambda(x_{i-1}, x_i) < \varepsilon.$$

By the certified under-report envelope,

$$D^\star(x_{i-1}, x_i) < \widehat{D}_\lambda(x_{i-1}, x_i) + \rho_-(\lambda) < \varepsilon + \rho_-(\lambda) \leq \underline{\delta}_Q^{\text{chain}}(\lambda) \leq \delta_Q^{\text{chain}}.$$

Hence the chain has target bottleneck strictly below δ_Q^{chain} , contradicting the definition of the query margin. $\square$

Definition E.15 (Partition certificate-only and policy-restricted sets). For fixed λ , define

$$I_\lambda^{\text{part,sep,cert}} := \left(0, \underline{\delta}_Q^{\text{chain}}(\lambda) - \rho_-(\lambda)\right],$$

and, when Q_0 is declared,

$$I_\lambda^{\text{part,class,cert}} := (0, \infty) \cap \left[\rho_+(\lambda), \underline{\delta}_Q^{\text{chain}}(\lambda) - \rho_-(\lambda)\right].$$

The package-conditioned partition sets are

$$\mathcal{A}_{Q_{\text{sep}}}^{\text{part,sep,cert}} := \left\{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{part,sep}}(\lambda) \cap I_\lambda^{\text{part,sep,cert}}, \quad \text{EdgeCert}(\lambda, \varepsilon)\right\},$$

and, when Q_0 is declared,

$$\mathcal{A}_{Q_0, Q_{\text{sep}}}^{\text{part,class,cert}} := \left\{(\lambda, \varepsilon) : \varepsilon \in \mathcal{E}_{\text{adm}}^{\text{part,class}}(\lambda) \cap I_\lambda^{\text{part,class,cert}}, \quad \text{EdgeCert}(\lambda, \varepsilon)\right\}.$$

Policy uses $\mathcal{E}_{\text{adm}}^{\text{part,sep},p}(\lambda)$ or $\mathcal{E}_{\text{adm}}^{\text{part,class},p}(\lambda)$, respectively, intersects the algebraic slice with $[p, \infty)$, and retains the same edge-certification condition.

With compatible certified margins, the algebraic inclusions

$$I_\lambda^{\text{part,sep,cert}} \subseteq I_\lambda^{\text{pair,sep,cert}}, \quad I_\lambda^{\text{part,class,cert}} \subseteq I_\lambda^{\text{pair,class,cert}}$$

hold. The corresponding package-conditioned set inclusions require, in addition, that partition-admissible configurations are pairwise-admissible under the declared task transports and that the partition edge certificate is available. No inclusion of full package-conditioned sets is inferred from the margin inequality alone.

Corollary E.2 (Symmetric partition endpoint criteria). *Assume $\rho_+ = \rho_- = \rho$, the relevant algebraic slice lies inside the corresponding package-admissible threshold set, and $\text{EdgeCert}(\lambda, \varepsilon)$ holds throughout that slice. Then the full partition-classification slice is nonempty exactly when*

$$2\rho(\lambda) \leq \underline{\delta}_Q^{\text{chain}}(\lambda) \quad \text{and} \quad \underline{\delta}_Q^{\text{chain}}(\lambda) > 0.$$

The partition separation-only slice is algebraically nonempty exactly when

$$\underline{\delta}_Q^{\text{chain}}(\lambda) - \rho(\lambda) > 0.$$

Policy uses the corresponding lower endpoint p or $\max\{p, \rho(\lambda)\}$. These endpoint conditions are sufficient for package-conditioned feasibility only under the stated admissibility and edge-certification assumptions.

Proposition E.3 (Pairwise feasibility need not imply partition feasibility). *It is possible that*

$$\delta_Q^{\text{pair}} > 0 \quad \text{and} \quad \delta_Q^{\text{chain}} = 0.$$

Consequently positive pairwise-certified thresholds may exist while every valid certified lower bound on the partition margin is zero, so no positive partition-certified threshold exists.

Proof. Take $X_{\text{adm}} = [0, 1]$ with $D^*(x, y) = |x - y|$ and $Q_{\text{sep}} = \{(0, 1)\}$. Then $\delta_Q^{\text{pair}} = 1$. For every n , the chain $0, 1/n, 2/n, \dots, 1$ has bottleneck $1/n$, so $B^*(0, 1) = 0$ and $\delta_Q^{\text{chain}} = 0$. $\square$

Remark E.5 (Practical scope of partition semantics). Positive partition-level feasibility requires a positive ideal bottleneck margin $\delta_Q^{\text{chain}} > 0$ and, operationally at an invoked mechanism, a positive certified lower margin $\underline{\delta}_Q^{\text{chain}}(\lambda) > 0$. This holds, for example, in finite metric domains with distinct required pairs, in uniformly gapped domains, and in other settings where the required pairs are separated by a positive bottleneck gap. If D^* is a metric and the relevant

component of X_{adm} is path-connected in the topology induced by D^* , every two points in that component admit chains with arbitrarily small maximal edge; hence $B^*(x, y) = 0$ for pairs in that component and $\delta_Q^{\text{chain}} = 0$ whenever the query lies within it.

At the opposite boundary, if D^* is an ultrametric, then its strong triangle inequality propagates along every finite chain and gives

$$D^*(x, y) \leq \max_i D^*(x_{i-1}, x_i).$$

The one-edge chain supplies the reverse inequality after taking the infimum, so

$$B^*(x, y) = D^*(x, y), \quad \delta_Q^{\text{chain}} = \delta_Q^{\text{pair}}.$$

Provided the required full edge-domain certificate is available, pairwise and partition feasibility then coincide geometrically. The margin incurs no additional loss, although the certification burden may remain stronger under partition semantics because the under-report envelope must hold on every pair that may serve as a chain edge. Ultrametric hierarchies and finite or uniformly gapped tree structures are therefore natural hosts of positive partition-level margins; unrestricted path-connected tree metrics are not.

Remark E.6 (Contact with finite-resolution repair non-uniqueness). The directional pairwise feasibility result and the Universal Closure-Restoring Complement theorem are not identified. Their precise contact is the non-transitivity of finite-resolution indistinguishability. In the present module, transitive closure can merge required-distinct pairs through threshold chains. In the universal-complement setting, the related obstruction appears as failure of finite-resolution adequacy to remain closed under joins, permitting incomparable maximal adequate repairs with no coarsest element [19]. A theorem equating the two constructions is not claimed.

E.6 Numerical step size as a fixed-target instance

A standard numerical differentiation or discretization method controlled by $\lambda = h > 0$ may admit a symmetric error envelope of the schematic form

$$\rho(h) = C_{\text{tr}} h^r + C_{\text{ro}} \frac{u}{h^s}, \quad r, s > 0,$$

where the first term is truncation or discretization error and the second is finite-precision amplification [5]. Both terms belong to the mechanism certificate; neither is the query demand.

For a pairwise separation-only query, the positive thresholds satisfy

$$0 < \varepsilon \leq \delta_Q^{\text{pair}}(h) - \rho(h),$$

so the algebraic slice is nonempty exactly when $\delta_Q^{\text{pair}}(h) - \rho(h) > 0$; the package-conditioned slice additionally requires intersection with $\mathcal{E}_{\text{adm}}^{\text{pair,sep}}(h)$. For a full two-sided classifier with a declared null set, the thresholds satisfy

$$\rho(h) \leq \varepsilon \leq \delta_Q^{\text{pair}}(h) - \rho(h),$$

and the algebraic slice is nonempty exactly when

$$2\rho(h) \leq \delta_Q^{\text{pair}}(h) \quad \text{and} \quad \delta_Q^{\text{pair}}(h) > 0.$$

The package-conditioned full-classification slice additionally requires intersection with $\mathcal{E}_{\text{adm}}^{\text{pair,class}}(h)$. For partition output, replace $\delta_Q^{\text{pair}}(h)$ by $\delta_Q^{\text{chain}}(h)$ and require the chain-edge certificate.

The U-shaped error curve is the geometry of the mechanism envelope $\rho(h)$: decreasing h reduces truncation error but eventually increases rounding amplification. When the minimum is attained, an error-minimizing step

$$h_{\text{err}} \in \arg \min_{h>0} \rho(h)$$

does not by itself determine the report threshold. Mechanism-error minimization, threshold selection within a fixed- h slice, and joint resource–error optimization are distinct problems.

E.7 Certificate-relative infeasibility and localization

Definition E.16 (Algebraic, package-conditioned, and policy-induced non-licensing). Fix one of the four invoked semantics/task combinations. For a given mechanism or family:

- an *algebraic certificate obstruction* occurs when the relevant certificate inequality slice is empty;
- *package-conditioned non-licensing* occurs when the algebraic slice is nonempty but its intersection with the relevant package-admissible threshold set (and, for partition semantics, the edge-certification condition) is empty;
- *policy-induced non-licensing* occurs when a pre-policy licensed set is nonempty but the corresponding policy-refined set is empty.

The union of the first two cases is the certificate-relative failure of the completed package family to license a threshold, but their causes must be reported separately.

Certificate-relative failure certifies only that the declared interface family, current mechanism and margin certificates, fixed target discrepancy, report semantics, package admissibility clauses, and task clauses are jointly unable to license a positive threshold. It does not prove that no sharper certificate or improved mechanism could do so. Nor does it identify a missing coordinate, prove closure failure, establish complement demand, or select a repair. Possible responses include extending the mechanism family, improving conditioning or calibration, defining a new discrepancy or representation package, changing report semantics, modifying the query, or reporting the query as unservable under the current certified family.

Proposition E.4 (Positive homogeneity of the algebraic feasibility inequalities). *Let $c > 0$ and apply the common unit rescaling*

$$D^*, \hat{D}_\lambda, \delta_Q^{\text{pair}}(\lambda), \delta_Q^{\text{chain}}(\lambda), \rho_\pm, \varepsilon, p \mapsto cD^*, c\hat{D}_\lambda, c\delta_Q^{\text{pair}}(\lambda), c\delta_Q^{\text{chain}}(\lambda), c\rho_\pm, c\varepsilon, cp.$$

Then the algebraic pairwise and partition feasibility inequalities are invariant under the coordinate map $(\lambda, \varepsilon) \mapsto (\lambda, c\varepsilon)$.

Proof. Every algebraic endpoint and defining inequality is multiplied by the same positive constant. $\square$

Remark E.7 (When full certified-set membership is scale-covariant). Full package-conditioned membership is preserved only when the package-admissibility clauses, certificate object, query-margin certificate, policy clauses, report spaces, and required transports are themselves carried by the declared unit transport. Positive homogeneity of the inequalities alone does not license cross-scale package membership. Thus a mere numerical change of units cannot repair an algebraic obstruction, while a claim about the full certified set additionally requires transport of the surrounding package clauses.

Remark E.8 (No automatic residual-complement bridge). Neither an algebraic certificate obstruction nor package-conditioned non-licensing implies complement demand. Residual-complement claims require additional fiber, evolution, future-readout, and activation conditions. The present output may be used as evidence only after those independent clauses are verified.

E.8 Robustness, selection, and adaptive control

Before comparing outputs across completed configurations, the family must declare either a common report space $\mathcal{Y}$ whose identity maps are explicitly adopted as the transports, or separately declared transports into $\mathcal{Y}$. A common codomain alone is not a transport declaration.

Definition E.17 (Configuration robustness). Fix an invoked certified configuration set $\mathcal{A}$ and let

$$R : \mathcal{A} \rightarrow \mathcal{Y}.$$

The report is configuration-robust if R is constant on all of $\mathcal{A}$, component-robust if it is constant on a declared ordered or connected component, and configuration-sensitive otherwise.

A robustness scan is licensed only under a certificate valid jointly over the scanned set; adaptive scans require sequential validity. If R is sensitive, the report should provide a certified scan, transition boundaries between stable regions, or a declared selector.

Selection of one configuration is substantive only when the certified set has more than one element, the deliverable requires a single configuration, and either the report or operational cost varies over the set. For fixed λ , possible declared selectors include the lowest certified threshold when that minimum is attained, the highest task-sufficient threshold when attained, or a loss-minimizing threshold. If an endpoint is only an infimum or supremum, a selected interior value and its gap from the unattained bound must be reported. Across mechanisms, threshold order does not automatically order λ ; a family-level selector must declare either a two-stage mechanism/threshold criterion or a joint objective $J(\lambda, \varepsilon)$.

Definition E.18 (Adaptive resolution meta-protocol). An adaptive protocol is declared by

$$\bar{V} := (\mathfrak{R}_{\text{res}}^{\text{cert}}, \mathcal{Q}^{\sigma, \tau}, [p], \hat{e}, \text{Cert}, \mathsf{T}, \text{Ctrl}, \text{Accept}, \text{Reject}, \text{Stop}),$$

where

$$\mathfrak{R}_{\text{res}}^{\text{cert}} := \{\mathcal{R}_{\lambda}^{\text{cert}}\}_{\lambda \in \Lambda_{\text{res}}}.$$

Here $[p]$ is the optional separate policy refinement, and T declares the transports between visited package reports, including any explicitly adopted identity transports. Writing $c_n = (\lambda_n, \varepsilon_n)$, the update rule is

$$c_{n+1} = \text{Ctrl}(c_n, \hat{e}_n),$$

and every visited configuration must lie in the applicable package-conditioned certified set and, when policy is invoked, in its policy-restricted subset. Under partition semantics, the sequential certificate must also cover every visited threshold-edge domain. The object $\bar{V}$ governs a path through ordinary task packages $V_{\lambda_n, \varepsilon_n}^{\sigma, \tau}$ or their policy-refined counterparts and does not redefine the top-level package tuple. Fixed-threshold adaptation is the special case $\varepsilon_n \equiv \varepsilon_0$.

Protocol E.2 (Locked adaptive package path). Adaptive variation is a declared path through completed packages. It is not undeclared package drift when the certified interface refinements, task clauses, optional policy clause, report semantics, margin certificates, estimator, sequential mechanism certificate, transports, update law, acceptance/rejection rules, and stopping rule are declared before the substantive diagnostic. What is locked is $\bar{V}$ and the admissible path, not necessarily constant values of λ or ε . Post-hoc modification in response to the substantive result defines a new meta-protocol.

E.9 Reporting states and non-claims

Condition	Licensed report
Certificate or certification domain undeclared	Object-level threshold claim is not licensed.
Certified query-margin lower bound undeclared	Operational certified set is unavailable; only ideal geometry may be stated.
Pointwise certificate reused for a scan	Joint scan claim is not licensed.
Adaptive path without sequential validity	Adaptive certificate claim is not licensed.
Algebraic certificate slice empty	Current mechanism or margin certificates leave no algebraically admissible positive threshold.
Package-conditioned set empty but algebraic slice nonempty	Package admissibility, transport, or partition edge-certification clauses block the otherwise algebraically feasible configuration.
Full classification set empty but separation-only set nonempty	Directional positive distinctions can be protected, but the same threshold is not certified as a two-sided null/separation classifier.
Directional pairwise certified but partition certified set empty	Chain-separation / transitive-closure obstruction.
Pre-policy licensed set nonempty but policy-restricted set empty	Policy-induced non-licensing.
Exactly one licensed configuration	Unique configuration relative to family, task clauses, certificates, policy clauses, and report semantics.
Multiple configurations, robust report	Report the certified set; point selection is optional.
Multiple configurations, sensitive report	Report a certified scan, stable-region boundaries, or a declared selector.
Adaptive control	Report the meta-package, sequential certificate, and realized configuration path.
Post-hoc protocol change	New package; comparison requires declared transport.

Table 5: Certificate-relative resolution-feasibility reporting states for the fixed-target module.

This appendix does not claim a universal threshold, a derivation of the external error certificate, an automatic bridge to residual complements, or a complete theory of coarse-graining families in which the semantic target changes with λ . Its contribution is the typed location of standard robust-separation geometry inside the declared-package framework, together with the distinction between separation-only and full classification tasks, the distinction between directional pairwise and partition output semantics, and the associated certification/control requirements.

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